RENAL
MCQ
AL HARRIF



وَمَا تَوْفِيقِي إِلَّا بِاللَّهِ

استعن بالله ولا تعجز

اللَّهمَّ لا سَهلَ إلَّا ما جَعَلتَه سَهلًا، وأنتَ تَجعَلُ الحَزْنَ إذا شِئتَ سَهلًا

ملحوظة : هذه المذكرة شغل خاص بتيم الحريف ولا نسامح فى سرقتها او اخذ اى اسئلة منها

القناة الخاصة بالحريف

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Department	Lectures	Pages	Answers
Anatomy	Anatomy of the kidney and ureters	3	13
	Anatomy of the urinary bladder and urethra	13	23
	Development &congenital anomalies of the renal system	23	26
HistoLogy	Renal system	27	34
Microbiology	UTI	35	48
Parasitology	Renal system	49	52
Pharmacology	Diuretics	52	55
	Nephrotic drugs	55	57
Pathology	Congenital anomalies of the kidney	57	61
	Urinary outflow obstruction -Diseases of the urinary bladder	61	65
	Glomerular Diseases	65	74
	Disease of renal tubules & renal interstitium	74	79
	Tumors of the Urinary System	79	88
Biochemistry	Urinary stones	88	90
Physiology	Functional organization of the Kidney and renal blood flow	90	111
	Mechanism of urine formation	112	138
	Renal handling	139	152
	Renal Acid-Base Control Mechanisms	153	158
	Micturition and Micturition reflex	158	160

Anatomy of the kidney and ureters

1-The length of the ureter is about.... {Formative2024}

a) 10 inches (25 cm) b) Three inches c) 15 cm d) 6 meters

2- Which muscle does not form the posterior relation of the kidney: {Summar2024}

a. the psoas major b. the quadratus lumborum

c. transversus abdominis d-trapezius

3-Which muscle does not form the posterior relation of the kidney: {Formative2024}

a) transversus abdominis b) iliacus

c) the quadratus lumborum d) the psoas major

4- Which statement concerning relations of the kidneys is correct? {Final2021}

a. Diaphragm is directly related to the superior pole of both kidneys.

b. The left kidney is related to the duodenum.

c. The kidneys are posteriorly related only to psoas major, quadrates lumborum and the transverses abdominis.

d. The right kidney is related to the descending colon.

5- Renal coverings are the followings Except: {Final2021}

a. Renal fibrous capsule b. Renal pyramids.

c. Renal fascia d. Peri-renal and para-renal fat

6- Right ureter is crossed by following vessels Except: {Final2021}

a. Right supra renal artery. b. Right colic artery.

c. Testicular/ovarian artery. d. Ileo-colic artery.

7- If you cut the kidney from the capsule, the first part will be {Final2023}

a. Cortex b. Nephron c. Pelvis d. Sinus

8- Regarding the anterior relations of left kidney all are true except: {Final2023}

a. Liver b. Pancreas c. Ileum d. Descending colon

9- One of the following structures lie in front of the left kidney:

a. The stomach b. The right lobe of the liver

c. The second part of the duodenum d. The right colic flexure

10- All the following structures are anterior relation to the right kidney EXCEPT: {Summar2022}{3rd trial2022}{Formative2022}

a- Right supra renal gland b- Second part of duodenum

c- Right lobe of the liver d- Spleen

11- The right kidney is related posteriorly to:

a. Right lobe of liver b. Right supra renal gland

c. Right colic flexure d. Right psoas major muscle

12- What type of organ is the kidney? {Final2021}

a. Retroperitoneal organ b. Hollow organ unpaired organ

c. Intraperitoneal organ d. Parenchymal organ

13- What are the main differences between the relationships of the right and left renal veins: {Final2021}

a. Left renal vein - has posteriorly relationships with aorta

b. Right renal vein - has posteriorly relationships with aorta

c. Left renal vein - has anteriorly relationships with aorta

d. Right renal vein - has anteriorly relationships with aorta

14- In renal hilum ureter is characterized by the followings Except: {Final2021}

a. Most posterior structure. b. Most dilated structure

c. Most inferior structure d. Most lateral.

15- Left testicular vein drains in: {Final2021}

a. Inferior vena cava b. Left renal c. Left internal iliac d. Left common iliac.

16- Right testicular vein is drained in to: {Summar2022}{Formative2022}

a- Inferior vena cava b- Right renal vein

c- Right Internal iliac vein d- Right common iliac vein

17-left suprarenal and gonadal veins both drain to

A. superior mesenteric vein B. left renal vein C. common iliac vein D. IVC

18-The renal arteries originate at which of the following vertebral level?

a. T12 b. L1 c. L2 d. L3 e. L4

19-The covering of the kidney include the following EXCEPT:

A.Fibrous capsule. B.Thoracolumbar fascia. C. Perirenal fat.

D.Renal fascia. E. Pararenal fat.

20-Which statement concerning the relations of the kidneys is correct?

a. The kidneys are posteriorly related to quadratus lumborum

b. The right kidney is related to the descending colon

c. The pancreas is related to the left kidney above the stomach

d. The left kidney is related to the duodenum

21- Regarding the kidney, which one of the following statements is CORRECT ?

a. The right kidney is slightly higher than the left kidney

b. At the hilum of the kidney, the renal artery lies anterior to the renal vein

c. It is related posteriorly to the transvers colon

d. The left kidney is related anteriorly to the body of the pancreas

22-Left kidney is: {Merit}

a- lower than right kidney b- higher than right kidney

c- right & left ventricles. d- left atrium.

23- The most important relation(s) of the right kidney is(are): {Merit}

a- right lobe of the liver b- 2nd part of duodenum .

c- right colic flexure d- all of the above

24- The most important relation(s) of the left kidney is(are): {Merit}

a- stomach b- pancreas c- left colic flexure d- all of the above

25- Posterior relations of the kidneys are: {Merit}

a- muscles of anterior abdominal wall b- muscles of posterior abdominal wall

c- muscles of posterior pelvic wall d- muscles of the back

26-The medulla of the kidney is described as:

A. Light-colored and superficial B. Darker reddish-brown and deep to the cortex

C. Funnel-shaped and continuous with the ureter D. Transparent and fluid-filled

27-Which of the following statements about the vascularisation of the kidneys is false:

a. The left artery is longer than the right one

b. The renal artery is subdivided in segmental, lobar, interlobar, arcuate and interlobular arteries

c. The left renal vein is longer than the right one

d. The renal veins open into the inferior cava vein

28-Which kidney structure is funnel-shaped and directly continues with the ureter?

A. Cortex B. Renal capsule C. Renal pelvis D. Medulla

29-The hilum of the right kidney contains the following important structures except which?

a. Part of the right suprarenal gland b. Branches of the renal artery

c. The renal pelvis d. Tributaries of the renal vein

30-Which kidney surface is covered with peritoneum?

A. Anterior surface B. Posterior surface C. Both surfaces D. Neither surface

31-Which of the following statements is TRUE regarding the position of the kidneys?

A. Both kidneys lie at exactly the same vertebral level

B. The right kidney lies slightly lower than the left kidney

C. The left kidney lies lower due to the spleen

D. Both kidneys are retroperitoneal and at the same height

32-The kidney is typically described as a:

A. Oval-shaped organ B. Bean-shaped organ

C. Spherical organ D. Cylindrical organ

33-The renal pelvis is located at the:

A. Cortex surface B. Hilum of the kidney

C. Apex of the renal pyramids D. Renal papilla

34-The outer region of the kidney, which is light in color, is the:

A. Renal pelvis B. Medulla C. Cortex D. Renal sinus

35-The anterior surface of the kidney faces:

A. Posteromedial B. Anterolateral C. Medial D. Lateral

36-Which kidney surface lacks peritoneal covering?

A. Anterior surface B. Posterior surface C. Superior pole D. Inferior pole

37-Which hormone-related function is performed by the kidneys?

A. Increasing heart rate B. Releasing hormones that regulate blood pressure

C. Producing thyroid hormones D. Releasing adrenaline

38-Which of the following matches the correct pairing of kidney region and color?

A. Cortex – dark reddish brown B. Medulla – light colored

C. Cortex – light colored D. Renal pelvis – dark gray

39-Which kidney covering is directly attached to the kidney surface?

A. Perirenal fat B. Pararenal fat C. Fibrous capsule D. Renal fascia

40-Which structure encloses both the kidneys and the suprarenal glands?

A. Perirenal fat B. Pararenal fat C. Renal fascia D. Fibrous capsule

41-Which of the following is a primary function of the kidneys?

A. Production of bile B. Removing waste products and drugs from the body

C. Secretion of digestive enzymes D. Regulation of body temperature

42-Which of the following is NOT a function of the kidneys?

A. Balancing the body's fluids B. Removing drugs from the body

C. Regulating blood pressure D. Producing digestive enzymes for fat breakdown

43-Which of the following correctly lists the coverings of the kidney from inner to outer?

A. Renal fascia → Pararenal fat → Fibrous capsule → Perirenal fat

B. Fibrous capsule → Perirenal fat → Renal fascia → Pararenal fat

C. Perirenal fat → Fibrous capsule → Pararenal fat → Renal fascia

D. Fibrous capsule → Renal fascia → Perirenal fat → Pararenal fat

44-The pararenal fat is found:

A. Deep to the fibrous capsule B. Between the cortex and the medulla

C. Outside the renal fascia D. Between the renal pelvis and ureter

45-The perirenal fat is located:

A. Inside the renal cortex B. Between the fibrous capsule and renal fascia

C. Outside the renal fascia D. Between the cortex and medulla

46-The hilum of the kidney is located on which border?

A. Lateral border B. Superior border C. Medial border D. Inferior border

47-At which vertebral level is the hilum of the kidney typically located?

A. C7 B. T12 C. L1 D. L5

48-The ureter begins at the level of which vertebra?

A. T12 B. L1 C. L2 D. L4

49-The lymphatic vessels of the kidney primarily drain into the:

A. Axillary lymph nodes B. Internal iliac lymph nodes

C. Lateral aortic (para-aortic) lymph nodes D. Superficial inguinal lymph nodes

50- The site where the ureter is normally constricted:

a. At its middle b. Above the renal pelvis

c. At the pelvi-ureteric junction d. At the level of T12 vertebra

51-The abdominal part of the ureter is approximately how long?

A. 5 cm B. 10 cm C. 12.5 cm D. 25 cm

52-The pelvic part of the ureter is approximately how long?

A. 5 cm B. 5 inches (12.5 cm) C. 10 inches (25 cm) D. 7 inches (17.5 cm)

53-The abdominal ureter descends opposite the tips of which vertebrae?

A. T12–L2 B. L1–L3 C. L2–L5 D. L4–S1

54-Which artery supplies the pelvis of the ureter?

A. Testicular/ovarian artery B. Renal artery

C. Internal iliac artery D. Inferior vesicle artery

55-The abdominal ureter enters the pelvis by crossing:

A. The external iliac artery B. The bifurcation of the common iliac artery

C. The inferior vena cava D. The pelvic floor

56-Medially, the right ureter is related to which structure?

A. Abdominal aorta B. Inferior vena cava

C. Inferior mesenteric vein D. Portal vein

57-Medially, the left ureter is related to which structure?

A. Inferior vena cava B. Superior mesenteric artery

C. Inferior mesenteric vein D. Abdominal aorta

58-Posteriorly, the ureter is related to which muscle?

A. Quadratus lumborum B. Psoas major C. Iliacus D. Rectus abdominis

59-Which nerve crosses behind the ureter posteriorly?

A. Femoral nerve B. Obturator nerve

C. Genitofemoral nerve D. Lateral cutaneous nerve of thigh

60-Which of the following is NOT a source of arterial supply to the ureter?

A. Renal artery B. Common iliac artery

C. Inferior vesicle artery D. Superior mesenteric artery

61-The lower most part of the ureter is supplied by:

A. Internal iliac artery B. Inferior vesicle artery

C. Common iliac artery D. Renal artery

62-The parasympathetic fibers of the ureter arise from which spinal segments?

A. S1–S3 B. S2–S4 C. T10–L2 D. L3–L5

63-The sympathetic fibers of the ureter arise from which spinal segments?

A. T1–T5 B. T6–T9 C. T10–T12 and L1 D. L2–L5

64-Not included in the formation of urinary system: {Final2019}

A. Kidneys B. Ureters C. Urinary bladder D. Urethra E. Uterus

65-The ……. convey(s) urine from the kidneys toward the bladder{اسئلة القسم} {Final2020}

A) urethra B) nephrons C) prostate gland D) ureters

66-The functional unite of the kidney is {Summar2021}

a-Nephrons b-collecting tubule c-renal artery d-renal veins

67-The ……. glands lie on top of the kidneys. {اسئلة القسم}

A) pituitary B) growth C) adrenal D) thyroid

68-On the upper end of each kidney the……. rests on it. {Final2020}{Summar2021}

a-adrenal gland b-stomach c-spleen d-liver

69-The adrenal gland lies: {Final2019}

a.Anterior to the kidneys. b. Posterior to the kidneys. c - Superior to the kidneys.

d- Inferior to the kidneys. e- All of the above

70-The urinary system has the following functions except: {Final2019}

a-Eliminates wastes from the body. b- Regulates blood volume.

c-Regulates blood pressure. d- Control levels of electrolytes.

e-None of the above

71-The kidneys lie at: {Final2019}

a- The anterior abdominal wall b-The posterior abdominal wall

c-Anterior thoracic wall d-Posterior thoracic wall

e-Superior mediastinum

72-Regarding the kidney followings are true EXCEPT: {Final2019}

A. Related to muscles of anterior abdominal wall B. Lies in upper part of abdomen

C. Filter urine from the blood D. Has cortex, medulla and sinus

E. Its size 12 x 6 x 3 Cm

73-The outer layer of the kidney is the ……. {اسئلة القسم}

A.renal medulla B.renal pelvis C.renal cortex D.Non of the above

74-The urinary system is also called: {اسئلة القسم}

A.The renal system B.The gastric system

C.The Reproductive system D. The Respiratory system

75-The organs of the urinary system are: {اسئلة القسم}

A. The paired kidneys, the paired testes, the urinary bladder, and the urethra

B. The paired kidneys, the paired ovaries, the urinary bladder, and the urethra.

C. The paired fallopian tubes, the paired ureters, the urinary bladder, and the urethra.

D. The paired kidneys, the paired ureters, the urinary bladder, and the urethra.

76-Each kidney contains approximately: {اسئلة القسم}

A.1 hundred nephrons. B.1 thousand nephrons.

C.1 billion nephrons. D.1 million nephrons.

1. A	2. D	3. B	4. A	5. B
6. A	7. A	8. A	9. A	10.D
11.D	12.A	13.A	14.D	15.B
16.A	17.B	18.C	19.B	20.A
21.D	22.B	23.D	24.D	25.B
26.B	27.A	28.C	29.A	30.A
31.B	32.B	33.B	34.C	35.B
36.B	37.B	38.C	39.C	40.C
41.B	42.D	43.B	44.C	45.B
46.C	47.C	48.C	49.C	50.C
51.C	52.B	53.C	54.B	55.B
56.B	57.C	58.B	59.C	60.D
61.B	62.B	63.C	64.E	65.D
66.A	67.C	68.A	69.C	70.E
71.B	72.A	73.C	74.A	75.D
76.D	77.	78.	79.	80.

Anatomy of the urinary bladder and urethra

1-The neck of urinary bladder in males is supported by..... {3rd tral2024}

a. pubocervical b. puboprostatic

c. median umbilical ligament d.lateral umbilical ligament

2-Vasculature of the urinary bladder in female are from..... {Formative2024}

a) inferior vesical artery and vaginal artery b) superior vesical artery and vaginal artery

c) Inferior vesical artery and superior vesical artery d) Superior vesical artery only

3- Which is anatomical structure that mark vesicle trigone? {Final2021}{3rd trial2022} {Final2023}

a. The ureter opening at the antero- inferior angle

b. The urethra opening at the postero- inferior angle

c. The ureters openings one side & another of inferior ridge

d. The urethra orifice at the apex of the trigone.

4-The length of penile part of male urethra is about..... {Formative2024}

A) 15cm B) 20 cm C) 25 cm D) 4cm

5- All of the following are parts of the urethra except {Final2023}

a. Prostatic b. Spongy c. Premembranous d. Membranous

6-Which of the following is the widest and most dilatable part of the male urethra? {Merit}

a. Prostatic urethra b. Urethral orifice c. Penile urethra d. Membranous urethra

7- Regarding male urethra the followings are true Except: {Final2021}

a. Prostatic part (3Cm.) passes through the prostate gland.

b. Ejaculatory ducts is formed by union of vas deferens and ureter

c. Membranous part is the shortest part (one Cm.)

d. Penile part is the longest part (16 Cm.)

8- Average length of female urethra is: {Final2021}{Formative2022}{Final2022}

a. about 4 Cm b. about 10 Cm. c. about 16 Cm. d. about 25 Cm

9-The length of the male urethra is approximately:

A. 8–10 cm B. 12–15 cm C. 18–20 cm D. 25–30 cm

10-All are parts of male urethra EXCEPT: {3rd trial2022}

a-Premembranous b-Preprostatic c-Spongy d-Membranous

11-Which of the following describes the apex of the bladder? {Merit}

a. Connected to the umbilicus by median umbilical ligament

b. Rests on upper surface of the prostate

c. Lies behind lower margin of symphysis pubis d. Points posteriorly

12- Vesicle trigone of the urinary bladder presents at the {Summar2022}{Formative2022}

a- Anterior wall b- Posterior wall c- Neck of the urinary bladder d- Superior wall

13-Which of the following statements describes the location of the vesicle trigone? {Merit}

a. At the base of the bladder b. At the anterio-inferior wall of the bladder

c. At the superior wall of the bladder d. At the neck of the bladder

14-Space which located anterior to the urinary bladder: {Formative2022}

a. Retropubic b. Vesicogenital c. Vesico rectal d. Perotineal

15- Female patient has a tumor at the level of posterior wall of urinary bladder. An ultrasound it's performed and it seems that the tumor invades posterior structures. Name what structure can be involved: {Final2021}

a. Rectum and anal canal b. Vagina c. Prostate d. Ileal loops.

16- If a patient has a urinary bladder tumor and both kidneys have hydronephrosis (that means that urine cannot be drained into the urinary bladder). What would be the most probable location of the tumor in the urinary bladder? {Final2021}

a. Superior part of the vesicle trigone b. Superior wall of the bladder

c. Base of the bladder d. The urethral orifice

17- Name the space located anterior to the urinary bladder: {Final2021}{Summar2022}

a. Vesico genital b. Vesico rectal c. Retropubic d. Peritoneal

18- Arterial supply of the urinary bladder is done by branches from the following arteries Except: {Final2021}

a. Superior vesical arteries b. Inferior vesical arteries in male and vaginal arteries in females

c. Obturator arteries d. Branches of the external iliac artery

19- Not related to male bladder posteriorly: {Final2021}

a. Rectum b. Seminal vesicles,

c. Termination of ureters d. Termination of the vas deferens.

20-Which branch of the internal iliac artery is most associated with arterial supply to the bladder?

A. Internal pudendal. B. Hypogastric. C. Superior vesical. D. Lateral sacral

21- Which structure does NOT form part of the boundary defining the trigone of the bladder?

a) Left Ureteric Orifice. b) Right Ureteric Orifice. c) Urachus. d) Urethral Orifice

22-The median umbilical ligament is a remnant of which embryonic structure?

A. Ureter B. Urachus C. Allantois D. Cloaca

23-In males, the ligament connecting the bladder to the pubic bone is called:

A. Pubovesical ligament B. Puboprostatic ligament

C. Median umbilical ligament D. Lateral ligament of bladder

24-In females, the ligament connecting the bladder to the pubic bone is called:

A. Puboprostatic ligament B. Pubovesical ligament

C. Median umbilical ligament D. Round ligament

25-The urinary bladder receives its arterial blood supply primarily from:

A. Renal arteries B. Superior and inferior vesical arteries

C. Internal pudendal artery only D. Testicular or ovarian arteries

26-The length of prostatic part of male urethra is about.....

A) 15cm B) 1 cm C) 25 cm D) 3cm

27-Which part of the male urethra is the widest?

A. Preprostatic B. Prostatic C. Membranous D. Spongy (penile)

28-Which part of the male urethra is longest?

A. Preprostatic B. Prostatic C. Membranous D. Spongy (penile)

29-All of the following are TRUE about the preprostatic urethra EXCEPT:

A. It is about 1 cm long B. It is associated with the internal urethral sphincter

C. It passes through the prostate D. It extends from the bladder base to the prostate

30- The membranous part of the male urethra is located in which of the following structures:

a. External urethra sphincter b. Urinary bladder c. Prostate d. Bulb of the penis

31-Which part of the urethra passes through erectile tissue of the penis?

A. Membranous urethra B. Prostatic urethra

C. Spongy urethra D. Preprostatic urethra

32-The internal urethral sphincter lies through

A. Penile part of the urethra B. Membranous part of the urethra

C. Preprostatic part of the urethra D. Prostatic part of the urethra

33-All of the following are parts of the male urethra EXCEPT:

a. Penile b. Prostatic c. Isthmus d. Membranous

34-The male urethra begins at:

A. External urethral orifice B. Internal urethral orifice of the bladder

C. Prostate apex D. Neck of the bladder

35-The female urethra begins at:

A. External urethral orifice B. Internal urethral orifice of the bladder

C. Vestibule of the vagina D. Urogenital diaphragm

36-The preprostatic urethra extends from:

A. External urethral orifice to the penis B. Base of the bladder to the prostate

C. Prostate to the membranous urethra D. Seminal colliculus to the external urethral meatus

37-The male urethra is divided into how many parts?

A. 2 B. 3 C. 4 D. 5

38-The preprostatic urethra is approximately how long?

A. 0.5 cm B. 1 cm C. 2–3 cm D. 4–5 cm

39-The membranous urethra is approximately how long?

A. 1 cm B. 2 cm C. 3–4 cm D. 5 cm

40-The spongy urethra is approximately how long?

A. 5 cm B. 10 cm C. 15 cm D. 20 cm

41-Which of the following is a part of the male urethra?

A. Prostatic B. Membranous C. Spongy (penile) D. All of the above

42-Which sphincter is associated with the preprostatic urethra?

A. External urethral sphincter B. Internal urethral sphincter

C. Anal sphincter D. Puboprostatic sphincter

43-The spongy urethra is surrounded by which structure?

A. Corpus cavernosum B. Corpus spongiosum C. Prostate D. Bladder

44-The prostatic urethra is surrounded by:

A. Bladder neck B. Penis C. Prostate gland D. Urethral sphincter only

45-The female urethra passes through which structure?

A. Penis B. Prostate C. Urogenital diaphragm D. Bladder neck

46-All of the following are TRUE about the prostatic urethra EXCEPT:

A. It is the widest part of the urethra B. It is about 3 cm long

C. It is surrounded by the prostate D. It passes through the penis

47-The membranous urethra passes through which structure?

A. Penis B. Prostate C. Urogenital diaphragm D. Bladder neck

48-Which sphincter completely surrounds the membranous urethra?

A. Internal urethral sphincter B. External urethral sphincter

C. Anal sphincter D. Preprostatic sphincter

49-The spongy urethra is enlarged at the tip of the penis to form the:

A. External urethral sphincter B. Navicular fossa C. Urethral crest D. Seminal colliculus

50-The external urethral orifice in females opens into:

A. Vestibule of the vagina B. Clitoris C. Labia majora D. Urethral crest

51-The female urethra passes just below which structure?

A. Clitoris B. Uterus C. Labia majora D. Vaginal orifice

52-The membranous urethra receives its nerve supply from:

A. Inferior hypogastric plexus B. Perineal nerve C. Vagus nerve D. Middle rectal nerve

53-The prostatic urethra is supplied by:

A. Perineal nerve B. Inferior hypogastric plexus

C. Pudendal nerve only D. Sympathetic chain

54-The internal urethral sphincter is controlled by:

A. Somatic nerves B. Autonomic nerves C. Perineal nerve D. Vagus nerve

55-The external urethral sphincter is supplied by:

A. Sympathetic nerves B. Autonomic nerves

C. Perineal nerve D. Inferior hypogastric plexus

56-The prostatic and membranous parts of the male urethra receives its blood supply from:

A. Inferior vesical B. middle rectal C. internal pudendal D. All of above

57-The prostatic part of the male urethra drains lymph into:

A. Superficial inguinal lymph nodes B. Internal iliac lymph nodes

C. External iliac lymph nodes D. Deep cervical lymph nodes

58-The spongy (penile) urethra drains lymph into:

A. Internal iliac lymph nodes B. Superficial inguinal lymph nodes

C. Deep pelvic lymph nodes D. Para-aortic lymph nodes

59-The spongy (penile) urethra is supplied by:

A. Superior vesical artery B. Urethral branches from internal pudendal artery

C. Inferior vesical artery D. Renal artery

60-Which of the following arteries does NOT contribute to the male urethra?

A. Inferior vesical artery B. Middle rectal artery

C. Internal pudendal artery D. Renal artery

61-The spongy (penile) urethra is supplied by:

A. Inferior hypogastric plexus B. Perineal nerve

C. Sympathetic fibers from T10–L2 D. Pudendal artery

62-The perineal nerve supplies all of the following EXCEPT:

A. Membranous urethra B. Spongy urethra

C. External urethral sphincter D. Prostatic urethra

63-Which part of the male urethra is supplied by autonomic nerves?

A. Spongy urethra B. Membranous urethra

C. Internal urethral sphincter D. External urethral sphincter

64- The external urethral sphincter lies in the:

a) Pelvis b) Deep perineal pouch c) Superficial perineal pouch d) Pelvic inlet

65- The internal urethral sphincter is formed of which type of muscle?

a) Striated muscle fibers b) Smooth muscle fibers

c) Cardiac muscle d) Mixed muscle fibers

66- The external urethral sphincter is controlled by:

a) Autonomic nervous system b) Somatic nervous system

c) Enteric system d) Sympathetic only

67- Which of the following is true about the internal urethral sphincter?

a) Acts voluntarily b) Well developed in females

c) Prevents retrograde ejaculation in males d) Surrounds the membranous urethra

68- The external urethral sphincter is made of:

a) Smooth muscle b) Striated skeletal muscle c) Elastic fibers d) Mixed fibers

69- Contraction of the internal urethral sphincter during ejaculation:

a) Allows semen to enter bladder b) Prevents semen from entering bladder

c) Pushes urine out d) Relaxes urethra

70- Which sphincter is voluntary?

a) Internal urethral sphincter b) External urethral sphincter c) Both d) None

71-The ……. extend(s) from the urinary bladder to an external opening. {اسئلة القسم}

A) ureter B) renal arteries C) urethra D) glomerulus

72-Followings are parts of male urethra EXCEPT: {Final2019}

A-Prostatic B. Fundus C. Membranous D. Preprostatic

73-Which of the following is NOT considered a part of the male urethra? {Final2019}

a-Prostatic b-Membranous c-Vasapore

d- Sponge e-A and B

74-The tube that carry the urine from the kidney to the urinary blader {Mid2025}

A-ureters B-urethra C-nephrons D-penis

75-Urethra of female lies .....to vagina orifice {Mid2025}

A-Anterior B-posterior C-medial D-lateral

76- vagina orifice of female lies .....to Urethra

A-Anterior B-posterior C-medial D-lateral

77-..... Not content of Urinary system {Final clinical}

a. kidney b. ureter c. urethra d. vas deference

78-Arrange the following structures in the correct sequence in which urine passes through them to the external environment: (1) ureter, (2) renal pelvis, (3) calyx, (4) urinary bladder, (5) urethra. {اسئلة القسم}

A.5,4,3,2,1 B.5,3,1,4,2 C.3,1,2,4,5 D.3,2,1,4,5

79-The renal .....is continuous with the ureter. {اسئلة القسم}

A.glomerulus B.cortex C.pelvis D.medulla

80-The female urethra is about..... cm long: {اسئلة القسم}

a.4. b.10. c.15 d. 20

81-Which of the following parts is not found in the urinary system {اسئلة القسم}

A.Urethra B.Ureter C.Rectum D.Bladder

1. B	2. B	3. D	4. A	5. C
6. A	7. B	8. A	9. C	10.A
11.A	12.B	13.A	14.A	15.B
16.A	17.C	18.D	19.D	20.C
21.C	22.B	23.B	24.B	25.B
26.D	27.B	28.D	29.C	30.A
31.C	32.C	33.C	34.B	35.B
36.B	37.C	38.B	39.B	40.C
41.D	42.B	43.B	44.C	45.C
46.D	47.C	48.B	49.B	50.A
51.A	52.B	53.B	54.B	55.C
56.D	57.B	58.B	59.B	60.D
61.B	62.D	63.C	64.B	65.B
66.B	67.C	68.B	69.B	70.B
71.C	72.B	73.C	74.A	75.A
76.B	77.D	78.D	79.C	80.A
81.C	82.	83.	84.	85.

Development &congenital anomalies of the renal system

1- All are derivatives of mesonephric duct in females except {Summar2024}

a. Gartner's duct b. Back of urethra

c. vesicourethral canal d. The ureteric bud and its derivatives

2-All are derivatives of mesonephric duct in males except {Formative2024}

a) Edipidymis b) Vasa efferentia c) Ejaculatory duct d) Vas deferens

3- The ductus deferens is derived from the:

a. Ureteric bud b. Mesonephric ducts

c. Metanephric cap (blastemal) d. Paramesonephric ducts

4- Urinary bladder develops from: {Merit}

a- vesicourethral canal b- mesonephric ducts

c- allantois (urachus) d- all of the above

5- The permanent kidney develops from: {Merit}

a- blastema (metanephrogenic cap) only b- ureteric bud only

c- blastema & ureteric bud d- epopheron & ureteric bud

6-Development of the kidney is from..... {Formative2024}

a) Endoderm b) Ectoderm c) paraxial mesoderm d) the intermediate mesoderm

7-Remnant of allantois is called ... {Formative2024}

a) Meckel's diverticulum b) Urachus c) epoophoron d) Oophoron

8- Derivatives of the anterior portion is the primitive urogenital sinus are the followings Except: {Final2021}

a. Urinary bladder. b. Ureter. c. Pelvic urethra. d. Definitive urogenital sinus.

9- Abnormal fusion of lower poles of both kidneys leads to: {Final2021}

a. Renal dysplasia b. Congenital poly cystic kidney.

c. Pelvic kidney. d. Horseshoe kidney

10- Not related to bladder defects: {Final2021}

a. Hypospadias b. Urachal fistula. c. Urachal cyst. d. Urachal sinus.

11-Abnormal fusion of upper and lower poles of kidney is called: {3rd trial2022}

a-Congenital polycystic kidney b-Horseshoe kidney

c-Rosette shaped kidney d-Pelvic kidney

12- Horse-shoe kidney is a: {Formative2022}{Merit}{شتاء}

A. Senile kidney B. Congenitally malformed kidney

C. Hypertensive kidney D. Atherosclerotic kidney

13- The kidney fused at both upper & lower poles is called: {Final2023}

a. Pelvic kidney b. Rosette shaped kidney c. Horse-shoe kidney

14- Diaphragm develops from all EXCEPT: {Formative2022}

a. Septum transversum b. ventral mesentery of esophagus

c. Pleuro-peritoneal membranes. d. Thoracic wall mesoderm

15- The commonest anomaly of urethra (ventral opening of urethra) is called ……{3rd tral2024}

a. epispadius b. hypospadias c. stenosis d. rectovesical fistula

16-Choose the correct order for developmental stages of urinary system.

a) Metanephros, Mesonephros, Pronephros. b) Mesonephros, Metanephros, Pronephros.

c) Metanephros, Pronephros, Mesonephros d) Pronephros, Mesonephros, Metanephros.

17-Which of the followings is a derivative of ureteric bud:

a) Collecting tubules b) Distal convoluted tubules

c) Loop of Henle d) Proximal convoluted tubules

18-In hypospadias the opening of the urethra lies on:

a) Dorsal surface of penis b) Ventral surface of penis c) Perineum d)Tip of penis.

19-Apex of urinary bladder is derived from:

A. Proximal part of urachus B. Distal part of urachus

C. Vesicourethral canal D. Urogenital sinus

20- Urethra develops from: {Merit}

a- vesicourethral canal only b- urogenital sinus only

c- vesicourethral canal & urogenital sinus d- trigone

21- The first kidney during the development is called: {Merit}

a- pronephros b- mesonephros c- metanephros d- epopheron

22- The second kidney during the development is called: {Merit}

a- pronephros b- mesonephros c- metanephros d- epopheron

23- The third kidney during the development is called: {Merit}

a- pronephros b- mesonephros c- metanephros d- epopheron

24- Urinary system develops from: {Merit}

a- outer mesoderm b- lateral mesoderm

c- paraxial mesoderm d- intermediate mesoderm

25-The ureteric bud give rise to all the following derivatives except:

a. Renal pelvis b. Collecting tubules c. Major calyces d. Bowman's capsule

26-A 4-year-old girl was still in diapers because she was continually wet. The pediatrician saw urine coming from the infant's vagina. An intravenous urogram showed two renal pelves and two ureters on the right side. One ureter was clearly observed to enter the bladder, but the termination of the other one was not clearly seen. A pediatric urologist examined the child under general anesthesia and observed a small opening in the posterior wall of the vagina. What congenital anomalies would be suspected?

a. Bifid ureter b. Ectopia vesica c. Ectopic kidney d. Ectopic ureter

1. C	2. B	3. B	4. D	5. C
6. D	7. B	8. B	9. D	10.A
11.B,C	12.B	13.B	14.B	15.B
16.D	17.A	18.B	19.A	20.C

21.A	22.B	23.C	24.D	25.D
26.D	27.	28.	29.	30.

HistoLogy

Renal system

1-What type of epithelium lines the renal pelvis, providing stretchability to accommodate urine volume changes? {سؤال في المحاضرة}

a.Simple squamous epithelium

b.Stratified squamous epithelium

c.Transitional epithelium (urothelium)

d.Pseudostratified columnar epithelium

2-Which layer primarily provides contractile motion to propel urine from the renal calyces to the ureter? {سؤال في المحاضرة}

a.Mucosa with transitional epithelium

b.The muscularis layer with smooth muscle

c.Serosal layer

d.Submucosal glandular layer

3-during bladder distension, how do the superficial cells of the transitional epithelium change? {سؤال في المحاضرة}

a.They become more cuboidal and folded

b.They thicken and increase keratinization

c.They flatten and their urothelial plaques spread out

d.They detach and are shed into the lumen

4-Which feature allows the transitional epithelium in the ureter and bladder to accommodate large volumes of urine? {سؤال في المحاضرة}

a.Presence of cilia

b.Multiple layers with flexible superficial cells

c.Thick keratinized layer

d.Continuous secretion of mucus

5-Podocyte {Formative2024}

a) lines the parietal layer of Bowman's capsule b) epithelial cell

c) lines the glomerular capillaries d) non branched cell

6-The juxta-glomerular apparatus is formed from three cells: {Formative2024}{Merit}

a) Juxta-glomerular cells, macula densa and lacis cells.

b) Macula densa, Bowman's capsule and lacis cells

c) Juxta-medullary cells, podocytes, and macula densa.

d) Macula densa, mesangial cells and lacis cells.

7- The juxtaglomerular apparatus secretes: {Summar2024}

a. Renin. b. Erythropoietin. c. Vitamin D. d. Angiotensin II.

8- Podocyte {3rdtral2024}

a lines the parietal layer of Bowman's capsule b. non branched cell

c. lines the glomerular capillaries d. line the visceral layer of Bowman's capsule

9- What type of cells are podocytes? {Summar2024}

a. Epithelial cells b. Muscle cells c. Connective tissue cells d. Nervous cells

10- What is the main function of the renal corpuscle in the kidney? {3rdtral2024}

a. Filtration of blood b. Secretion of hormones

c. Reabsorption of water d. Excretion of urine

11- What type of cells line the thin segment of the loop of Henle? {Summar2024}

a. Simple cuboidal b. Simple columnar c. Simple squamous d. Transitional

12- Which structure is responsible for secreting renin in the kidney? {Summar2024}

a. Proximal convoluted tubule b. Juxtaglomerular cells

c. Macula densa d. Bowman's capsule

13-Which of the following structures are not related to juxtaglomerular apparatus? {Formative2024}

a) Modified smooth muscle of afferent arteriole b) Macula densa

c) Extraglomerular mesangial cells d) Intraglomerular mesangial cells

14-Which of the following structures is related to juxtaglomerular apparatus: {3rd trial2022}{Final2023}{3rd trial2024}

a-Macula densa b-Podocyte c-Endothelial cells of glomeruli

d-Loop of Henel e. Intra raglomerular mesangial cells

15-The lining epithelium of penile urethra is. {Formative2024}

a) Stratified squamous. b) Stratified columnar epithelium.

c) Stratified cuboidal. d) Transitional epithelium

16-Regarding the lining epithelium of the glomerular capillaries. Which of the following is true: {3rd trial2022}{Final2023}{3rd tral2024}

a-Continuous endothelium b-Non-fenestrated endothelium with diaphragm

c-Fenestrated endothelium with diaphragm d-Fenestrated endothelium without diaphragm

17-Stratified squamous epithelium is related to: {3rd trial2022}{Final2023}

a-Prostatic urethra b-Membranous urethra

c-Initial part only of membranous urethra d-Terminal part of penile urethra

18-Macula densa is present in {Formative2022}

a. Proximal tubules b. Distal tubules c. Lope of Henle d. Collecting tubules

19-Macula densa is ……..

A. Modified region of tubular epithelium of DCT

B. Modified region of tubular epithelium of PCT

C. Modified region of tubular epithelium of loop of Henle

D. Modified region of tubular epithelium of collecting tubule

20-Urinary bladder is lined by : {Formative2022}

a. Simple squamous epithelium b. Simple cuboidal epithelium

c. Simple columnar epithelium d. Transitional epithelium

21-Proximal convoluted tubules are characterized by {Formative2022}

a. Apical cilia b. Apical microvilli

c. Basophilic cytoplasm d. Apical secretory granules

22-The structure of juxta glomerular apparatus include: {Formative2022}

a. afferent arteriole b. renal glomeruli c. collecting ducts d. lope of Henle

23- Podocyte is : {Final2021}

a. epithelial cell b. lines the visceral layer of Bowman's capsule

c. non branched cell d. a & b

24-Which of the following structures is related to podocytes? {Summar2022}

a- Glomerular capillaries b- Mesangial cells

c- Parietal layer of Bowman's capsule d- Proximal convoluted tubules

25-Which type of mesangial cells is related to juxtaglomerular apparatus? {Summar2022}

a- Intraglomerular mesangial cells

b- Extraglomerular mesangial cells

c- Both intra and extraglomerular cells

d- No mesangial cell present in juxtaglomerular apparatus

26- Medullary rays represent: {Final2021}

a. Collecting tubules b. Collecting ducts c. Distal tubules d. Loop of Henel

27- Regarding EM characters of proximal convoluted tubule, which is wrong? {Final2021}

a. No apical microvilli b. Numerous apical microvilli

c. Basal infoldings d. Basal mitochondria

28- Which of the following vessels are not related to glomerular circulation: {Final2021}

a. Vasa recta vera b. Capsular plexus c. Afferent arterioles d. Both a &b

29-The lining epithelium of membranous urethra is : {Final2021}{Final2022}

a. Stratified columnar epithelium b. Stratified squamous

c. Transitional epithelium d. None of the above

30- The terminal part of female urethra is lined with {Final2021}

a. Simple squamous epithelium b. Stratified columnar

c. Stratified squamous d. Transitional

31- Prostatic urethra is lined with {Summar2022}

a- Stratified columnar epithelium b- squamous epithelium

c- epithelium d- cuboidal epithelium

32-What type of epithelium lines the prostatic urethra?

a. Transitional (urothelium) b. Simple columnar

c. Stratified squamous d. Simple squamous

33-Blood in the renal arcuate arteries flows next into which vessels?

a. Efferent arterioles b. Interlobar arteries c. Glomerular capillaries d. Afferent arterioles

34-Which region of the kidney contain glomeruli?

a) Cortex. b) Medulla. c) Calyx. d) Pelvis. e) Papillae.

35-Which of the following is TRUE regarding renal lobe?

a) Is separated from other lobes by connective tissue septa.

b) Is visible grossly as a rounded bulge on the kidney surface.

c) Consists of one medullary pyramid with the associated cortex.

d) Consists of one medullary ray and its associated cortex

36- Characteristics of the distal convoluted tubule are: {Merit}

a. Cell borders are more distinict than in proximal tubules.

b. Lined by cuboidal cells.

c. Their walls appear to have more cells than those of proximal tubules of equal size.

d. All of the above.

37- Podocytes are present in: {Merit}

a. The wall of the blood capillaries. b. In the pars nervosa of the pituitary gland.

c. The alveoli of the lung. d. The epithelial lining of the bowman's capsule.

38- The proximal convoluted tubule of the kidney is lined by: {Merit}

a. Cuboidal epithelium with a brush border. b. Cuboidal epithelium without a brush boder.

c. Ciliated pseudostratified columnar epithelium. d. Non of the above.

39- Podocytes are: {Merit}

a. Glomerular endothelial cells. b. Parietal epithelial cells of Bowman.

c. Mesangial cells. d. Visceral epithelial cells of Bowman's capsule.

40-The glomerular filteration barrier is formed from …….layers

A. One layer B. Four layers C. Two layers D. Three layers

41-What is the foot process on podocytes?

A. Visceral layer of Bowman's capsule. B. Parietal layer of Bowman's capsule.

C. Pedicels. D. Juxtaglomerular cells

42-What is a renal pyramid and its associated cortex referred to?

A. Medulla. B. Lobe. C. Renal columns. D. Nephron

43-The cortex of the kidney is dark brown and granular due to the presence of:

A. Medullary pyramids. B. Collecting tubules

C. Malpighian corpuscles D. Medullary arrays

44-Juxtamedullary nephrons contain ……. loops of Henle.

A. Shorter. B. Equal. C. Longer. D. Larger

45-The glomerular blood capillaries are:

a. Continuous capillaries b. Fenestrated capillaries

c. A network of arteriovenous anastomosis d. Sinusoidal blood capillaries

46-Regarding the podocytes all are wrong Except:

a. Have primary and secondary processes b. Line the parietal layer of Bowman's capsule

c. Are simple columnar cells d. Have microvilli

47-Visceral layer of Bowman's capsule is formed of:

a. Podocytes b. Simple cubical cells c. Endothelial cells d. Simple squamous cells

48- The glomerular filtration barrier is composed of all the following Except:

a) fenestrated capillary endothelium. b) macula densa .

c) basement membrane. d) podocytes.

49-The glomerular filtration barrier consists of the followings except:

a. The fenestrated endothelium b. Continous endothelium of the glomerular capillaries

c. The basement membrane d. The filtration slit membrane

50-Regarding the Bowman's capsule; all of the followings are true except:

a. Formed of a parietal and visceral layers b. Shows Bowman's space or capsular space

c. It is a double walled cup d. Share in the formation of Juxtaglomerular apparatus

51-Juxta-glomerular cells are found in:

a. The media of afferent arterioles b. The Intima of afferent arterioles

c. The Intima of efferent arterioles d. The wall of DCTs

52-Juxtaglomerular cells are a modification in:

a) Distal convoluted tubule. b) Proximal convoluted tubule.

C)Loop of Henle. d)Afferent arteriole.

53-A brush border is present in the:

a)Glomerulus. b)Proximal convoluted tubule. c)Loop of Henle. d)Collecting tubule

54- On examination of the section of the kidney by light microscope, which cell that has a brush border?

a- Cells of PCT B-Cells of DCT c-Podocyte d-Principal cell of the collecting tubule.

55-.....is/are ion transporting cell(s)

a.Cells of DCT b. Cells of the parietal layer of Bowman capsule

c-Extra mesengial cells d.intramesengial cells

56-What is the entry point to the kidney for the renal artery, renal vein, lymphatics and nerves called?

a. Renal column b. Renal capsule c. Renal pyramid d. Renal hilus

1. C	2. B	3. C	4. B	5. B
6. A	7. A	8. D	9. A	10.A
11.C	12.B	13.D	14.A	15.B
16.D	17.D	18.B	19.A	20.D
21.B	22.A	23.D	24.A	25.B
26.A	27.A	28.D	29.A	30.C
31.A	32.A	33.B	34.A	35.C
36.D	37.D	38.A	39.D	40.D
41.C	42.B	43.C	44.C	45.B

46.A	47.A	48.B	49.B	50.D
51.A	52.D	53.B	54.A	55.A
56.D	57.	58.	59.	60.

Microbiology

UTI

1-Which of the following is the most common causative agent of UTI? {3rd tral2024}

a. Staphylococcus aureus b. Klebsiella pneumoniae

c. Escherichia coli d. Pseudomonas aeruginosa

2-Which part of the urinary system is most commonly affected by UTIs? {3rd tral2024}

a. Kidneys b. Ureters c. Bladder and urethra d. Renal pelvis

3-What is the most effective strategy to prevent catheter-associated UTIs? {3rd tral2024}

a. Avoiding catheterization when unnecessary b. Increasing water intake

c. Drinking cranberry juice d. Daily use of antibiotics

4- Which one of the following is not a lower urinary tract infection? {Summar2024}

a. Cystitis b. Prostatitis c. Pyelonephritis d. Urethritis

5-The most common cause of UTI in the community is..... {Formative2024}

a) S. aureus b) Enterobacter c) Klebsiella d) E. coli

6-A 37-year-old woman with a history of urinary tract infections comes to the emergency department with burning on urination along with frequency and urgency. She says her urine smells like ammonia. The cause of her urinary tract infection is likely to be {Formative2024}

a) E. aerogenes b) E. coli c) Citrobacter freundii d) Proteus mirabilis

7- Acute haemorrhagic cyctitis in children is mostly caused by..... {Final2021} {Summar2024}{Formative2022}

a. Proteus mirabilis b. Staphylococcus aureus c. Chlamydia d. Adenoviruses

8- A 60-years-old woman is hospitalized following a stroke and develops a high grade fever with chills. She is catheterized due to urinary incontinence. Blood cultures and Gram stain are performed by the laboratory. The organisms isolated are gram-positive cocci that are catalase-negative and capable of growth in 6.5% sodium chloride. Which of the following is the most likely causal agent? {Final2021}

a. Staphylococcus aureus b. Staphylococcus saprophyticus

c. Enterococcus faecalis d. Streptococcus pyogenes

9- Which is the most common route of bacterial entry to the urinary tract? {Final2021} {Formative2022}

a. Ascending infection b. Blood borne spread

c. Lymphatic spread d. Direct extension from other organs

10- A 45-years-old woman presents the emergency department with intense pain in her lower back and a burning sensation upon urination. A urine culture was taken and plated on MacConkey agar. Gram-negative rods that did not ferment lactose were identified. Which virulence factor of the causal agent is most important to pathogenesis? {Final2021} {Formative2022}{أسئلة القسم}

a. Coagulase b. Urease c. Catalase d. Exotoxin

11- Significant bacteriuria: {Summar2022}{3rd trial2022}{Summar2024}

a- Presence of ≥ 10^3 CFU/ml of urine b- Presence of ≥ 10^4 CFU/ml of urine

c- Presence of ≥ 10^5 CFU/ml of urine d- Presence of ≥ 102^6 CFU/ml of urine

12- Bacteruria is detected if. {Final2022}

a. Bacteria ≥ 10^5

13- Which of the following is Not an infection term used for UTI? {Summar2022}{3rd trial2022}

a- Endometriosis b- Cystitis c- Pyelonephritis d- Urethritis

14- A 40-year-old women has dysuria, urine analysis showed an Alkaline urine and infection with a urea splitting organism, What is the type of stone? {Final2023}

a. Phosphate b. Struvite c. Calcium stone d. Uric acid

15- Mg ammonium PO4 (struvite) stones are associated with……PH. {Final2021}

a. Acidic b. Basic c. Neutral d. none of the above

16- Sterile pyuria can be caused by all except: {Final2023}

a. Streptococcus saprophyticus b. Mycoplasma hominis

c. Chlamydia d. Mycobacterium Tb

17- Complications of UTI include the followings ECXEPT: {Formative2024}

a. Recurrent infections b. Urethral stricture

c. Permanent kidney damage d. Vesico-ureteric reflux

18- A 60-year-old woman with uti the organisms isolated are gram-positive cocci that are catalase-negative and capable of growth in 6.5% sodium chloride. Which of the following is the most likely causal agent? {Formative2022}

a. Staphylococcus aureus b. Staphylococcus saprophyticus

c. Enterococcus faecalis d. Streptococcus pyogenes

19- A young woman presents with recurrent urinary tract infections caused by the same Proteus mirabilis strain. What is the major concern? {Formative2024}

a. She does not take her medication.

b. She is pregnant as pregnant patients are more susceptible to UTIs.

c. She has a bladder or kidney stone.

d. Her partner is infected.

20- Significant bacteriuria means that bacterial count is : {أسئلة القسم}

A - 1000 organisms per ml of urine . B- 10,000 organisms per ml of urine .

C- 100,000 or more organisms per ml of urine .

21-Positive Urine Culture is diagnostic when bacterial cell count is

A. =>10^6 CFU/mL B. =>10^7 CFU/mL C. =>10^5 CFU/mL D. =>10^9 CFU/mL

22- Precautions during collection of urine sample for laboratory diagnosis in case of UTI include : {أسئلة القسم}

A - Cleaning and drying of external genitalia . B- Middle urine sample is needed .

C - Collection in sterile container . D- All of the above .

23- If renal tuberculosis is suspected the urine sample used for laboratory diagnosis should be : {أسئلة القسم}

A- Early morning sample B - 3 urine samples are collected on consecutive days .

C- Entire sample is collected . D- All of the above .

24 - Sterile pyuria means : {أسئلة القسم}

A- Presence of pus in urine (more than 10 cells / HPF) .

B- Presence of pus in urine in absence of bacterial growth on ordinary media .

C- Presence of pus in urine with bacterial growth on ordinary media .

25 - Sterile pyuria can occur with {أسئلة القسم}

A - Renal tuberculosis . B- Anaerobes C - Mycoplasma . D - All of the above .

26- The most common cause of a urinary tract infection (UTI) is : {أسئلة القسم}

A - Escherichia coli (E. coli) B- Staphylococcus aureus (S. aureus)

C- Chlamydia . D - Mycoplasma .

27- Urine cultures are specially indicated in the following situations : {أسئلة القسم}

A - Complicated UTI including pyelonephritis B- Transplant and diabetic patients .

C - Pregnancy . D- All of the above .

28- Bacterial count 100.000 bacteria per ml of urine or more : {أسئلة القسم}

A- Indicates UTL B- Regarded as contamination . C- Is of no significance .

29- Bacterial count less than 10.000 bacteria per ml of urine : {أسئلة القسم}

A- Indicates UTI . B- Regarded as contamination . C- Is of no significance .

30- A count of 10.000 organisms per ml of urine is considered significant if: {أسئلة القسم}

A- The organism is gram positive and only one type .

B- The organism is gram positive and many types ,

C- The organism is gram negative and many types .

31- Antibiotic sensitivity test is especially indicated in case of : {أسئلة القسم}

A - Urine with significant bacteriuria , particularly from patient with recurrent UTI .

B - Complicated UTI . C - A and B. D- None of the above .

32- On Macconkey medium E. coli gives : {أسئلة القسم}

A - Metallic green colonies . B - Rose pink colonies .

C - Pale colonies . D- None of the above .

33 - On EMB E. Coli gives : {أسئلة القسم}

A - Metallic green colonies B- Pink colonies

C - Pale colonies . D- None of the above

34- On triple sugar iron (TSI) E. coli gives : {أسئلة القسم}

A - Yellow slant and yellow butt B- Yellow slant and red burt .

C - Red slant and red butt . D- None of the above .

35- E. coli gives the following IMVC reaction result : {أسئلة القسم}

A - (++ --) B - (- ++) C - (+ - + -) D - (++++)

36 - On Macconkey medium proteus gives : {أسئلة القسم}

A- Metallic green colonies B- Rose pink colonies .

C - Pale colonies D- None of the above .

37- On nutrient agar proteus gives : {أسئلة القسم}

A- Colonies that grow and produce swarming cover all plate B- Rose pink colonies .

C - Pale colonies . D- None of the above .

38- Proteus gives positive : {أسئلة القسم}

A - Urease test B - H2S production test C - A and B D- None of the above ,

39- On triple sugar iron (TSI) proteus gives {أسئلة القسم}

A- Yellow slant and yellow butt . B- Yellow slant and red butt .

C- Red slant and black butt . D- None of the above .

40- The following bacteria can grow on bile esculin azide : {أسئلة القسم}

A- Enterococci B- Staphylococcus aureus. C- Chlamydia D- None of the above .

41- On bile esculin azide enterococci gives : {أسئلة القسم}

A- Black colonies B - Rose pink colonies . C - Pale colonies D- None of the above

42- Candida species can grow on the following media : {أسئلة القسم}

A - Nutrient agar B - Com meal agar

C- Sabaroud dextrose agar D- All of the above .

43 - On Sabaroud dextrose agar C. albicans gives : {أسئلة القسم}

A - Large , cream colored colonies with yeasty odour B- Black colonies .

C - pink colonics D- None of the above .

44- The following test (s) can be used to confirm diagnosis of c. albicans : {أسئلة القسم}

A - Germ tube test . B - Chlamydospore formation test

C - A and B D- None of the above .

45- A 21 year newly married woman developed a urinary tract infection (UTI) . At the time she sought medical advice , she was febrile and complained of painful urination and flank pain . Her urine appeared " cloudy " . Urine culture yields a lactose - fermenting , indole - positive , Gram negative bacillus . The infectiveness of the organism responsible for this urinary tract infection is associated with specific ? {أسئلة القسم}

A - exotoxins B - K antigens c - Metabolic properties d - P fimbriae e . Plasmids

46- Which is a common UTI risk factor in adults ? {أسئلة القسم}

A- Enlarged prostate B- Catheter usage C- Diabetes D- All of the above

47- What are signs and symptoms of a UTI in adults ? {أسئلة القسم}

A- Frequent urge to urinate B- Pain during urination

C- Milky / cloudy urine . D- All of the above

48- Bacteria from the intestines (most commonly) often infect the urinary tract by ascending from the perineal area into the lower urinary tract . {أسئلة القسم}

A- Escherichia coli B - Bacillus anthracis

C - Staphyloccus aureus D- Streptococcus pyogenes , Group A

49- Infections are among the most common urinary tract infections . (multiple response) { أسئلة القسم}

A- Urethritis B- Prostatitis C – Cystitis D- Nosocomial (hospital acquired)

50-Uncomplicated UTI is usually considered in

A. Premenopausal adult women with no structural or functional abnormality of the urinary tract

B. Pregnant women with no structural or functional abnormality of the urinary tract

C. Women with risk factors that could lead to more serious complications

D. Middle aged men

51-Most urinary tract infections involve

A. The kidneys B. The lower urinary tract

C. The upper urinary tract D. The whole urinary tract

52-Symptoms of urinary tract infection are all of the following except:

A. Increased frequency B. Suprapubic pain or discomfort C. Hematuria D. Diarrhea

53-Typical treatment for urethritis is:

A. Ceftriaxone plus azithromycin for 7 days B. Nitrofurantoin for 3 days

C. Ciprofloxacin 500 mg for 3 days D. Azithromycin for 7 days

54- Females are more susceptible to urinary tract infection because: {Merit}{ثناء}

A. The immune system is weaker in females B. High oestrogen level

C. Short urethra D. Low testosterone level

55-What is the most common pathogen responsible for urinary tract infections (UTIs)? {Merit}

A. Staphylococcus aureus B. Escherichia coli (E. coli)

C. Klebsiella pneumonia D. Enterococcus faecalis

56-Why are women at greater risk of developing UTIs compared to men? {Merit}

A. Due to their longer urethra B. Due to their higher estrogen levels

C. Due to their shorter urethra D. Due to their weaker immune system

57-What is a common symptom of lower urinary tract infection (cystitis) characterized by pain or discomfort during urination? {Merit}

A. Hematuria B. Urinary frequency C. Dysuria D. Flank pain

58-How do bacteria primarily enter the urinary tract? {Merit}

a. Through the bloodstream b. Through the kidney

c. Through the urethra d. Through the bladder

59-What is the primary bacteria responsible for causing UTIs? {Merit}

a. Staphylococcus aureus b. Streptococcus pneumoniae

c. Escherichia coli d. Klebsiella pneumoniae

60-What condition, commonly associated with weakened immune systems, increases vulnerability to UTIs? {Merit}

a. Diabetes b. Hypertension c. Asthma d. Hyperthyroidism

61-Which of the following factors can create conditions for infection by providing a reservoir for bacteria in the urinary tract? {Merit}

a. Complete bladder emptying b. Structural abnormalities

c. High fluid intake d. Regular exercise

62-Which symptom is more indicative of upper urinary tract infection (pyelonephritis) rather than lower urinary tract infection? {Merit}

A. Hematuria B. Cloudy or foul-smelling urine C. Dysuria D. Flank pain

63-What is a systemic sign of infection commonly associated with pyelonephritis? {Merit}

A. Urinary frequency B. Hematuria C. Fever and chills D. Pelvic discomfort

64-What gastrointestinal symptom may be present in individuals with pyelonephritis? {Merit}

a. Hematuria b. Nausea and vomiting c. Urinary frequency d. Hematuria

65-What colony count threshold is commonly used for diagnosing a UTI in symptomatic individuals? {Merit}

a. 10^2 CFU/mL b. 10^3 CFU/mL c. 10^4 CFU/mL d. 10^5 CFU/mL

66-What is the purpose of collecting a clean-catch midstream urine sample for urine culture? {Merit}

a. To minimize contamination from bacteria present in the urinary tract

b. To maximize the presence of bacteria present on the skin

c. To increase the likelihood of fungal growth in the urine sample

d. To decrease the chances of identifying bacterial causes of UTIs

67-What is the purpose of culturing a urine sample in the laboratory? {Merit}

a. To analyze the chemical composition of the urine

b. To identify the presence of white blood cells in the urine

c. To encourage bacterial growth for identification

d. To assess the presence of fungal and viral causes of UTIs

68-How are bacteria identified after they have grown in the culture medium? {Merit}

a. Through antibiotic sensitivity testing

b. By performing a microscopic examination

c. By analyzing the chemical composition of the urine

d. Using various techniques such as microscopy and biochemical tests

69-Which of the following is not a bacterial cause of UTIs? {Merit}

a. Escherichia coli b. Klebsiella oxytoca c. Candida albicans d. Proteus mirabilis

70-Which virulence factor contributes to the formation of biofilms, aiding bacterial persistence within the urinary tract? {Merit}

a. Fimbriae b. Flagellum c. Capsular lipopolysaccharide d. Haemolysin

71-The most common organism causing UTI by ascending route is …….. {مهم}

a. E. Coli b. S.aureus

72-The most common organism causing UTI by hematogenous route is ……..{مهم}

a. E. Coli b. S.aureus

73-Staphylococcus saprophyticus UTIs are particularly common in: {مهم}

A. Elderly men B. Sexually active young women

C. Children under 5 years D. Pregnant women only

74-Viable bacterial count in urine is performed using: {مهم}

A. Centrifuged urine B. Uncentrifuged urine

C. Only suprapubic aspirate D. Only catheterized urine

75-A bacterial count of ≥ 10^5 CFU/ml in urine indicates: {مهم}

A. Contamination B. Urinary tract infection

C. Normal flora D. Asymptomatic bacteriuria

76-A bacterial count of < 10^4 CFU/ml is generally regarded as: {مهم}

A. Urinary tract infection B. Contamination

C. Significant bacteriuria D. Pyelonephritis

77-When urinary tuberculosis is suspected, how many successive morning urine samples are usually examined? {مهم}

A. One B. Three C. Five D. Seven

78-Hospital-acquired UTIs account for approximately what percentage of overall nosocomial infections?

A. 10% B. 25% C. 40% D. 60%

79-Which condition is classified as a lower UTI?

A. Perinephric abscess B. Pyelonephritis C. Cystitis D. Intra-renal abscess

80-Which of the following does NOT define a complicated UTI?

A. Pregnant patient with UTI B. UTI in a child

C. Healthy adult female with simple cystitis D. UTI after recent urinary tract surgery

81-All of the following is a risk factor for UTI among adults EXCEPT:

a. Pregnancy b. Enlaged prostate c. Circumcision

d. Diabetes e. Using a catheter

82-Which of the following is not a common cause of UTI among men?

a. Complete emptying of bladder b. Enlarged prostate

c. Kidney stone d. Using catheter

83-UTI as a result of Proteus mirabilis facilitates of the formation of kidney stones because the organism does which of the following?

a. Ferments many sugars b. Exhibit swarming motility

c. Secretes many exotoxins d. Produces a potent urease

84-Which of the following criteria qualifies a patient as having recurrent UTI?

A. Two symptomatic UTIs in 3 months B. Four symptomatic UTIs in 12 months

C. One symptomatic UTI per year D. Asymptomatic bacteriuria

85-Which of the following is NOT a factor that facilitates UTIs in women?

A. Short urethra B. Vaginal colonization by fecal flora

C. Proximity of urethra to anus D. Longer urethra

86-UTI is considered complicated in which of the following patients?

A. A healthy adult woman with no comorbidities B. A child with a urinary tract infection

C. A healthy adult man with a simple cystitis D. None of the above

87-Which of the following increases the risk of complicated UTI?

A. Structural or functional urinary tract abnormality B. Obstruction of urine flow

C. Recent urinary tract instrumentation D. All of the above

88-A patient with poorly controlled diabetes develops UTI. This infection is considered:

A. Uncomplicated B. Complicated C. Always asymptomatic D. Non-bacterial

89-Which of the following is TRUE about relapse in UTI?

A. It is caused by a different pathogen

B. It occurs within 2 weeks after therapy and is caused by the same species

C. It occurs after 3 months D. It only happens in children

90-Recurrent UTI is defined as:

A. ≥1 UTI per year B. ≥2 symptomatic UTIs within 6 months

C. >3 symptomatic UTIs within 12 months following treatment D. ≥5 UTIs in 5 years

91-Reinfection in urinary tract infections is defined as:

A. Recurrent UTI caused by the same pathogen

B. Recurrent UTI caused by a different pathogen at any time

C. Asymptomatic bacteriuria D. UTI in a child only

92-Upper urinary tract infections include all Except:

A. Pyelonephritis B. Cystitis C.intra-renal abscess D.perinephric abscess

93-Lower urinary tract infections include:

A. Pyelonephritis B. Perinephric abscess

C. Cystitis, urethritis, and prostatitis D. Renal abscess

94-The most common cause of gonococcal urethritis is:

A. E. coli B. N. gonorrhoeae C. Chlamydia trachomatis D. S. aureus

95-Which bacterium is most commonly associated with struvite stone formation?

A. Escherichia coli B. Proteus mirabilis C. Staphylococcus aureus

96-After Escherichia coli, the second most common cause of community-acquired urinary tract infections is:

A. Proteus mirabilis B. Klebsiella pneumoniae

C. Staphylococcus saprophyticus D. Pseudomonas aeruginosa

97-The preferred type of urine specimen for routine UTI culture is:

A. First morning urine B. Mid-stream urine

C. Catheterized urine only D. Random urine without cleaning

98-The presence of bacteria in urine without pyuria most likely indicates:

A. Acute UTI B. Contamination of the sample

C. Pyelonephritis D. Chronic kidney disease

99-Sterile pyuria refers to:

A. Presence of bacteria in urine without pus

B. Presence of pus in urine without bacterial growth on ordinary media

C. Normal urine D. Bacteriuria with fever

100-Which of the following is a cause of sterile pyuria?

A. Renal tuberculosis B. E. coli cystitis

C. Acute pyelonephritis with bacteriuria D. Proteus mirabilis stones

1. C	2. C	3. A	4. C	5. D
6. D	7. D	8. C	9. A	10.B
11.C	12.A	13.A	14.B	15.B
16.A	17.D	18.C	19.C	20.C
21.C	22.D	23.D	24.B	25.C
26.A	27.D	28.A	29.B	30.A
31.C	32.B	33.A	34.A	35.A
36.C	37.A	38.C	39.C	40.A
41.A	42.D	43.A	44.C	45.C
46.D	47.D	48.A	49.	50.A
51.B	52.D	53.A	54.C	55.B
56.C	57.C	58.C	59.C	60.A
61.B	62.D	63.C	64.B	65.D
66.A	67.C	68.D	69.C	70.A
71.A	72.B	73.B	74.B	75.B
76.B	77.C	78.C	79.C	80.C

81.C	82.A	83.D	84.B	85.D
86.B	87.D	88.B	89.B	90.C
91.B	92.B	93.C	94.B	95.B
96.C	97.B	98.B	99.B	100. A

Parasitology

Renal system

1-The infective stage of Schistosoma haematobium is: {Formative2024}

a) egg with lateral spine b) egg with terminal spine

c) encysted metacercaria d) cercaria

2-The infective stage of plasmodium falciparum is: {3rd trial2022}

a-Promastigote b-Encysted metacercaria c-Sporozoite d-Sporulated oocyst

3- All the following parasites can affect renal system Except: {Final2021}

a. Schistosoma haematobium b. Schistosoma mansoni c. Leishmania tropica

d. Plasmodium falciparum e. Leishmania donovani

4- Acute Renal failure is caused by {Final2023}

a. Plasmodium oval b. Plasmodium vivas

c. Plasmodium malaria d. Plasmodium falciparum

5- all the following statements are true except: {Formative2022}

a) Only two of the malaria parasites, namely, P. malariae and P. falciparum, are associated with renal disease

b) Plasmodium malariae can cause nephrotic syndrome and acute renal failure

c) Plasmodium falciparum can cause acute renal failure

d) Plasmodium falciparum cause renal failure due to impaired blood flow in the microcirculation due to adhesiveness of erythrocytes and hypovolemia

e) Non of the above

6- all the following statements are true except: {Formative2022}

a) Renal involvement in Schistosoma haematobium and S. mansoni occurs due to precipitation of immune complexes in the glomerular vascular bed

b) Schistosoma haematobium can cause carcinoma of the bladder

c) Both Schistosoma haematobium and Schistosoma mansoni can cause nephrotic syndrom

d) Schistosoma mansoni can cause hydronephrosis

e) Schistosoma haematobium can be diagnosed by finding egg in urine And Schistosoma mansoni can be diagnosed by finding egg in stool

7-all the following parasites can affect renal system except: {Formative2022}

a) Schistosoma haematobium b) Schistosoma mansoni c) Leishmania major

d) African trypanosome e) Leishmania donovani

8- Schistosoma Hematobium Causes all the following EXCEPT {Final2022}

a.Dysentry

9-An Egyptian 12 years old child presented by frequent micturition, dysuria and terminal hematuria with mild anemia. Urine examination showed egg with microscopic blood.-The causative parasite is:

a-Fasciolopsis Buski b-Schistosoma haematobium.

c-Schistosoma mansoni. d-Schistosoma japonicum.

10-Nephrotic syndrome occurs with

A. P.vivax B. P.malariae C. P.falciparum D. P.ovale

11-The infective stage of Schistosoma haematobium is:

a- Egg b- Furococercus cercaria c- Adult stage d- Encysted metacercaria

12-Nephrotic syndrome occurs with: {Merit}

a-P.malariae b- P.ovale c-P.falciparum d-P.vivax

13-In urinary schistosomaisis, terminal haematuria occurs in: {Merit}

a-stage of invasion b-stage of migration c-stage of egg extrusion d-chronic stage

14-Female Anopheles mosquito transmit:

A. Trench fever B. Pediculosis C. Human Malaria D. Q fever

15-Plasmodium falciparum intermediate host is:

a- Female Anopheles mosquitoes b- male Anopheles mosquitoes

c-Male and Female Anopheles mosquitoes d-Man

16-Habitat of Schistosoma hematobium adults in:

A-Vesical plexus of veins. B-Tributaries of superior mesentric vein.

C-Tributaries of inferior mesenteric vein D-Lumen of the small intestine

17-The infection with schistosomiasis is through

a. Ingestion of polluted water b. Eating infected meat c. Skin penetration

d. Blood transifusion e. Inhalation

18-Urinary myiasis is defined as:

A. Bacterial infection of the bladder

B. Infestation of the urinary tract by fungal spores

C. Infestation of the urinary tract by larvae of dipterous flies

D. Viral infection of the urinary tract

19-Which of the following dipterous flies are most commonly involved in urinary myiasis?

A. Anopheles and Culex B. Fannia and Musca domestica (housefly)

C. Glossina and Chrysops D. Sandfly and blackfly

20-Urinary myiasis is considered:

A. Obligatory myiasis B. Facultative myiasis

C. Accidental myiasis D. Nosocomial myiasis

1. D	2. C	3. C	4. D	5. B
6. D	7. C	8. A	9. B	10.B
11.B	12.A	13.C	14.C	15.D
16.A	17.C	18.C	19.B	20.C

Pharmacology

Diuretics

1-The most important therapeutic indication of carbonic anhydrase inhibitor agents is: {3rdtral2024}

a-Congestive heart failure b-Renal insufficiency c-Cirrhosis of liver d-Glaucoma

2- The following drug inhibits the action of aldosterone. {Summar2024} {3rdtral2024}

a. Amiloride b. Triamterene c. Eplerenone d. Furosemide

3-The diuretic which allow excretion of 8% of filtered sodium is: {Final2021}

a. Spironolactone b. Furosemide c. Hydrochlorothiazide d. Diamox

4-…….acts on the collecting duct and can block reabsorption of only 3% of filtered sodium

a. Mannitol b. Triamterene c. Acetazolamide d. Hydrochlorothiazide

5- The following drug acts on thick ascending limb of loop of Henle: {Final2021}

a. Ethacrynic acid b. Indapamide c. Spironolactone d. Chlorthalidone

6- Which of the following diuretics is most effective in treatment of glaucoma? {Final2021}{Final2023}{Summar2024}

a. Indapamide b. Mannitol c. Amiloride d. Furosemide

7-The diuretic of choice for the treatment of a patient with chronic renal failure is: {3rd trial2022}{Formative2022}

a-Hydrochlorothiazide b-Bumetanide c-Furosemide d-Indepamede

8- Drug of choice for treatment acute pulmonary edema: {Final2023}

a. furosemide b. Chlorthalidone c. Indapamide d. spironolactone

9- Furosemide may produce an increase in blood levels of {Summar2022}

a- Potassium b- Calcium c- Magnesium d- Glucose

10- Increased blood glucose is associated with which of the following Drugs? {Final2023}

a. Amiloride b. Eplerenone c. Triamterene d. Bumetanide

11-The primary site of action of thiazide diuretics is: {3rd trial2022}{Formative2022}

a-Proximal tubule b-Ascending limb of loop of Henle

c-Distal convoluted tubules d-Collecting ducts

12- The diuretic which used in patient with creatinine clearance below 25 ml/min is: {Final2021}{Formative2024}

a. Hydrochlorothiazide b. Spironolactone c. Amiloride d. Furosemide

13- Which Drug is used if Clearance of creatinine is 20 ? {Final2022}

a. Furisamide

14- Thiazide diuretics may produce an increase in blood levels of all the following Except: {Final2021}{Summar2024}

a. Sodium b. Uric acid c. Calcium d. Glucose

15-Thiazide diuretics may produce an increase in blood levels of: {3rd tral2024}

a. Potassium b. Glucose c. Urea d. Magnesium

16- Most effective drug in ttt of hypertension..... {Final2022}

a.Thiazide

17- The diuretic that can cause gynecomastia and menstrual disturbance on long- term use: {Summar2022}

a- Amiloride b- Metolazone c- Spironolactone d- Acetazolamide

18-Diuretic inhibits the Na K 2 Cl transport

a. manitol b. acetazolamide c. thiazide d. furosemide

19-The drug which acts by competitively blocking NaCl contetransporter in the distal tubules is:

A. Acetazolamide (Diamox) B. Furosemide (Lasix)

C. Hydrochlorothiazide (Hydro Diuril) D. Spironolactone (Aldactone)

20- Which one of the following is NOT therapeutic use for furosemide? {Merit}

A. Acute pulmonary edema B. Hyperkalemia

C. Hypertension with renal dysfunction D. Hypercalciuria

21-Furosemide (lasix) produces diuresis by inhibiting NaCl reabsorption in:

a)PCT. b)ALH. c)DLH. d)All segments of nephron

22-Which one of the following drugs used in dialysis disequilibrium syndrome? {Merit}

A. Mannitol B. Furosemide C. Dorzolamide D. Thiazides

23-The following drug inhibits epithelial Sodium channels in collecting duct: {Merit}

A. Spironolactone B. Amiloride C. Hydrochlorothiazide D. Mannitol

24-Thiazides are used for all the following except: {Merit}

A. Chronic hypertension B. Generalized edema

C. Nephrogenic diabetes insipidus D. Hypercalcemia

25-Nephrogenic diabetes insipidus is treated by the following: {Merit}

a. Furosemide b. Hydrochlorothiazide c. Spironolactone d. Mannitol

26-Spironolactone is used with thiazides to: {Merit}

A. Prevent hypokalemia B. Prevent hyponatremia

C. Prevent hypercalcemia D. Prevent hyperglycemia

27-All the following are uses of thiazides except: {Merit}

A. Hypercalcemia B. Chronic hypertension

C. Idiopathic hypercalciuria D. Nephrogenic diabetes insipidus

28-Which of the following diuretics is K+ sparing:

a)Carbonic anhydrase inhibitor (acetazolamide). b)Loop diuretics (frusemide).

c)Spironolactone. d)ADH antagonist

1. D	2. C	3. C	4. B	5. A
6. B	7. C	8. A	9. D	10.D
11.C	12.D	13.A	14.A	15.B
16.A	17.C	18.D	19.C	20.D
21.B	22.A	23.B	24.D	25.B
26.A	27.A	28.C	29.	30.

Nephrotic drugs

1-NSAID can induce nephrotoxicity by: {Formative2024}{Final2022}

a) Constriction of afferent arteriole b) Dilatation of efferent arteriole

c) Dilatation of afferent arteriole d) Constriction of efferent arteriole

2- Which of the following drug may causes alteration of intraglomerular hemodynamics? {Summar2024}

a. Cocaine b. NSAIDS c. Statin d. Rifampicin

3-All the following drugs have been implicated in acute interstitial nephritis except {Formative2022}

a. Cimetidine b. Methicillin c. Cocaine d. NSAIDs

4- Which of the following may contribute to altered intraglomerular hemodynamics {Summar2022}

a- Paracetamol b- Penicillin c- Prazosin d- Cocaine

5-Which of the following may associate with renal tubular toxicity: {3rd trial2022}

a-Allopurinol b-Methicillin c-Aminoglycoside d-NSAIDs

6-Which of the following mechanisms is primarily involved in the pathogenesis of aminoglycoside-induced nephrotoxicity? {Final2022}{3rd tral2024}

a. Altered intraglomerular hemodynamics b.Acute interstitial nephritis

c.Rhabdomyolysis d. Tubular cell toxicity

7- The pathogenesis of NSAIDS induced nephrotoxicity may involve which of the following? {Formative2022}

a) Altered intraglomerular hemodynamics b) acute interstitial nephritis

c) Chronic interstitial nephritis d) All of the above

8- Rhabdomyolysis is associated with {Final2023}

a. NSAID b. Methicillin c. Fluoxetine d. Thiazide

9- Which of the following may associate with Rhabdomyolysis? {Final2021}

a. Thiazide b. Methicillin c. Statin d. NSAIDS e. All of the above

10- All the following drugs have been implicated in acute interstitial nephritis Except {Final2021}

a. Cimetidine b. Methicillin c. Fluoxetine d. NSAIDs

11-The most common manifestation induced nephrotoxicity (DIN) is a decline in: {Formative2022}

a) Glomerular filtration rate b) serum creatinine value

c) Blood urea nitrogen level d) None of the above

1. A	2. B	3. C	4. A	5. C
6. D	7. D	8. C	9. C	10.C
11.A	12.	13.	14.	15.

Pathology

Congenital anomalies of the kidney

1-……..is a congenital anomaly in which there is complete absence of one kidney. {Summar2024}

a. Renal agenesis. b. Renal hypoplasia. c. Ectopic kidney. d. Polycystic kidney disease.

2-Congenital anomalies of the kidney do not include:

a. horse-shoe kidney b. ectopic kidney c. polycystic kidney

d. contracted kidney e. hypoplastic kidney

3- The following is not correct about autosomal dominant poly-cystic kidney disease: {Final2021}

a. The pattern of inheritance is autosomal- dominant pattern.

b. Stones and hemorrhage may be seen inside the renal cysts.

c. Most cases are due to mutation of polycystic kidney disease-2(PKD2) gene.

d. The cysts develop in all segments of renal tubules and glomerular capsules.

e. All of the above

4- The following is correct about autosomal recessive poly-cystic kidney disease: {Formative2022}{Mid2021}

a. Outer surface is nodular.

b. Usually accompanied by multiple biliary cysts in enlarged portal tracts.

c. Most cases appear in 5 decade of life. d. All the above. e. None of the above

5- Adult polycystic kidney is characterized by all EXCEPT: {Formative2022}{Merit}{ثناء}

A. Inherited as an autosomal recessive trait B. Commonly leads to renal failure

C. Bilateral D. Shows multiple cysts filled with serous or haemorrhagic fluid

6-Adult polycystic kidney disease characterized by all the following except

a. The cysts communicate with the renal pelvis

b. The kidney is enlarged and formed of multiple cysts

c. Associated with cysts in liver

d. Age of onset between 30. 40ys

7- Which is FALSE about AD-PKD ? {Final2022}

a. More in females

8- All the following are congenital anomalies of kidney EXCEPT: {Final2022}

a. Ectopica Vescica

9-Complete absence of one kidney is called {Merit}

A. Agenesis of the kidney B. Hypoplasia of the kidney

C. Ectopic kidney D. Horse shoe kidney

10-Failure of one kidney to develop to the normal size is called: {Merit}

A. Agenesis of the kidney B. Hypoplasia of the kidney

C. Ectopic kidney D. Horse shoe kidney

11-Adult polycystic kidney is characterised by all EXCEPT:

a. Inherited as an autosomal dominant trait

b. Shows multiple cysts communicating with the pelvis

c. The cysts are lined by cuboidal epithelium

d. Bilateral e. Commonly leads to renal failure

12-ADPKD is best described as:

A. An acquired cystic kidney disorder

B. A hereditary condition with progressively enlarging cysts destroying renal parenchyma

C. A neonatal lethal disease D. A mild disorder that never leads to renal failure

13-ADPKD ultimately leads to:

A. Complete recovery B. Renal failure C. Hyponatremia only D. Liver cirrhosis

14-ADPKD accounts for approximately what percentage of patients requiring dialysis or renal transplantation?

A. 1% B. 5% C. 10% D. 50%

15-ADPKD affects:

A. Only males B. Only females

C. Males more than females D. Males and females equally

16-ADPKD involves:

A. Unilateral kidneys B. Bilateral kidneys

C. Only the left kidney D. Only the right kidney

17-The expanding cysts in ADPKD primarily cause renal damage by:

A. Infecting the nephrons directly B. Compressing and destroying renal parenchyma

C. Reducing blood supply only D. Blocking the ureters

18-Which of the following may occur inside renal cysts in ADPKD?

A. Hydatid cysts B. Stones and hemorrhage C. Tuberculosis D. Fungal colonies

19-In Autosomal Dominant Polycystic Kidney Disease (ADPKD), renal cysts may develop in:

A. Only proximal tubules B. Only distal tubules C. Only collecting ducts

D. All segments of renal tubules and the glomerular capsule

20-The cysts in AR-PKD primarily arise from:

A. Proximal tubules B. Distal convoluted tubules C. Collecting tubules D. Glomeruli

21-In the early stages, the fluid inside ADPKD cysts is derived from:

A. Lymphatic drainage B. Glomerular filtrate C. Blood vessels D. Interstitial fluid

22-When an enlarged cyst loses its attachment to the nephron, the cyst fluid is produced by:

A. Glomerular filtration B. Blood vessels

C. Their lining epithelial cells D. The surrounding interstitium

23-Which microscopic feature may sometimes be seen in the cyst lining of ADPKD?

A. Keratinization B. Focal areas of hyperplasia

C. Necrotic debris always D. Smooth muscle proliferation

24-The epithelial lining of ADPKD cysts is usually:

A. Columnar epithelium B. Stratified squamous epithelium

C. Cuboidal or flattened epithelium D. Transitional epithelium

25-Autosomal Recessive Polycystic Kidney Disease is also known as:

A. Adult-type PKD B. Classic infantile AR-PKD

C. Juvenile nephronophthisis D. Medullary sponge kidney

26-AR-PKD primarily affects:

A. Adolescents B. Adults C. infants and children D. Elderly

27-On cut section, AR-PKD kidneys show:

A. Large solitary cysts B. Numerous small cysts in cortex and medulla

C. Hemorrhagic cysts only D. Only medullary involvement

1. A	2. D	3. C	4. B	5. A
6. A	7. A	8. A	9. A	10.B
11.B	12.B	13.B	14.C	15.D
16.B	17.B	18.B	19.D	20.C
21.B	22.C	23.B	24.C	25.B
26.C	27.B	28.	29.	30.

Urinary outflow obstruction - Diseases of the urinary bladder

1-The following type of renal stones occurs in the urinary bladder: {Formative2024}{ثناء}

a) Phosphate stones b) Urate stones c) Oxalate stones d) Cystine stones

2- The following type of renal stone is multiple and spiky {Formative2022}{Summar2022} {Final2023}

a. Oxalate stones b. Phosphate stones c. Cystine stones d. Urate stones

3- The following type of renal stones usually occurs on top of infection: {Mid2021}{Formative2024}{ثناء}

a. Oxalate stones b. Urate stones c. Phosphate stones

d. Cystine stones e. None of the above

4- Urinary tract obstruction predispose to: {Final2021}

a. Urinary tract infection. b. Urinary calculi. c. Hydronephrosis.

d. All of the above. e. None of the above

5- The following is not a type of renal stones: {Final2021}{Summar2024}{Formative2024}{ثناء}

a. Oxalate stones b. Urate stones c. Cholesterol stones d. Phosphate stones

6-Renal calculi is a predisposing factor for: {3rd trial2022}

a-Chronic glomerulonephritis b-Chronic pyelonephritis

c-Acute tubular injury d-Acute drug induced nephropathy

7- The following is NOT a cause of renal calculi: {Formative2024}{ثناء}

A. Vitamin D deficiency B. Urinary tract infection

C. Urinary tract obstruction D. Hyperparathyroidism e. Vitamin A deficiency

8-Most common stone in Urinary Bladder? {Final2022}

a. phosphate b. Oxalate c. Cystine

9-The following are the common causes of renal calculi except: {قسم}

A) Cystitis. B) Hyperparathyroidism. C) Bilharzial urethritis.

D) Urinary tract obstruction. E) Vitamin D Deficiency.

10-The patient with renal stone in the urinary bladder presented with……. {قسم}

A. Dysuria, hematuria and supra pubic pain. B. Polyurea, proteinuria and loin pain.

C. Polyurea, proteinuria and generalized edema. D. Oliguria, hematuria and generalized edema.

E. Oliguria, proteinuria and hypertension.

11-There are ……types of renal stones. {قسم}

A.Three. B. Seven. C. Five. D. Two. E. Four.

12-Struvite stone is composed of…… {قسم}

A. Magnesium-ammonium-calcium phosphate. B. Oxalate-ammonium-calcium phosphate.

C. Cysteine- magnesium-calcium phosphate D. Magnesium-ammonium- potassium oxalate.

E. Magnesium-ammonium-calcium oxalate.

13-……..is the most common type of stones. {قسم}

A.Potassium. B. Calcium. C. Cysteine. D. Oxalate. E. Uric acid.

14-Hydronephrosis means …… {قسم}

A. An enlarged kidney filled with pus. B. An enlarged kidney filled with blood.

C. An enlarged kidney filled with urine. D. An enlarged kidney with a tumor.

E. A congenital polycystic kidney.

15-Stag horn stone is characterized by: {قسم}

A. Multiple small brown in colour. B. Small brown with sharp surface.

C. Green small multiple. D. Large, irregular surface.

E. Oval, white, smooth surface

16- Which pathological changes occur in hydronephrosis? {Merit}

a. Dilatation of renal pelvis & calyces b. Hypertrophy of renal parenchyma

c. Atrophy of renal calyces d. Dilatation of urinary bladder

17- What is the meaning of hydroureter? {Merit}

a. Dilatation of ureter b. Dilatation of renal pelvis & calyces

c. Hydropic degeneration of ureter d. Dilatation of urethra

18- One of the following is considered a congenital cause for hydronephrosis; {Merit}

a. Atresia of urethra b. Agenesis of one kidney

c. Ectopic kidney d. Hypoplasia of kidney

19- Which of the following can lead to hydronephrosis? {Merit}

a. Outflow obstruction of urine b. Hydroureter

c. Renal vein thrombosis d. Dilatation of urethra

20- Which of the following is an acquired cause for hydronephrosis? {Merit}

a. BPH b. Aberrant renal artery c. Valve formation d. Hyperplasia of kidney

21- In hydronephrosis, unilateral obstruction occurs when there is; {Merit}

a. Blockage at the ureter or above b. Agenesis of one kidney

c. Atresia of urethra d. Paralysis of urinary bladder

22- One of the following is a common route of infection in acute cystitis: {Merit}

a. From urethra b. Direct form adjacent organ

c. Metastasis d. Drinking contaminated water

23- …… is a predisposing factor for acute cystitis. {Merit}

a. Urine stasis b. Cigarette smoking

c.Atrophy of renal calyces & pelvis d.Hypoplasia of kidney

24- The following clinical features occurs in acute cystis: {Merit}

a. Dysuria, urgency, frequency & hematuria b. Fever & frequency

c. Fever, nausea, vomiting & pain d. Abdominal colic and hematuria

25- …….. can be a complication of chronic non-specific cystis. {Merit}

a. Calculi formation b. Bladder hypertrophy

c. Hyperplasia of renal pelvis d. Acute renal failure

26- The following type of renal stones usually has a stag horn appearance: {ثناء}

A. Oxalate stones B. Urate stones C. Phosphate stones D. Cystine stones

27- The following type of renal stones is radio-translucent: {ثناء}

A. Oxalate stones B. Urate stones C. Phosphate stones D. Cystine stones

28- The following type of renal stones is multiple and spiky: {ثناء}

A. Oxalate stones B. Urate stones C. Phosphate stones D. Cystine stones

29- The following type of renal stones is commonly associated with haematuria: {ثناء}

A. Oxalate stones B. Urate stones C. Phosphate stones D. Cystine stones

30- The following type of renal stones is very rare: {ثناء}

A. Oxalate stones B. Urate stones C. Phosphate stones D. Cystine stones

1. A	2. A	3. C	4. D	5. C
6. B	7. A	8. A	9. E	10.A
11.E	12.A	13.B	14.C	15.D
16.A	17.A	18.A	19.A	20.A
21.A	22.A	23.A	24.A	25.A
26.B	27.B	28.A	29.A	30.D

Glomerular Diseases

1- …….. is an example for primary glomerulonephritis: {Mid2021}

a. Diabetic glomerulosclerosis. b. Glomerulonephritis due to systemic lupus erythematosus.

c. Membranous glomerulonephritis. d. All of the above e. None of the above.

2- Membranous glomerulonephritis is: {3rd trial2022}{Summar2024}

a. The commonest cause of acute nephritic syndrome in adults.

b. The commonest cause of acute nephritic syndrome in children.

c. The commonest cause of nephrotic syndrome in children.

d. Characterized by dense IgG deposition between basement membrane and overlying visceral epithelial cells.

3- Which is FALSE about membranous GN(glomerulonephritis)? {Final2022}

a. Markedly Hypercellularity

4-The following is not-correct in nephrotic syndrome: {Formative2024}{Summar2022}

a) Massive proteinuria exceed 3 gm/24 hours. b) Hypoalbuminemia.

c) Hyperlipidemia. d) Tea-colored urine.

5-Nephrotic syndrome is characterized by all except: {قسم}

A)Heavy proteinuria. B)Hyperlipidemia. C)Hypoproteinemia.

D)Generalized edema. E)Hypertension.

6-Nephritic syndrome is characterized by all EXCEPT: {Merit}{ثناء}

A. Hematuria B. Heavy proteinuria C. Oliguria D. Hypertension

7-Nephrotic syndrome is NOT characterized by: {Merit}{ثناء}

A. Heavy proteinuria B. Hypoproteinaemia

C. Oedema of the face only D. Hyperlipidaemia

8- The following is not correct about Minimal Change Glomerular Disease: {Final2021}

a. The immune complexes are deposited between visceral epithelial cells and basement membrane.

b. The proximal convoluted tubules show lipid deposition in the lining cells.

c. The visceral epithelial cells show diffuse loss of their foot processes.

d. It is the commonest cause of nephrotic syndrome in children.

e. It leads to proteinuria.

9- The following is correct about minimal change glomerular disease: {Formative2024}

A. It is the commonest cause of nephrotic syndrome in adults.

B. It is the commonest cause of nephrotic syndrome in children.

C. It is the commonest cause of acute nephritic syndrome in adults.

D. It is the commonest cause of acute nephritic syndrome in children.

10- Kimmelstiel-Wilson nodule is characterized by the following: {Final2021}

a. It is a localized sclerotic nodule located at one side of the glomerular capillary tuft.

b. It is seen in diabetic glomerulosclerosis.

c. It has the same staining properties as Mesangeal matrix.

d. All the above.

e. None of the above

11- In diabetic glomerulosclerosis; Capsular drops means: {Summar2022}

a- Accumulation of plasma proteins between visceral and parietal epithelial cells.

b- Accumulation of plasma proteins between capillary endothelium and glomerular basement membrane.

c- Localized sclerotic nodule located at one side of the glomerular capillary tuft and has the same staining properties as Mesangeal matrix.

d- Immune complexes deposited between visceral epithelial cells and basement membrane.

12- Diabetic nephropathey is not characterized by:

a. Diffuse glomerulosclerosis

b.Nodular glomerulosclerosis (Kimmel Steil-Wilson lesion)

c.Thickening of the glomerular basement membrane and loss of foot processes

d.Crescent formation

13- Fusion of the foot processes of the podocytes is characteristic of: {Merit}{شتاء}

A. Acute diffuse proliferative glomerulonephritis B. Rapidly progressive glomerulonephritis

C. Membranous glomerulonephritis D. Minimal change glomerulonephritis

14-The main lesion in minimal change glomerulonephritis is:

A- Loss of foot processes of viseral epithelial cell B- Mesangeal proliferation

C- Glomerular parital epithelial cell proliferation D-Sub epithelial electron dense deposit

15- In minimal change glomerulonephritis there is… {Merit}

A. Proliferation of the mesangial cells. B. Proliferation of the epithelial cells.

C. Proliferation of the endothelial cells. D. Loss of the foot process of the podocytes

16- The diagnostic histologic feature of rapidly progressive glomerulonephritis is {Final2023}{ثناء}

a. Hypercellular glomeruli b. Segmental necrosis of the glomeruli

c. Epithelial crescents d. Protein casts in the tubules

E. Thick-walled blood vessels

17- Epithelial crescents are characteristic of: {Merit}{ثناء}{قسم}

A. Acute diffuse proliferative glomerulonephritis B. Rapidly progressive glomerulonephritis

C. Membranous glomerulonephritis D. Membrano-proliferative glomerulonephritis

18- Rapidly progressive glomerulonephritis (crescentic GN), the cresent is due to proliferation of:

A- Glomerular parietal cells. B. Glomerular capillary.

C. Glomerular mesangeal cells. D. Glomerular mesangeal matrix.

19- Acute diffuse proliferative glomerulonephritis is NOT characterised by: {Formative2022}{ثناء}

A, Hypercellular glomeruli B. Infiltration of the glomeruli by polymorphs

C. Mostly occur in children D. Epithelial crescents in the glomeruli

20- The histological alterations of glomerulonephritis include all except: {Formative2022}

A. Hypercellularity. B. Crescent formation.

C. Cyst formation D. Thickening of glomerular basement membrane.

21- All of the following cause Nephrotic syndrome except {Final2023}

a. Minimal change disease b. Membranoproliferative Glomerulonephritis

c. Acute diffuse proliferative Glomerulonephritis d. Membranous Glomerulonephritis

22- Renal damage in ..... occurs through gradual progressive degeneration of renal papillae with interstitial tissue inflammation. {Mid2021}

a. Post-streptococcal glomerulonephritis b. Analgesic nephropathy

c. Chronic pyelonephritis d. Good-Pasteur renal disease

e. Membranous glomerulonephritis

23- The most common cause of nephrotic syndrome in children is: {Formative2022}

A. Acute diffuse proliferative glomerulonephritis B. Membranous glomerulonephritis

C. IgA nephropathy D. Minimal change glomerulonephritis

24- In acute diffuse glomerulonephritis; {Formative2022}

A. Mechanism of glomerular injury is circulating immune complexes trapped in the glomeruli.

B. The formed antibodies react with glomerular antigens instead of bacterial antigens.

C. The most presenting feature is nephrotic syndrome.

D. All cases progress to chronic glomerulonephritis.

25-The following is NOT a cause of nephrotic syndrome: {Merit}{ثناء}{قسم}

A. Acute diffuse proliferative glomerulonephritis B. Membranous glomerulonephritis

C. Membrano- proliferative glomerulonephritis D. Minimal change glomerulonephritis

26- The pathogenesis of glomerular diseases is: {Merit}

A. Immune mediated. B. Inflammatory reaction. C. Hypertension related.

D. Proliferation of the mesangial cells. E. A tumor of the renal glomeruli.

27-A child suffered from upper respiratory tract infection. Two weeks later, he presented with loin pain, edema of the face with puffiness of the eyelids. His renal biopsy will show the picture of: {Merit}{ثناء}

A. Acute diffuse proliferative glomerulonephritis. B. Membranous glomerulonephritis.

C. Membranoproliferative glomerulonephritis. D. Chronic glomerulonephritis

E. Focal glomerulonephritis.

28- Subepithelial electron dense deposits are characteristic of: {ثناء}{قسم}

A. Acute diffuse proliferative glomerulonephritis B. Rapidly progressive glomerulonephritis

C. Membranous glomerulonephritis D. Membrano-proliferative glomerulonephritis

29- A child complains of fever and hematuria 2 weeks after an attack of acute tonsillitis, the most possible diangosis is: {ثناء}

A. Rheumatic fever. B. Acute diffuse glomerulonephritis.

C. Wilm's tumor. D. Chronic glomerulonephritis. E. None of the above.

30-The most common cause of nephrotic syndrome in adults is : {قسم}

A.Minimal change glomerulonephritis. B.Membrano-proliferative glomerulonephritis.

C.Membranous glomerulonephritis. D.Acute diffuse proliferatrive glomerulonephritis.

31-The most common cause of nephrotic syndrome in children is : {قسم}

A.Minimal change glomerulonephritis. B.Membrano-proliferative glomerulonephritis.

C.Membranous glomerulonephritis. D.Acute diffuse proliferatrive glomerulonephritis

32-The following is "not-correct " in nephrotic syndrome: {قسم}

A.Massive proteinurea exceed 3 gm/ 24 hours. B.Tea-colored urine.

C.Hyperlipidaemia. D.Hypoalbuminaemia.

33-The histological alterations of glomerulonephritis include all "except": {Final2022}{قسم}

A.Hypercellularity. B.Crescent formation.

C.Cyst formation. D.Thickening of glomerular basement membrane.

34-Oliguria, hypertension and edema are features of {قسم}

A.Nephrotic syndrome. B.Acute renal failure C.Chronic pyelonephritis.

D. Chronic renal failure. E. Nephritic syndrome

35- Not a manifestation of nephritic syndrome:

a. Mild hypertension B. Haematuria C- Uremia. D. Hypercholesterolaemia

36-Hypoproteinemia, heavy proteinuria and edema are features of {قسم}

A.Nephrotic syndrome. B.Acute renal failure. C.Chronic pyelonephritis.

D. Chronic renal failure. E. Nephritic syndrome.

37-Edema in nephrotic syndrome is caused by {قسم}

A. Increased hydrostatic pressure of blood. B. Proteinuria and ↑ osmotic pressure of blood

C. Hypoproteinemia and ↓osmotic pressure of blood. D. Acute renal failure and oliguria.

E. Chronic renal failure and ↓ glomerular filtration.

38-Acute Post streptococcal GN characterized by all the following except

a. Subepithelial deposits along GBM

b. Crescent formation

c. Characterized by glomerular diffuse cellular proliferation and leucocytic infiltration

d. Occurs mostly in children

39- The diagnostic histological feature of rapidly progressive GN is {Merit}

A. Hypercellular glomeruli. B. Necrosis in the glomeruli.

C. Protein casts in the renal tubules D. Epithelial crescents.

E. Thick-walled blood vessels.

40-The aetiology of rapidly progressive glomerulonephritis may be: {ثناء}

A. Post-streptococcal B. Systemic lupus erythematosus.

C. Formation of antiglomerular basement membrane antibody D. All of the above

41-A woman suffering from systemic lupus erythematosus is now complaining of hematuria, renal biopsy showed generalized glomerular hypercellulatiry, thickening of the glomerular capillary walls and epithelial crescents. The most possible, diagnosis is: {ثناء}{Merit} بصمجها

A. Focal glomerulonephritis. B. Membranous.

C. Diffuse proliferative glomerulonephritis. D. Chronic glomerulonephritis.

42- Diabetic nephropathy predisposes to: {Merit}{ثناء}

A. Glomerulosclerosis B. Renal arteriolar sclerosis

C. Pyelonephritis D. Papillary necrosis E. All of the above

43- Acute diffuse proliferative glomerulonephritis follows infection by: {ثناء}{قسم} تاكد

A. Streptococcus viridans B.Streptococcus haemolyticus C. Streptococcus foecalis

D. Staphylococcus aureus E. E. Coli

44- Acute diffuse proliferative glomerulonephritis is due to: {ثناء}{قسم}

A. Blood-borne infection B. Ascending infection

C. Deposition of immune complexes in the glomeruli

D. Autoimmune reaction E. None of the above

45- The following is NOT a cause of haematuria: {ثناء}

A. Schistosomiasis of the bladder B. Renal stones C. Nephrotic syndrome

D. Tumors of the urinary system E. Congenital polycystic kidney

46- Oedema of the nephritic syndrome appears in: {ثناء}{قسم}

A. Face B. Lower limbs C. Back D. All-over the body

47- Oedema in the nephritic syndrome is due to: {ثناء}

A. Hypoproteinaemia B. Excessive reabsorption of water and salt by the tubules

C. Increased hydrostatic pressure of the blood

D. All of the above E. None of the above

48- Oliguria in the nephritic syndrome is due to: {ثناء}

A. Decreased blood volume B. Decreased renal blood flow C. Decreased water intake

D. All of the above E. None of the above

49- Hypoproteinaemia in the nephrotic syndrome is due to: {ثناء}

A. Loss of proteins in urine B. Decreased protein intake

C. Decreased absorption of proteins in the intestine

D. Escape of proteins to the interstitial tissue E. None of the above

50- Membrano-proliferative glomerulonephritis is characterised by all EXCEPT: {ثناء}

A. Hypercellular glomeruli B. Increased mesangial matrix

C. Infiltration of the glomeruli by polymorphs

D. Thickening of the glomerular capillary walls E. Protein casts in the tubules

51- The following changes occur in minimal change glomerulonephritis: {ثناء}

A. Proliferation of the mesangial cells B. Proliferation of the mesangial cells

C. Proliferation of the capsular epithelium D. All of the above

E. None of the above

52- The following disease is diagnosed only by electron microscopy: {ثناء}

A. Acute diffuse proliferative glomerulonephritis B. Rapidly progressive glomerulonephritis

C. Membranous glomerulonephritis D. Membrano-proliferative glomerulonephritis

E. Minimal change glomerulonephritis

53- A child suffering from post streptococcal glomerulonephritis may die due to: {ثناء}

A. Acute renal failure. B. Complications of hypertension. C. Chronic renal failure.

D. Any of the above. E. None of the above.

54- Upper respsiratory tract streptococcal infection may be followed by: {ثناء}

A. Rheumatic fever. B. Acute diffuse proliferative glomerulonephritis.

C. Rapidly progressive glomerulonephritis.

D. Any of the above. E. Non of the above.

55-Blood casts in urine are seen in:

a. acute diffuse proliferative GN b. chronic diffuse GN

c. minimal change GN d. pyelonephritis

56-All of the following characterised the nephrotic syndrome EXCEPT:

a. oedema b. heavy proteinuria c. hyperlipidemia

d. glomerular lesions are always seen by light microscopy

1. C	2. D	3. A	4. D	5. E
6. B	7. C	8. A	9. B	10.D
11.A	12.D	13.D	14.A	15.D
16.C	17.B	18.A	19.D	20.C
21.C	22.B	23.D	24.A	25.A
26.A	27.A	28.D	29.B	30.C
31.A	32.B	33.C	34.E	35.D
36.A	37.C	38.B	39.D	40.D
41.C	42.E	43.B	44.C	45.C
46.A	47.B	48.B	49.A	50.C
51.E	52.E	53.D	54.D	55.A
56.D	57.	58.	59.	60.

Disease of renal tubules & renal interstitium

1- Thyrodization of renal tubules is a histologic feature of: {3rdtral2024}

a. Hydronephrosis b. Pyonephrosis c. Acute pyelonephritis d. Chronic pyelonephritis

2-…….is a bacterial inflammation of interstitial tissues of renal pelvis, renal medulla, and renal cortex. {Summar2024}

a. Glomerulonephritis. b. Pyelonephritis.

c. Ischemic acute tubular necrosis. d. Toxic acute tubular necrosis.

3-An important microscopic feature of chronic pyelonephritis is: {Formative2024}{Merit}

a) Cortex is not differentiated from medulla b) Adherent renal capsule

c) Shedding of renal tubular epithelium d) Lymphoid follicle formation in interstitium

4- In chronic pyelonephritis; the kidneys are characterized by: {Final2021}

a. Enlarged size b. Smooth outer surface c. Adherent capsule

d. Cortex is well-differentiated from medulla e. Normal serum urea and creatinine

5- The histological feature of Chronic Pyelonephritis is: {Final2023}

a. Adherent Capsule b. Sheded mucosal cells

c. Lymphoid follicle in the interstitium d. No differentiation between cortex & medulla

6- Chronic pyelonephritis is characterize by EXCEPT:

a) The kidney is reduced in size and the capsule is thick and adherent.

b) The surface shows irregular depressions due to cortical scarring.

c) Some tubules are atrophic, others are dilated and contain large hyaline casts (colloid casts).

d) The blood vessels are thin and dilated.

7- Female came to the clinic with flank pain, dysuria and Neutrophil casts, the most possible diagnosis is {Final2023}

a. Acute cystitis b. Acute urethritis c. Acute prostatitis d. Acute Pyelonephritis

8- Ischemic acute tubular necrosis occurs due to: {Summar2022}{Formative2024}

a- Consumption of high dose of NSAI drugs b-Expsure to toxic poisons

c- Severe crush injuries d- Renal deposition of circulating immune complex

9- Which of the following is a feature of acute tubular necrosis? {Formative2022}

A. Adherent renal capsule B. Shedding of tubular epithelial lining

C. Normal serum urea and creatinine D. Commonly caused by E coli infection

10- Multiple small cortical abscesses could be seen in {Formative2022}

A. Acute pyelonephritis B. Chronic pyelonephritis

C. Hydronephrosis D. Acute post-streptococcal glomerulonephritis

11- Multiple focal necrosis is present in {Final2022}

a. Acute tubular injury b. Chronic pyelonephritis c. Acute pyelonephritis

12- ……is a Bacterial inflammation of the interstitial tissues of the renal pelvis, medulla and cortex. {Merit}

A. Pyelonephritis B. Glomerulonephritis C. Acute tubular necrosis D. Hydronephrosis

13- End arteritis obliterans occurs in: {Merit}

A. Acute glomerunephritis B. Acute tubular injury

C. Acute pyelonephritis D. Chronic pyelonephritis

14- The commonest causative organism of pyelonephritis is ……: {Merit}

A. typhoid bacilli B. E. coli C. staphylococcus aureus D. streptococcus foecalis

15- Thyroidization is a histologic feature of: {Merit}{ثناء}

A. Hydronephrosis B. Pyonephrosis C. Acute pyelonephritis D. Chronic pyelonephritis

16- Factors predisposing to pyelonephritis include all EXCEPT: {Merit}{ثناء}

A. Hypertension B. Urinary tract obstruction

C. Urinary tract instrumentation D. Schistosomiasis

17- The risk factors for pyelonephritis include all except……: {Merit}

A. Instrumentation of the urinary tract B. Urinary tract obstruction

C. Diabetes mellitus D. Antibiotics

18- Routes of infection in pyelonephritis: {Merit}{ثناء}

A. Ascending infection from the urinary bladder B. Lymphatic spread from the intestine

C. Blood spread from a septic focus D. All of the above

19- In acute pyelonephritis the interstitial tissue is mainly infiltrated by……. {Merit}

A. polymorphonuclear leucocytes B. lymphocytes C. RBCs D. Fibroblasts

20-A 40 years old woman presented with chronic renal failure-pyelography showed dilation of the pelvicalyceal system. Renal biopsy revealed tubular atrophy, periglomerular fibrosis and interstitial inflammatory cells. The most possible diagnosis is: {Merit}{ثناء}

A. Renal stones. B. Chronic glomerulonephritis.

C. Chronic pyelonephritis. D. Membranous glomerulonephritis

E. Diffuse proliferative glomerulonephritis

21-Bacteria reach the kidney in chronic pyelonephritis through …… {قسم}

A. Lymphatic spread from the colon. B. Direct spread from the uterus.

C. Ascending from the bladder. D. Ascending from the small intestine.

E. Lymphatic spread from the liver.

22-…….. predisposing factor for chronic pyelonephritis. {قسم}

A.Hypertension. B.Diabetes Mellitus. C.Contracted kidney.

D. Obesity. E. Atherosclerosis.

23-Chronic pyelonephritis is caused by obstruction of urinary tract or……. {قسم}

A.Stone in the bladder. B.Stone in the ureter. C.A tumor in the urinary bladder.

D. Vesico-ureteral reflux. E. A tumor in the ureter

24-Chronic pyelonephritis is characterized histologically by: {قسم}

A. Crescent formation and hyalinized glomeruli.

B. Acute inflammatory cell infiltration and necrosis.

C. Cystically dilated calyces and atrophic renal cortex.

D. Diffuse interstitial chronic inflammatory cell infiltration.

E. Invasion of the upper pole by polyhedral clear cells.

25-A 40-Years-old female presented with chronic renal failure. Radiography showed dilatation of the pelvicalyceal system. Renal biopsy showed tubular atrophy, periglomerular fibrosis and interstitial. What is the possible diagnosis? {قسم}

A) Renal stone. B) Chronic pyelonephritis. C) Membranous glomerulonephritis.

D) Chronic glomerulonephritis. E) Post infectious glomerulonephritis.

26- Anuria due to decrease in the glomerular filtrate, Blockage of the tubules by the necrotic cells, and complete reabsorption of the glomerular filtrate by the peritubular plexus is caused by:

A-Toxic tubular necrosis. B- Ischemic tubular necrosis

C- Obstruction by tumor D- Obstruction by stone

27-Nausea/Vomiting and Flank pain are symptoms of

A. Pyelonephritis B. Cystitis C. Prostatitis D. Urethritis

28-Acute pyelonephritis is characterized by all the following except

a. Show thyroidization of tubules

b. acute supp inflammation of the interstitium causing destruction of the tubules

c. Main etiology is ascending urinary tract infection

d. The cut surface shows small yellow. white abscesses with a hemorrhagic rim

29-Chronic pyelonephritis is a common cause of:

a. systemic hypertension b. the nephrotic syndrome c. both d. neither

1. D	2. B	3. D	4. C	5. C
6. D	7. D	8. C	9. B	10.A
11.A	12.A	13.D	14.B	15.D
16.A	17.D	18.D	19.A	20.C
21.C	22.B	23.D	24.D	25.B
26.B	27.A	28.A	29.A	30.

Tumors of the Urinary System

1-The most common malignant renal tumor is: {3rd tral2024}

a. Oncocytoma b. Angiomyolipoma

c. Clear cell renal cell carcinoma d. Sarcomatoid renal cell carcinoma

2-Schistosomiasis of the bladder predisposes to: {3rd tral2024}

a. Nephritic syndrome b. Stagehorn stone of kidney

c. Hydronephrosis d. Bladder carcinoma

3-The most common pediatric tumor if kidney: {3rd tral2024}

a. Chromophobe carcinoma b. Papillary renal cell carcinoma

c. Wilms tumor d. Transitional cell carcinoma

4-…….. characterize chromophobe RCC. {Summar2024}

a. Amyloid-containing stroma and papillary structures.

b. Amyloid-containing stroma and solid sheets.

c. Papillary structures containing foamy histocytes.

d. Papillary structures with pseudostratified epithelial lining.

5- Wilms tumor is synonym for: {Final2021}

a. Clear renal cell carcinoma. b. Chromophobe renal cell carcinoma.

c. Nephroblastoma. d. Angiomyolipoma.

e. Sarcomatoid renal cell carcinoma.

6- Nephroblastoma is called: {ثناء}

A. Warthin's tumor B. Wilm's tumor C. Benign mixed tumor

D. Pancoast's tumor E. Yolk sac tumor

7-Wilms tumor is: {3rd trial2022}{قسم}

a-Renal cell carcinoma. b-Chromophobe RCC. c-Nephroblastoma. d-Papillary RCC.

8-Hypernephroma is:

a. a renal pelvis carcinoma b. a renal cell carcinoma

c. a teratoma of the kidney d. an adrenal gland carc

9-Renal cell carcinoma cells are most commonly:

a. basophilic b. spindle cells c. acidophilic d. clear cells

10- Patients with renal cell carcinoma may present with: {Final2021}

a. Canon ball lung metastasis. b. Total hematuria. c. Renal (flank) mass.

d. Renal (flank) pain. e. All of the above.

11- Which of the following is a character of papillary Renal Cell carcinoma? {Final2023}

a. Amyloid material is present in the stroma b. Commonly is Multifocal and bilateral

c. Polyhedral cells with clear cytoplasm d. Triphasic

12- Mass that has heterogenous consistency in child? {Final2022}

A. Nephroblastoma

13- The most common type of renal cell carcinoma is:

A- Clear cell type b. Papillary type c. Chromophobe cell type D. Sarcomatoid type

14-The following is NOT a cause of hematuria: {Merit}

A. Schistosomiasis of the bladder B. Renal stones

C. Nephrotic syndrome D. Tumours of the urinary system

15- A 3-years-old child with his mother complained that she noticed a large swelling in his abdom for 2 weeks. She recognized that the mass gradually increased in size with loss of weight and generalized weakness of her child. On physical examination the child was underweight; pale wit a large soft tissue mass occupied the center of the abdomen. Sonography revealed a renal mas 7x8cm, firm, heterogeneous in consistency. What is the diagnosis? {Formative2022}{قسم}

A. Renal cell carcinoma. B. Nephroblastoma

C. Hydronephrosis. D. Chronic pyelonephritis.

16- A biopsy from the mass showed: {Formative2022}{قسم}

A. Branching papillary processes with central fibrous care covered by neoplastic cells.

B. Neoplastic spindle cells with hyperchromatic nuclei and eosinophilic cytoplasm arranged in bundles with frequent mitosis and necrosis.

C. Blastemal cells, mesenchymal spindle cells and aborted glomeruli.

D. Interstitial chronic inflammatory cell infiltration with endarteritis obliterans and renal tubular pink casts.

17-A 56-years-old man suffered from dysuria and hematuria for 3 months. Urine analysis showed RBCs, neutrophils and necrotic tissue. Transurethral resection (TUR) biopsy revealed cells nests and solid sheets of neoplastic epithelial cells invading the submucosa of the bladder with areas of ulceration. What is the diagnosis? {قسم}

A. Adenocarcinoma. B. Transitional cell carcinoma. C. Bilharzial cystitis.

D. Squamous cell carcinoma. E. Ulcerative cystitis.

18-What is the most serious complication of this lesion for the patient? {قسم}

A. Infection. B. Lung metastasis. C. Fistula.

D. Spread to the rectum. E. Spread to the urethra.

19-Renal cell carcinoma had all these criteria except: {قسم}

A. Golden yellow in colour. B. The cross section showed areas of hemorrhage and necrosis

C. Occupied the upper or lower pole. D. Had well-defined margins with a fibrous capsule.

E. Is called a conventional RCC.

20-Histopathology of conventional renal cell carcinoma is characterized by: {قسم}

A. Nests of polyhedral cells with clear cytoplasm and round nuclei.

B. Branching papillary processes with foamy histiocytes.

C. Spindle cell with marked nuclear pleomorphism and necrosis.

D. Sheets of neoplastic cells with round nuclei surrounded by clear halo.

E. Nests of columnar cells with deep pink cytoplasm and small nuclei.

21- The most frequent histological type of RCC is: {قسم}

A. ccRCC B. Papillary RCC C. Chromophobe RCC D. Oncocytoma

22- The followings charectrizes ccRCC, except: {قسم}

A. Polygonal neoplastic epithelial cells. B. Abundant clear cytoplasm.

C. Numerous thin-walled blood vessels. D. Abundant stromal deposition.

23- ……. Is the most frequent histological type of RCC, only after ccRCC. {قسم}

A. Wilm's tumor. B. Papillary RCC C. Chromophobe RCC D. Oncocytoma

24- The following doesn't charecterize type I papillary RCC. {قسم}

A. Foamy macrophages in the papillary cores. B. Hemosidrin-laden macrophages.

C. Pseudostratified epithelial lining. D. Psamomma bodies.

25- ……. Is present in type II papillary RCC. {قسم}

A. Foamy macrophages in the papillary cores. B. Hemosidrin-laden macrophages.

C. Pseudostratified epithelial lining. D. Psamomma bodies.

26- ……. Is seen in ccRCC {قسم}

A. High vascular stroma. B. Psamomma bodies.

C. Pseudostratified epithelium. D. Amyloid stroma.

27- Psamomma bodies are seen in …….. {قسم}

A. ccRCC. B. Chromophobe RCC. C. Papillary RCC. D. Nephroblastoma.

28- Amyloid stroma is seen in …….. . {قسم}

A. ccRCC. B. Chromophobe RCC. C. Papillary RCC. D. Nephroblastoma.

29- The following is in-correct about Wilm's tumor. {قسم}

A. It occurs in infants and young children. B. It may show bilateral involvment.

C. It may occur in extrarenal sites. D. It is more common in males.

30- Bilateral renal involvement may be found in …… . {قسم}

A. ccRCC & Chromophobe RCC. B. Chromophobe RCC & Papillary RCC.

C. Oncocytoma Chromophobe RCC. D. Papillary RCC & Nephroblastoma.

31- The clinical presentation of Wilm's tumor is …… . {قسم}

A. Painful hematuria. B. Painless hematuria. C. Abdominal pain. D. Abdominal mass.

32- The cell of origin of Wilm's tumor is …… . {قسم}

A. Endothelial cell. B. Stem cell.

C. Renal tubular epithelial cell. D. Fibroblastic cell

33- The most frequently occurring primary UB malignancy is ……. . {قسم}

A. Squamous cell carcinoma. B. Urothelial carcinoma. C. Lipoma. D. Leiomyoma.

34- The following is " incorrect" about UB carcinoma. {قسم}

A. The most important predisposing factor in Egypt is bilharziasis.

B. The most frequently occurring UB carcinoma is squamous cell carcinoma.

C. The most frequently occurring UB carcinoma is urothelial carcinoma.

D. It is more common in men.

35- In non-invasive urothelial carcinoma; the neoplastic urothelial cells …… . {قسم}

A. Are restricted to the surface urothelium. B.Invade the lamina propria but not the muscles.

C. Invade the lamina propria and the muscles. D. Reached the perivesical fat.

36-A 64-years old man complained of Lt Loin pain and hematuria for 7 days. MRI imaging showed a round well defined renal mass about 5x5cm, firm, yellow, with area of necrosis and hemorrhage. The mass infiltrated the renal capsule. What is the diagnosis? {قسم}

A.Renal stone. B.Chronic pyelonephritis. C.Renal hydronephrosis.

D. Renal cell carcinoma. E. Chronic glomerulonephritis.

37-A 56-years-old-man complained of hematuria and suprapubic pain for 3 months. A cystoscopy showed a urinary bladder papillary mass about 2x3cm. {قسم}

A.Bilharzial cystitis. B.Cystitis cystica. C.Squamous cell carcinoma.

D. Urothelial carcinoma. E. Cystitis glandularis

38-Histological features of renal cell carcinoma include all except: {قسم}

A) It may be papillary, tubular or solid pattern of growth.

B) The neoplastic cells may have clear or eosinophilic cytoplasm.

C) Chronic inflammatory cell infiltration.

D) The neoplastic cells may be sarcomatoid.

E) The stroma is scanty and vascular.

39-Wilms's tumor consists of: {قسم}

A)Blastema cells. B) Immature glomeruli and tubules.

C) Loose fibrous tissue stroma. D) Fibrous and muscular tissue.

E) Germ cells.

40- Gross features of renal cell carcinoma include all EXCEPT: {ثناء}

A. It is commonly centred in the cortex B. It is surrounded by a true capsule

C. It is commonly well circumscribed D. Cut surface is golden yellow

E. Cut surface shows haemorrhage and necrosis

41- Histologic features of renal cell carcinoma include all EXCEPT: {ثناء}

A. It may show tubular or papillary formations B. The cells have clear or granular cytoplasm

C. The cells contain mucin D. The cell may show sarcomatoid appearance

E. The stroma is very vascular

42- The most important prognostic factor in renal cell carcinoma is: {ثناء}

A. The patient's age B. The tumor size C. The degree of nuclear atypia

D. The degree of papillary formation E. The amount of intracellular glycogen

43- Wilm's tumor consists of: {ثناء}

A. Blastematous tissue B. Mesenchymal tissue C. Epithelial tissue

D. All of the above E. None of the above

44- The blastomatous tissue consists of: {ثناء}

A. Spindle-shaped cells B. Small primitive cells C. Small muscle cells

D. Malignant epithelial cells E. Neuroendocrine cells

45- RCC is defined as: {Merit}

a. Adenocarcinoma arising from tubular epithelium

b. Transitional cell carcinoma arising from tubular epithelium

c. Embryonic tumour arising from primitive cells

d. Adenocarcinoma arising from glomerular tufts

46- Papillary RCC is characterized grossly by being: {Merit}

a. Large, yellowish & circumscribed b. Bilateral & multifocal lesion

c. Fish flesh-like replacing most of kidney d. Small, hard & infiltrative lesion

47- The most common microscopic picture of RCC is: {Merit}

a. Clear cell b. Chromophobe c. Granular cell d. Papillary

48- Which of the following is right about the incidence of Wilms' tumour? {Merit}

a. It occurs commonly in young children, between 1-6 years old

b. It occurs most commonly from 6th to 7th decades

c. Above 40 years d. Any age

49- A 5-year-old-male child was presented with right-sided large palpable abdominal mass, pain & hematuria. Sonar revealed a large mass replacing most of the kidney. Microscopic examination of the excised mass showed a triphasic tumour that contains mixture of primitive epithelial & mesenchymal cells. What is your diagnosis? {Merit}

a. Wilms' tumour b. Renal Sarcoma

c. Undifferentiated carcinoma d. Chromophobe cell RCC

50- Wilms' tumour is characteristically mostly by being: {Merit}

a. Large, solitary & unilateral lesion replacing most of kidney

b. Multifocal & bilateral well-circumscribed lesion

c. Capsulated large nodule d. Papillary non-invasive lesion

51- Wilms' tumour is characterized by presence of: {Merit}

a. Mixture of primitive epithelial & mesenchymal cells b. Tumour cells with clear cytoplasm

c. Von Hansemann histocytes d. Tumour cells arranged in papillary pattern

52- Squamous cell carcinoma of the bladder commonly developed on top of: {Merit}

a. Bilharzial cystitis b. Bladder atrophy

c. Hyperplasia of urothelium d. Acute cystitis

53- Histologic types of bladder carcinoma include: {ثناء}

A. Squamous cell carcinoma B. Transitional cell carcinoma C. Adenocarcinoma

D. Sarcomatoid carcinoma E. All of the above

54- The most prognostic factor in bladder carcinoma is: {ثناء}

A. The patient's age B. The tumor size C. The depth of tumor invasion

D. The histologic grade E. The histologic type

55- Schistosomiasis of the bladder predisposes to: {ثناء}

A. Cystitis B. Stone formation C. Bladder carcinoma

D. All of the above E. None of the above

56- Leukoplakia of the urinary bladder means transformation of the bladder epithelium into: {ثناء}

A. Non-keratinized squamous epithelium B. Keratinized epithelium

C. Glandular epithelium D. Atrophic epithelium

E. Dysplastic epithelium

1. C	2. D	3. C	4. B	5. C
6. B	7. C	8. B	9. D	10.E
11.B	12.A	13.A	14.C	15.B
16.C	17.B	18.B	19.D	20.E
21.A	22.D	23.B	24.C	25.C
26.A	27.C	28.B	29.D	30.D
31.D	32.B	33.B	34.B	35.A
36.D	37.D	38.C	39.E	40.B
41.C	42.C	43.D	44.B	45.A
46.B	47.A	48.A	49.A	50.A
51.A	52.A	53.E	54.C	55.D
56.B	57.	58.	59.	60.

Biochemistry

Urinary stones

1-The following types of stones is associated with infection: {Formative2026}

a) Uric acid stones. b) Ca oxalate stones. c) Ca phosphate stones. d) Sturvite stones.

2-The following types of stones is radiobaque except: {Formative2026}

a)uric acid stones. b) cystine stones. c) Ca oxalate stones. d) Ca phosphate stones.

3- The following type of stones is radiolucent. {Summar2024}{3rd tral2024}

a. uric acid stones. b. Ca phosphate stones. c. Ca oxalate stones. d. cystine stones.

4-The following stones are radiolucent: {Formative2024}

a) Ca oxalate stones b) Uric acid stones

c) cystine stones. d) Ca phosphate stones

5-Most common cause of staghorn stone {Formative2022}

a-Calcium oxalate b-cystine stones c-Mg ammonium Po4 d-calcium phosphate

6- Uric acid stones are formed in {Final2023}

a. Alkaline Ph b. Acidic Ph c. Neutral Ph d. are not affected by Ph

7- Most common type of renal stones is {Final2023}

a. Ca containing stones b. Uric acid stones

c. Phosphate stones d. Calcium ammonium magnesium phosphate

8-The general appearance of calcium stones is: {3rd trial2022}{Summar2022}{Final2022}

a-Small, friable and yellowish. b-White, hard and radio-Opaque.

c-Small, faint and radio-opaque. d-Black, faint and radio-opaque

9- An alkaline PH favors the crystallization of: {Mid2021}

a. Calcium stones. b. Uric acid stones. c. All of the above. d. None of the above.

10-Acidic urine promotes which type of stone formation?

a) Cystine stones. b) Uric acid stones. c) Struvite stones. d) Both A & B.

11- Radiotransluscent renal stone is:

a) Calcium oxalate & phosphate stone b) Urate & uric acid stone

c) Cystine & xanthine stones. d) Triple stones or struvite stones

12-All of the following urinary stones are radio-opaque except:

a) Uric acid. b) Brushite. c) Calcium oxalate. d) Struvite.

13-Which of the following stones can't be diagnosed by plain X ray:

a. Calcium oxalate stones b. Calcium phosphate stones

c. Uric acid stones d. Struvite stones

14-Alkaline urinary pH can predispose to which of the following urinary stones:

a. Oxalate stones b. Struvite stones c. Cystine stones d. Uric acid stones

15-Urinary tract infection with Proteus vulgaris may predispose to:

a. Uric acid stone b. Calcium phosphate stone

c. Calcium oxalate stone d. Staghorn kidney stone

16-The most common kidney stone is:

a. urate b. carbonate c. phosphate d. oxalate

1. D	2. A	3. A	4. B	5. C
6. B	7. A	8. B	9. A	10.D
11.B	12.A	13.C	14.B	15.D
16.D	17.	18.	19.	20.

Physiology

Functional organization of the Kidney and renal blood flow

1- The juxtaglomerular apparatus secretes: {Summar2024}

a. Renin. b. Erythropoietin. c. Vitamin D. d. Angiotensin II.

2-Angiotensin II causes: {Formative2024}{Merit}

a) Stimulation of RBCs production b) Control of calcium excretion.

c) Formation of vitamin D d) Increase of peripheral resistance.

3-Renin is produced and secreted from granules located in the: {Formative2022}{VIP}

a. Macula densa b. Mesangial cells c. Intercalated cells

d. Afferent arteriole e. Interstitial cells

4- What molecule catalyses the formation of angiotensin I? {Final2021}

a. carbonic anhydrase. b. calcitriol. c. erythropoietin. d. rennin.

5- The cortical nephrons when compared to the juxtamedullary nephrons cortical nephrons have: {Summar2024}{Merit}

a. longer loops of Henle. b. greater numbers.

c. vasa recta. d. a major role in urine concentration

6- How are cortical nephrons different from juxtamedullary nephrons? {Final2021}

a. cortical nephrons lie almost entirely outside the renal medulla.

b. cortical nephrons have an associated vasa recta.

c. cortical nephrons have a longer tubule.

d. there are fewer cortical nephrons.

7-Cortical nephrons represent ……. of total kidney nephrons

A. 65% B. 15% C. 85% D. 50%

8-Which of the following is the main pressure that favours filtration in the glomerulus? {Formative2024}

a) Bowmans capsule colloid osmotic Pressure b) Bowmans capsule hydrostatic pressure

c) Glomerular capillary colloid osmotic pressure d) Glomerular capillary hydrostatic pressure

9- Which change tends to increase GFR? {Summar2024}

a. afferent arteriolar VC b. efferent arteriolar VD

c. afferent arteriolar VD d. Increased Bowman's capsule hydrostatic pressure.

10- Which of the following causes decrease in GFR? {Final2023}

a. Increase plasma colloid pressure b. Increase plasma hydrostatic pressure

c. Decrease afferent resistance d. Decrease pressure inside bowman's capsule

11- If the systemic blood pressure increased by 20 mmHg, {Final2023}

a. Glomerular capillary pressure increases by 20 mmHg b. GFR increases dramatically

c. Afferent arteriole constricts d. Efferent arteriole constricts

12- Glomerular filtration rate will be increased by: {Mid2023}{Final2021}{VIP}

a) Constriction of the afferent arteriole b) A decrease in afferent arteriolar pressure

c) Decrease resistance in efferent arteriole d) A decrease in the concentration of plasma protein

13-Amount of glomerular fluid filtered per day is: {Formative2024}{Merit}

a) 25 liters b) 200 liters c) 180 liters d) 125 liters

14-Normal glomerular filtration rate is: {Formative2022}

a. 25 ml/min. b. 75 ml/min. c. 100 ml/min. d. 125 ml/min.

15- Which of the following would cause an increase in both glomerular filtration rate (GFR) and renal plasma flow (RPF)? {3rdtral2024}

a. Hyperproteinemia b. A ureteral stone

c. Dilation of the afferent arteriole d. Constriction of the efferent arteriole

16- Renin secretion by the kidney is increased by increasing: {Mid2023}{VIP}

a) Angiotensin II b) Mean systemic blood pressure

c) Glomerular filtration rate d) Renal sympathetic nerve activity

17- A substance is being secreted by the renal tubules if its …… {Final2021}

a. Clearance rate is greater than 250 ml/minute.

b. Clearance rate is lower than 125 ml/minute.

c. Concentration falls along the proximal convoluted tubule.

d. Concentration in urine is lower than in plasma.

18- Glomerular filtration rate may be decreased by..... {Final2021}{VIP}

a. ADH. b. Renal oedema. c. Hypertension. d. Sympathetic stimulation.

19- Which of the following is a protein made by the kidney? {Summar2022}

a- Angiotensin b- Mesangial Matrix c- Creatinine d- Erythropoietin

20- Renal blood flow is kept constant by autoregulation in the mean systemic blood pressure range of..... {Final2021}{Formative2022}{VIP}

a. 0-90 mmHg. b. 60-180 mmHg. c. 200-250 mmHg. d. All of the above.

21-Renal blood flow is kept constant over a wide range of arterial pressures (about 60-180 mmHg). What is this range of pressure referred to as? {Formative2022}

a. the myogenic pressure threshold. b. the autoregulatory range

c. the threshold range. d. extrinsic regulation range.

22- Regarding tubuloglomerular feedback, which is true? {Final2021}

a. the sensor is macula densa cells.

b. operates via contraction of the mesangial cells.

c. acts to reduce GFR if the flow rate in the ascending loop of Henle falls.

d. GFR is modulated via contraction or dilation of the glomerular capillaries.

23-Which two factors are controlled by tubuloglomerular feedback? {Formative2022}

a. GFR and RBF b. RBF and tubular load.

c. GFR and tubular load. d. Filtration fraction and RBF.

24- If the glomerular filtration rate is too high, the macula densa sends a message to the afferent arteriole. What is the effect of this message? {Final2021}

a. granular cells of arteriole walls release rennin.

b. afferent arteriole dilates.

c. it inhibits the action of ATP and adenosine on the afferent arteriole.

d. afferent arteriole constricts.

25- The renal clearance of : {Final2021}

a. PAHA provides an estimate of glomerular filtration rate.

b. Chloride increases after an injection of aldosterone.

c. PAHA increases when the PAHA load exceeds the Tm for PAHA.

d. Urea is higher than that of inulin.

26- The renal clearance of a substance..... {Final2021}

a.Is inversely related to its urinary concentration, U.

b.Is directly related to the rate of urine formation, V.

c. Is directly related to its plasma concentration, P.

d. Must fall in the presence of metabolic poisons.

27-Renal blood flow {Formative2022}

a. Glomerular capillaries drain into peritubular veins.

b. Renal autoregulation is prevented by denervation.

c. Kidney's receive approximately 15% CO.

d. Angiotensin II causes greater constriction of the efferent than the afferent arteriole.

28- GFR increase by ..... {Final2022}

a. Increase of glomerular Co-efficient

b. Increase of afferent arteriole resistance

c. Decrease of efferent arteriole resistance

d. Decrease of afferent arteiole Blood Pressure

29- All of the following are criteria of a substance used for GFR measurement, EXCEPT {Summar2022}

a- Should not be secreted by the tubular cells. b- Should be secreted by the tubular cells.

c- not be reabsorbing by the tubular cells. d- Should not be toxic.

30- Which of the following is responsible for the high (GER)? {Formative2022}

a. low blood hydrostatic pressures in the glomerular capillaries.

b. High permeability of the glomerular capillaries

c. High hydrostatic pressure of the fluid in the Bowman's capsule.

d. High rate of water reabsorption across the walls of the proximal tubules.

31-The kidneys are: {Formative2022}

a. Help regulate blood volume. b. Help control blood pressure.

c. Help control PH. d. All of the above are correct.

32-Which of the following changes tends to increase glomerular filtration rate (GFR)? تاكد{Summar2022}

a- Increased afferent arteriolar resistance

b- efferent arteriolar resistance

c- Increased glomerular capillary filtration coefficient

d- Increased Bowman's capsule hydrostatic pressure

33-Renal blood flow is {VIP}

a. increased when renal sympathetic nerves are stimulated

b. Autoregulated when mean arterial blood pressure is between 60-180

c. Measured by urea clearance.

d. about 2 liter in a normal man

e. Decreased when the blood flow is at a site of macula densa decreased

34- Renal blood flow: {Merit}

a) is about 630 mL/minute..

b) can be measured by use of creatinine clearance..

c) remains fairly constant over a mean arterial pressure between 80-200 mmHg.

d) auto-regulation depends on intact sympathetic supply.

35- Increases in arterial pressure above 180 mmHg: {Merit}

a) Increases GFR b) Decreases GFR

c) No change in GFR d) First decreases then increase GFR

36- Autoregulation depends upon the concentration at macula densa of: {Merit}

a) Calcium chloride. b) Sodium chloride.

c) Potassium chloride. d) Magnesium chloride.

37- How does the juxtaglomerular apparatus respond when systemic blood pressure increases? {Merit}

a) The juxtaglomerular cells send a message to the afferent arteriole to dilate.

b) The macula densa sends a message to the efferent arteriole to constrict.

c) The macula densa sends a message to the afferent arteriole to constrict.

d) The granular cells release renin which causes systemic arterioles to constrict.

38- Four sections of the vasculature of the kidney tubule are listed below. Which one lists them in correct order of blood flow from left to right? {Merit}

a) Efferent arteriole, glomerulus, afferent arteriole, peritubular capillaries

b) Afferent arteriole, glomerulus, efferent arteriole, peritubular capillaries

c) Peritubular capillaries, afferent arteriole, glomerulus, efferent arteriole

d) Glomerulus, afferent arteriole, peritubular capillaries, efferent venule

39- Complete the following sentence correctly. Angiotensin II: {Merit}

a) Stimulates the adrenal glands to release aldosterone

b) Increases potassium reabsorption from the filtrate

c) Increases sodium (Na+) excretion at the kidneys

d) Reduces our thirst

40- Macula densa is present in: {Merit}

a) Collecting tubule b) Proximal convoluted tubule

c) Distal convoluted tubule d) Afferent arteriole

41- The glomerulus: {Merit}

a) Has both afferent and efferent arterioles

b) Contains capillaries that are at a higher hydrostatic pressure than the peritubular capillaries

c) Contains renin-secreting cells in its afferent arteriole.

d) is lined by fenestrated endothelium

e) all of the above.

42- The juxtamedullary nephrons are characterized by: {Merit}

a) Their glomeruli lie in the inner part of the cortex near the medulla.

b) Represent about 85% of the total number of nephrons

c) Their loops of Henle are short and penetrate to short distance in the medulla

d) Surrounded by U-shaped capillaries called the peritubular capillaries.

43- Juxta-medullary nephrons: {Merit}

a) make up more than 40% of the nephrons.

b) have their glomeruli in the renal medulla.

c) play an important role in creating hyperosmolarity of the renal medullary interstition.

d) are associated with afferent arterioles which only divide to form peritubular capillary network.

44- The juxtamedullary nephrons: {Merit}

a) constitute about 30% of nephrons in the human kidney.

b) have vasa recta that acts as a counter-current multiplier system..

c) are responsible for creating the hyperosmolarity of the medullary interstitium..

d) have proximal convoluted tubules, which extend into the renal medulla..

45- The juxtaglomerular apparatus secretes: {Merit}

a) Renin. b) Erythropoietin. c) Vitamin D. d) Angiotensin II.

46-Renin causes……

A. Decreased phosphate reabsorption B. Decreased water reabsorption

C. Angiotensin II formation D. Decreased sodium reabsorption

47- The juxtaglomerular cells of the JGA secrete: {Merit}

a) aldosterone. b) ADH. c) angiotensin II. d) renin. e) adrenaline.

48- Concerning the juxtaglomerular (JGA) apparatus which of the following is correct: {Merit}

a) it is located in the area of contact between the afferent arteriole, efferent arteriole, and DCT.

b) JG cells act as receptors for NaCl concentration.

c) macula densa cells secrete rennin.

d) its main function is to regulate glucose concentration in the plasma.

e) it is present in the juxtamedullary nephrons only.

49- Macula densa: {Merit}

a) is present in the collecting ducts. b) secretes renin.

c) responding to an increase in glucose conc. d) detects NaCl load.

e) is present in media of afferent arteriole.

50- Which of the following is not a function of the kidney: {Merit}

a) regulation of solutes concentration. b) production of renin.

c) production of angiotensin. d) activation of vitamin D.

e) urine formation.

51- All of the following are functions of the kidney EXCEPT: {Merit}

a) regulation of acid-base balance. b) production and release of renin.

c) conversion of angiotensin I into angiotensin II. d) activation of vitamin D.

e) regulation of the volume and ionic composition of body fluids.

52- Renin hormone is secreted: {Merit}

a) from juxtaglomerular cells. b) from macula densa cells. c) from epithelial cells.

d) in response to increased blood flow to the kidney. e) in response to increased NaCl.

53- Blood in route to glomeruli flows through which sequence of arteries? {Merit}

a. Interlobar-interlobular-arcuate. b. Renal-interlobar-interlobular.

c. Renal-interlobular-arcuate. d. Renal-interlobar-arcuate.

54- Which statement is correct about kidney circulation? {Merit}

a. Blood flows from the arcuate artery to the interlobar artery.

b. Blood flows from the arcuate vein to the interlobular vein.

c. Vasa recta are found in the cortex of the kidney.

d. Afferent arteriole connects with a glomerulus.

55- The functions of the kidney include: {Merit}

a) regulation of blood pressure.

b) regulation of the volume, composition, and pH of body fluids.

c) control of the rate of RBCs production.

d) all of the above are correct. e) only b & c are correct.

56-Regarding the juxtaglomerular apparatus: {VIP}

a. Lacis cells secrete renin.

b. The macula densa is modified afferent arteriole endothelium.

c. Renin is secreted in response to reduced tubular pressure.

d. Juxtaglomerular cells are located in the walls of the afferent arteriole of the glomerulus

57-The following substance leads to increased sodium reabsorption in kidney {VIP}

A. Aldosterone B. Vasopressin C. ANP D. PgE2 E. All of the above

58- Renal clearance of: {Merit}

a) inulin provides an estimate of GFR b) urea is higher than that of inulin

c) PAH falls when PAH load exceeds Tm for PAH d) of glucose is 180 ml / min

59-The GFR can be calculated by all of the following Except {VIP}

a. Ph clearance b. Creatinine clearance c. Urea clearance d. Inulin clearance

60-The following factors which increase they will increase the GFR Except {VIP}

a. Renal blood flow b. Afferent arterioles diameter

c. Efferent arterioles diameter d. Permeability of glomerular membrane

e. Capillary surface area

61-The substance that used to measure renal plasma flow should be {VIP}

a. Bound to plasma protein b. Easily reabsorbed c. Highly secreted

d. Metabolized by the body e. Stored only in the kidney

62-In a healthy adult man the GFR is expected to be {VIP}

a. 225 ml/min b. 325 ml/min c. 125 ml/min d. 25 ml/min e. 180 ml/min

63- Sympathetic stimulation of the kidneys: {Merit}

a) Decreases the GFR b) Increases the GFR

c) No change in GFR d) Increases urine formation

64- The first step in urine formation is: {Merit}

a) Glomerular filtration b) Tubular reabsorption.

c) Tubular secretion. d) Urine excretion.

65- All the plasma undergoes filtration EXCEPT: {Merit}

a) Water. b) Proteins. c) Glucose. d) Ions.

66-Which of the following is the main pressure that favors filtration in the glomerulus? {Merit}

a) Glomerular capillary hydrostatic pressure b) Glomerular capillary colloid osmotic pressure

c) Bowman's capsule hydrostatic pressure d) Bowman's capsule colloid osmotic Pressure

67- Average filtration fraction is approximately: {Merit}

a) 20% b) 25% c) 80% d) 60%

68-Normal GFR {VIP}

a. 25 ml/min b. 50 ml/min c. 75 ml/min d. 100 ml/min e. 125 ml/min

69- Which change tends to increase GFR? {Merit}

a) Increased afferent arteriolar resistance

b) Decreased efferent arteriolar resistance

c) Increased glomerular capillary filtration coefficient

d) Increased Bowman's capsule hydrostatic pressure

70-Which of the following is most likely to cause an increase in GFR {VIP}

a. Contraction of mesangial cells

b. Blockage of the ureters

c. Release of renin from Juxtaglomerular apparatus

d. Dilatation of afferent arterioles

71-GFR would be increased by {VIP}

a. A decrease of afferent arteriole measure

b. A decrease in renal blood flow

c. A decrease in efferent arteriole resistance

d. Decrease in the conc of plasma protein

e. Obstruction of ureters

72- If the glomerular capillaries hydrostatic pressure is 50 mm Hg, the hydrostatic pressure in the Bowman's space is 12 mm Hg, the colloid osmotic pressure is 30 mm Hg, what is the net pressure driving glomerular filtration .....mm Hg {Merit}

a) 8 b) 32 c) 48 d) 60

73- A negatively charged molecule is filtered with more difficulty compared to a positive one because: {Merit}

a) Presence of negatively charged protein on the filtering membrane

b) Negatively charged molecules are larger

c) Positively charged molecules are smaller

d) Urine is acidic

74- Appearance of proteins in urine in appreciable amounts: {Merit}

a) is directly related to plasma protein concentration

b) is greater following damage to the glomerular membrane

c) is increased by sympathetic stimulation

d) may be due to decreased reabsorption by the proximal tubule

75- Glomerular filtration rate is increased when: {Merit}

a) Plasma oncotic pressure is increased b) Glomerular hydrostatic pressure is decreased

c) Tubular hydrostatic pressure is increased d) Increased renal blood flow

76-The GFR will increase if {VIP}

a. Circulating blood volume increase b. The afferent arteriole resistance increase

c. Efferent arteriole resistance decrease d. The plasma protein conc decrease

77- In renal disease albumin is the first protein to appear in urine because: {Merit}

a) Of its high concentration in plasma

b) Has molecular weight slightly lesser than other proteins

c) High albumin: globulin ratio

d) Tubular epithelial cells are sensitive to albumin

78- Glomerular filtration rate increases by: {Merit}

a) Afferent arteriolar constriction b) Efferent arteriolar constriction

c) Hypotension d) Sympathetic stimulation

79- Glomerular filtration rate: {Merit}

a) is the volume of fluid filtered by all the glomeruli of one kidney/min.

b) is greatly decreased by lowering the mean arterial blood pressure to 90 mmHg..

c) is decreased when the efferent arterioles constrict.

d) is decreased as the number of functioning glomeruli decreases.

80- Use the values below to answer the following question: Glomerular capillary hydrostatic pressure = 47 mmHg. Bowman capsule hydrostatic pressure = 10 mmHg at what value of glomerular capillary oncotic pressure would glomerular filtration stop ..... mmHg {Merit}

a) 57 b) 47 c) 37 d) 10 e) 0

81- Glomerular filtration rate and renal blood flow will both be increased if: {Merit}

a) The afferent arteriole is dilated

b) The efferent and afferent arterioles are both constricted

c) Only the afferent arteriole is constricted

d) Only the efferent arteriole is dilated

82- Which of these factors oppose(s) filtration from the glomerulus? {Merit}

a) plasma oncotic pressure b) hydrostatic pressure in glomerular (Bowman's) capsule

c) plasma hydrostatic pressure d) both a and b

83- Which of the following decreases filtration from the glomeruli: {Merit}

a) an increase in the diameter of afferent arteriole.

b) a mild constriction of the efferent arteriole.

c) an increase in Bowman's capsule pressure.

d) an increase in renal blood flow.

e) an increase in filtration coefficient (K f).

84- GFR is increased when: {Merit}

a) afferent arteriole is constricted. b) efferent arteriole is severely constricted.

c) efferent arteriole is mildly constricted. d) efferent arteriole is dilated.

e) pressure in Bowman's capsule is increased.

85- Which of the following causes an increase in GFR: {Merit}

a) severe constriction of the efferent arterioles.

b) constriction of the afferent arterioles.

c) increased plasma osmotic pressure.

d) increased Bowman's capsule osmotic pressure.

e) decreased glomerular hydrostatic capillary pressure.

86- Which of the following would cause an increase in both renal blood flow and glomerular filtration rate? {Merit}

a) hyperproteinemia b) dilation of the afferent arteriole

c) dilation of the efferent arteriole d) stone in the ureter

87- Which of the following would cause an increase in GFR ? {Merit}

a) constriction of the afferent arteriole b) constriction of the efferent arteriole

c) increased plasma protein concentration d) constriction of the ureter

88- The glomerular filtrate: {Merit}

a) is different from plasma in its ionic composition. b) production rate is about 1 ml/min.

c) is hypotonic compared to plasma. d) contains a small amount of albumin

89-All of the following are functions of the kidney EXCEPT:

a) Regulation of acid base balance, b) Production and release of renin.

c) Conversion of angiotensin I into angiotensin II. d) Activation of vitamin D.

90-An 18-year-old male presents with muscle weakness, cramps, and tetany. Blood pressure is normal and no edema is present. Laboratory analysis reveals hypokalemic alkalosis, hyperaldosteronism, and high plasma renin activity. Which of the following statements about renin is true?

a)It is secreted by cells of the proximal tubule

b)Its secretion leads to loss of sodium and water from plasma

c)Its secretion is stimulated by increased mean renal arterial pressure

d)It converts angiotensinogen to angiotensin I

91-A patient with renal failure develops symptoms caused by the loss of a hormone produced by the kidney. Which of the following is the most likely diagnosis?

a)Oedema b)Hypertension c)Anaemia d)Acidosis

92-Renin in synthesized by

a) Granular cells (JG cells). b) Intercalated cells.

c) Macula densa cells. d) Mesangial cells.

93-Which of the following statements about renin is true?

a) It is secreted by the macula densa cells.

b) Its secretion is stimulated by an increased in arterial blood pressure.

c) Its secretion leads to expansion of ECF volume.

d) It converts angiotensin I to angiotensin II

94-Which of the following causes decreased renin release by the kidneys?

a) Decreased fluid and solute delivery to the macula densa.

b) Intravenous infusion of isotonic saline.

c) Narrowing of the renal artery.

d) Stimulation of renal sympathetic nerves

95-A decrease in the concentration of NaCl in the intraluminal fluid causes the juxtaglomerular apparatus to release which of the following hormones?

a) Aldosterone b) Adenosine c) Renin d) Angiotensinogen

96-A hypertensive patient is given an angiotensin-converting enzyme (ACE) inhibitor. Which of the following changes would be expected?

a) Plasma angiotensin I level will rise. b) Plasma angiotensin II level will rise.

c) Plasma bradykinin level will fall. d) Plasma renin level will fall

97-The juxtaglomerular apparatus:

a)Is located in the juxtamedullary mephron only.

b)Contains macula densa cells that act as baroreceptors stimulated by drop of ABP.

c)Contains juxtaglomerular cells that secrete renin.

d)Plays an important role in creating high osmolarity of the medullary interstitium

98-…….secretes renin

A. PCT B. Macula densa C. Juxtaglomerular cells D. lacis cells

99-Renal blood flow:

a)Is about 630 ml/min.

b)Can be measured by use of creatinine clearance.

c)Remains fairly constant over a mean ABP between 70-160 mmHg.

d)Auto-regulation depends on intact sympathetic supply

100-Renal autoregulation:

a)Is associated with increased renal vascular resistance when ABP is lowered from 100 to 80 mmHg.

b)Mainly involves changes in the caliber of efferent arterioles.

c)Maintains a normal renal blood flow during severe hypotension .

d)Minimizes the impact of changes in ABP on renal Na + excretion

101- Inulin clearance is one of the methods to measure:

a) GFR. b) RBF. c) RPF. d) Concentration of urine

102-Glomerular filtration rate (GFR) may be decreased by: ??

a) Constriction of efferent arterioles. b) Ingestion of diuretic drugs.

c) Intra-venous infusion of 1-2 liter of saline. d) Slight decrease in mean arterial pressure

103-Glomerular filtration rate:

a) Is the volume of fluid filtrated by all the glomeruli of one kidney/min.

b) Is greatly decreased by lowering the mean ABP to 90 mmHg.

c) Is decreased when the efferent arterioles constrict.

d) Is decreased as the number of functioning glomeruli decreases

104-The filtration fraction is increased by which of the following?

a) Increasing afferent arteriolar resistance b) Increasing efferent arteriolar resistance

c) Increasing plasma oncotic pressure d) Increasing the pressure within Bowman's capsule

105- The dimensions of renal clearance are

a) mg/mL. b) mg/min. c) mL plasma/min. d) mL urine/min

106-Which of the following is most likely to cause an increase in the glomerular filtration rate?

a)Contraction of mesangial cells b)Blockage of the ureter

c)Release of renin from the juxtaglomerular apparatus d)Dilation of the afferent arterioles

107-Filtration fraction equal:

a)20 – 30 % of RBF. b)40 – 45 % of RBF. c)10 – 15 % of GFR. d)15 – 20 % of RPF.

108-The normal GFR for an adult person is about:

a)125 ml%. b)125 ml/min. c)125 gm/ml. d)125 gm/min

109-Dilatation of efferent arterioles results in an increase in:

a)Glomerular blood flow. b)Glomerular capillary pressure.

c)GFR. d)Filtration fraction

110-Which of the following causes an increase in GFR?

a)Severe constriction of the efferent arterioles. b)Constriction of the afferent arterioles.

c)Increased plasma osmotic pressure. d)Increased glomerular hydrostatic capillary pressure

111-A 35-year-old male with polycystic kidney disease has a decrease in both glomerular filtration rate (GFR) and renal blood flow (RBF). The nephrologist wants to administer a drug that will increase both GFR and RBF. GFR and RBF would both increase if which of the following occurred?

a)The efferent and afferent arterioles are both dilated

b)The efferent and afferent arterioles are both constricted

c)Only the afferent arteriole is constricted

d)Only the efferent arteriole is constricted

112-The role of the kidney in homeostasis may include which of the following

a-regulation of extracellular fluid composition

b-regulation of red blood cell formation.

c-secretion of certain hormones, such as Ang-II, PGs & kinins in the regulation of blood pressure.

d- All of the above

113- All of the following about the nephron is true, except:

a- Has endocrine function

b- Is lined by a single layer of epithelial cells throughout its length

c- Is under endocrine control

d- Lies exclusively within the renal cortex

114-A patient with atherosclerosis shows signs of chronic renal failure attributed to poor renal perfusion and ischemic necrosis of the nephrons. Which of the following endogenous substances causes renal blood flow to decrease?

a)Nitric oxide b)Atrial natriuretic peptide c)Angiotensin II d)Dopamine

115-…….. stimulates the renin-angiotensin system.

A. Hypertension. B. Hypotension. C. Increase Na+ concentration in macula densa cells

116-The cells of the macula densa are sensitive to ……. concentration and rate of flow through the thick ascending limb / DCT.

A. Na+ B. K+ C. H+

117-About Angiotensin II, all are true EXCEPT

a. It stimulates aldosterone secretion.

b. It increases the reabsorbing force at the peritubular capillaries and increase sodium and water reabsorption.

c. It Stimulates sodium-potassium-ATPase pump at the luminal membrane

d. It Stimulates sodium-hydrogen exchange on the luminal membrane, especially in the proximal tubules.

118-The juxtaglomerular apparatus:

a. Functions as a sphincter around the distal tubules

b. Includes a long loop of Henle that dips into medulla

c. Participates in the control of aldosterone secretion through the renin-angiotensin system

d. Secrete angiotensin II

119-The juxtamedullary nephrons:

a. Have shorter loop of Henle

b. Contain Juxtaglomerular apparatus

c. Form more than 40% of the total nephrons

d. Responsible for concentration and dilution of urine

120- If the hydrostatic pressure in the glomerular capillaries is 50 mm Hg, the hydrostatic pressure in the Bowman's space is 12 mm Hg, the colloid osmotic pressure in the glomerular capillaries is 30 mm Hg, and there is no protein in the glomerular ultrafiltrate, what is the net pressure driving glomerular filtration?

a. 8 mm Hg b. 32 mm Hg c. 48 mm Hg d. 60 mm Hg

121- There is very little protein in the glomerular filtrate because:

a. All serum proteins are too large to fit through the glomerular pores

b. Of a combination of pore size and negative charges lining the pores

c. Of active reabsorption of filtered protein by the glomerular epithelial cells

d. Of metabolism of filtered proteins by the urinary flora

122-Urine formation is determined by the following equation

A. Excretion= Filteration – reabsorption + secretion

B. Excretion= Filteration – secretion + reabsorption

C. Excretion= Filteration +reabsorption + secretion

D. Excretion= Reabsorption + Filteration – secretion

1. A	2. D	3. D	4. D	5. B
6. A	7. C	8. D	9. C	10.A
11.C	12.D	13.C	14.D	15.C
16.D	17.A	18.D,B	19.D	20.B
21.B	22.A	23.A	24.D	25.C
26.B	27.D	28.A	29.B	30.B
31.D	32.B,C	33.B	34.C	35.A
36.B	37.C	38.B	39.A	40.C
41.E	42.A	43.C	44.C	45.A
46.C	47.D	48.A	49.D	50.C

51.C	52.A	53.D	54.D	55.D
56.D	57.A	58.A	59.A	60.C
61.C	62.C	63.A	64.A	65.B
66.A	67.A	68.E	69.C	70.D
71.D	72.A	73.A	74.B	75.D
76.D	77.B	78.B	79.D	80.C
81.A	82.D	83.C	84.C	85.D
86.B	87.B	88.D	89.C	90.D
91.C	92.A	93.C	94.B	95.C
96.A	97.C	98.C	99.C	100. D
101. A	102. A	103. D	104. B	105. C
106. D	107. D	108. B	109. A	110. D
111. A	112. D	113. D	114. C	115. B
116. A	117. C	118. C	119. D	120. A
121. B	122. A	123.	124.	125.

Mechanism of urine formation (filtration, Reabsorption , secretion)

1-The hormone that increases the reabsorption of Na+ in the DCT is: {Merit}{Formative2024}

a) Aldosterone b) Erythropoietin c) Thyroxin d) Angiotensin II

2-Sodium reabsorption occurs in all segments of the nephron except: {Merit}{Summar2024}

a) Proximal convoluted tubule. b) Distal convoluted tubule.

c) Descending limb of the loop of Henle. d) Ascending limb of the loop of Henle.

3-Which part of the nephron is impermeable to water? {Merit}{Formative2024}

a) Proximal convoluted tubule b) Distal convoluted tubule in the presence of ADH

c) Ascending limb of the loop of Henle d) Descending limb of the loop of Henle

4-Under what conditions will the kidney produce concentrated urine? If: {Merit} {Formative2024}

a) Glomerular filtration rate is low. b) Glomerular filtration rate is high.

c) Atrial natriuretic peptide is present in blood. d) Antidiuretic hormone is high in the blood.

5-Sodium is reabsorbed from all parts of the nephron except: {Merit}{Formative2024}

a) The Distal convoluted tubule. b) The Proximal convoluted tubule.

c) The descending limb of Loop of Henle. d) The ascending limb of Loop of Henle.

6-Which hormone acts on the distal tubule? {Merit}{Formative2024}

a) Vasopressin b) Thyroid hormone c) Corticosterone d) Adrenaline

7-Regarding glucose: {Merit}{Formative2024}

a) All filtered glucose will appear in urine

b) The tubular maximum for glucose (Tm is 475mg/ min)

c) Glucose is completely reabsorbed in the PCT

d) Glucose is reabsorbed through K+- glucose co-transport

8-Which of the following is most likely to be seen if a sample of fluid leaving the proximal tubule in a person with normal kidneys is analyzed? {Merit}{Formative2024}

a) It will be hypertonic compared with plasma b) It will have more K+ than plasma

c) It will have no amino acids d) It will have similar pH to that in ureter

9-In the kidney, parathyroid hormone influences phosphate metabolism as it: {Merit} {Formative2024}

a) Favours the tubular reabsorption of filtered phosphate

b) Inhibits the reabsorption of filtered phosphate

c) Decreases the tubular load by significantly decreasing the glomerular filtration rate

d) Increases the tubular load by decreasing calcium reabsorption.

10-Which of the following is normally not found in the urine: {Merit}{Summar2024}

A) Water. B) Waste products. C) Sodium. D) Albumin.

11-Reabsorption of water through the tubules occurs by {Merit}{Formative2024}

a) osmosis. b) active transport. c) facilitated diffusion. d) all of these.

12-What part of the nephron performs most of the reabsorption of materials from the filtrate? {Merit}{Summar2024}

a) The Bowman's capsule and glomerulus b) The loop of Henle (the nephron loop)

c) The distal convoluted tubule and collecting duct d) The proximal convoluted tubule

13-Which of the following is most permeable to water {Final2022}{Formative2022}

a. Thin ascending limb of the loop of Henle. b. Distal convoluted tubule.

c. Thin descending limb of the loop of Henle d. Cortical portion of the collecting tubule

14-in the kidney {Formative2022}

a. Glucose is secreted by the collecting duct

b. The renal threshold refers to glucose resorption from the urine.pc

c. Glucose is reabsorbed by secondary active transport

d. Glucose is reabsorbed in the loop of Henle.

15- Sodium reabsorption is coupled with the following substance in proximal convoluted tubule: {Formative2022}{VIP}

a. Hydrogen ion. b. Amino acid c. Glucose. d. All of the above

16-About……. from the filtered amount of Na+ is reabsorbed in PCTs

A. 25% B. 65% C. 1 0% D. 80%

17- Regarding Na+ reabsorption: {Merit}

a) Na+ reabsorption across proximal tubular cells is active and transcellular.

b) Na+ transport across the apical membrane provides energy for cotransport of H+

c) is increased by ADH .

d) can be blocked by lasix in the distal tubule .

18-Glucose is reabsorped from apical membrane of proximal convoluted tubule using: {Formative2022}{VIP}

a. Passive diffusion. b. Endocytosis.

c. Facilitated diffusion. d. Secondary active transport.

19-Na+ is reabsorbed from basolateral surface of the renal epithelial cells by:

a. Na+/H+ exchange b. Na-glucose cotransport

c. Na+-K+ pump d. Facilitated diffusion

20- The glucose started to appear in the urine before the transport maximum is reached because {Summar2022}{3rd trial2022}

a- Not all nephrons have the same transport maximum for glucose.

b- All of nephrons have the same transport maximum.

c- Filtered load of glucose is completely reabsorbed.

d- Filtered load of glucose is completely secreted

21- which of the following is true? {Final2023}

a. reabsorption of water is normally less in descending limb of loop of Henle than in Ascending loop

b. reabsorption of sodium is normally less in ascending limb of loop of Henle than in descending loop

c. Sodium is absorbed in all segments of Nephron except in thin descending loop of Henle

22- 25 year old male running, become very dehydrated with very high secretion of ADH, the maximum water reabsorption is in : {Final2023}

a. Cortical collecting tubule b. PCT c. DCT d. Loop of henle

23- In normal healthy subjects , urine : {Merit}

a) osmolality ranges from 300-400 m osmol/L b) specific gravity ranges from 1003-1060

c) phosphate content is increased by parathormone . d) pH rises after protein-rich meal.

24- Concerning water reabsorption by the nephrons: {Merit}

a) ADH increases water reabsorption from the PCT.

b) the thick part of the ascending limb of loop of Henle is impermeable to water.

c) the collecting ducts are the main sites for water reabsorption.

d) counter current mechanism increases water reabsorption from the DCT.

e) reabsorption of water decreases the osmolarity of urine.

25- The following is true about glucose reabsorption in the renal tubules {Final2023}{VIP}

a. Transport maximum is 80 mg/min

b. The transport maximum is dependent on the amount of glucose filtered

c. It's mainly by symport with k+

d. It's reduced with metabolic inhibitors, that reduce the activity of Na/K Atpase pump.

26- Regarding glucose reabsorption: {Merit}

a) it occurs by the process of secondary active transport.

b) the renal threshold is 300 mg/dL.

c) most of the glucose that is filtered through the glomerulus undergoes reabsorption by the proximal tubules and loop of Henle.

d) it is controlled by insulin.

27-Glucose reabsorption occurs in :

a. medullary collecting duct b. distal tubule c. loop of Henle d. proximal tubule

28- When Glucose level is more than Tmg, this means {Final2023}

a. Glucose excreted is equal to Glucose filtered

b. Glucose lever in artery equal Glucose level in vein

c. Excretion of Glucose Increases with Increasing its plasma concentration

29-The part of loop of Henle which is permeable to water is: {3rd trial2022}

a-Thick ascending limb b-Thin ascending limb

c-Thin descending limb d-All of them

30-Water reabsorption in the collecting duct {VIP}

a. Is dependent on aldosterone b. Is increased by ADH

c. Has the largest percentage of total water reabsorption

d. Is facilitated by Na reabsorption e. Independent of osmotic difference across tubules

31-which of the following substances will be more concentrated at the end of proximal tubule than at the beginning of proximal tubule {VIP}

a- glucose b- creatinine c- sodium d- bicarbonate

32- In a normal person, the concentration of glucose is higher in:

a) DCT than in PCT. b) Loop of Henle (LH) than in DCT.

c) First half of PCT than in second half of PCT. d) Bowman's capsule than in plasma

33- The thick ascending loop of Henle..... {Final2021}

a. Is impermeable to water. b. Has maximal permeability to NaCl.

c. Is relatively permeable to water. d. Is a site where there is no active transport of Na+

34- Glucose is reabsorbed from apical membrane of proximal convoluted tubule using..... {Final2021}

a. Passive diffusion. b. Endocytosis.

c. Facilitated diffusion. d. Secondary active transport.

35- About the proximal convoluted tubules, all are true EXCEPT:

a. reabsorb most of Na+ ions in glomerular filtrate

b. reabsorb most of Cl- ions in glomerular filtrate

c. reabsorb most of K+ ions in glomerular filtrate

d. contains JGCs which secrete renin.

36-The characteristic of loop of Henle include {VIP}

a. Descending limp is not permeable to water

b. Urine because more dilute as it reaches the tip of the loop

c. Ascending limb is permeable to water

d. Thin ascending limb urea is actively transported to interstitial fluid

e. Na is reabsorbed by Na/k/2cl in ascending limb

37-Hydrogen ion secretion in proximal convoluted tubules uses: {VIP}

A) passive diffusion B) proton pump C)H+,K+ exchange D)H+, Na+ exchange

38- Amino acids are almost complete reabsorbed from the glomerular filtrate via active transport in the..... {Final2021}

a. proximal tubule. b. loop of Henle. c. distal tubule. d. collecting duct.

39-About Na reabsorption from renal tubules the following is true except: {3rd tral2024}

a. two-thirds (65%) of the filtered sodium are reabsorbed in the PCT.

b. Na+ is reabsorbed via cotransport with amino acids.

c. Na + then passes to the interstitium by Na/K pump

d. aldosterone hormone increases sodium reabsorb from the PCT.

40-In the Thin descending limb of Henle: {3rd tral2024}

a. Fluid osmolarity is 1200 mosmole

b. It is permeable to water but poorly permeable to solutes.

c. - Impermeable to water but permeable to solutes.

d. is a diluting segment of the tubular system

41- Reabsorption of glucose in the renal tubular cells occur via: {Summar2024}

a. Primary Active transport b. Passive transport.

c. Secondary active transport with sodium d. Secondary active transport with potassium

42- All of the following is true regrading obligatory water reabsorption EXCEPT: {Summar2024}

a. occurs mostly in the proximal convoluted tubule.

b. represents 87% of renal water reabsorption.

c. depends on ADH. d. requires opening of aquaporin 1 channels.

43- the renal threshold of glucose is: {Summar2024}{Merit}

a. 100 mg/dl b. 125 mg/dl c.180 mg/dl d. 375 mg/dl

44- At plasma glucose concentration of 400 mg/dl:

a. Clearance of glucose is zero

b. Excretion rate of glucose equal the filtration of glucose

c. Reabsorption rate of glucose equals the filtration rate of glucose

d. Excretion rate of glucose increases with further increasing of plasma concentration of glucose

45- The maximum amount of substance that can be transported by renal tubules in I minute is: {Summar2024}{Merit}

a. Renal threshold. b. Load of substance.

c. Transport maximum. d. Filtration fraction.

46-The function of renal PCT include {VIP}

a. Reabsorption of all bicarbonate filtered at glomeruli

b. Production of renin

c. Reabsorption of 80% of filtered Hco3

d. Reabsorption of bicarbonate resulting in fixed acid secretion

e. hydrogen ion secreted by facilitated diffusion

47-Regarding water reabsorption: {VIP}

a. The ascending limb of the loop of Henle is impermeable to water.

b. Maximum urine osmolarity is about 800 mosm/L.

c. The CD is freely permeable to water even in the absence of ADH.

d. The DCT is permeable to water

48- The plasma concentration at which a substance appears in urine is: {Merit}

a) Renal threshold. b) Load of substance.

c) Transport maximum. d) Filtration fraction.

49- The amount of substance applied to renal tubules in one minute is: {Merit}

a) Renal threshold. b) Load of substance

c) Transport maximum. d) Filtration fraction

50- The glucose Transport Maximum is: {Merit}

a. 80 mg/min b. 120 mg/min c. 200 mg/min d. 375 mg/min

51- Which of the following is true regarding glucose transport in the renal tubules: {Merit}

a. Glucose appears in urine when its plasma concentration is between 120-170 mg/dl.

b. The glucose is not reabsorbed in the renal tubules in diabetic patients.

c. In nephrogenic glucosuria, glucose appears in urine when its plasma concentration reaches 180 mg/dl.

d. Glucose is a normal constituent of urine in healthy subjects.

e. Glucose could appear in the urine before reaching the Transport Maximum

52-A 58-year-old man is given a glucose tolerance test. In the test, the plasma glucose concentration is increased and glucose reabsorption and excretion are measured. When the plasma glucose concentration is higher than occurs at transport maximum (Tm), the {3rdtral2024}

a. clearance of glucose is zero

b. excretion rate of glucose equals the filtration rate of glucose

c. reabsorption rate of glucose equals the filtration rate of glucose

d. excretion rate of glucose increases with increasing plasma glucose concentrations

53- Glucose begins to appear in urine before reaching its Tm duo to {Final2022}

a. Every nephron don't have the same Tm

b. Tubular load of glucose is completely reabsorbed

c. Every nephron have the same Tm

54-The characteristics of the Loop of Henle include: {Formative2022}

a. Urine becomes more dilute as it reaches the tip of the loop

b. Ascending limb is permeable to water

c. In thin ascending limb urea is actively transported to interstitial fluid

d. Na is reabsorbed by Na-K ATPase in ascending limb

55-The functions of the renal proximal convoluted tubes include: {Formative2022}

a. Reabsorption of all bicarbonate filtered at the glomerulus

b. Production of renin

c. Reabsorption of 80% of filtered bicarbonate

d. Reabsorption of bicarbonate resulting in fixed acid secretion

e. Hydrogen ion secretion by facilitated diffusion

56-The reabsorption of Na+ in ascending loop of Henle: {3rd trial2022}

a-All passively. b-All actively.

c-Passive in thick, active in thin. d-Active in thick, passive in thin.

57- Sodium reabsorbed passively in: {Merit}

a) PCT. b) DCT. c) descending limb of loop of Henle.

d) thin part of the ascending limb of loop of Henle. e) CD.

58- Functions of Late distal tubule:

a. The principle cells reabsorb sodium and secrete potassium in hormonal independent.

b. The intercalated cells secrete hydrogen by second active transport.

c. Water reabsorption Only under the control of ADH

d. The tubular membranes of both segments are completely permeable to urea.

59- H+ is secreted: {Merit}

a) by the thin descending limb of loop of Henle b) by the P-cells in the collecting duct.

c) by the intercalated cells of the collecting duct. d) in response to ADH .

60-The principle cells in DCT are involved in……..secretion

A. Na+ B. K+ C. H+ D. Water

61- Facultative water reabsorption {Merit}

a) occurs mostly in the proximal convoluted tubule

b) represents 87% of renal water reabsorption

c) depends on ADH

d) requires opening of aquaporin I channels

62- The cells of the distal convoluted tubule: {Final2021}

a. Reabsorb about 50 per cent of the water filtered by the glomeruli.

b. Secrete hydrogen ions into the tubular lumen.

c. Reabsorb sodium in exchange for hydrogen or chloride ions.

d. Determine the final composition of urine.

63-A drug which inhibits carbonic anhydrase will do which of the following: {3rd trial2022}

a-Decrease bicarbonate formation and reabsorption in the kidney.

b-Increase plasma bicarbonate levels.

c-Increase blood PH. d-Decrease urinary loss of potassium ions.

64- For those substances that are actively reabsorbed or secreted, the highest limit of the amount that can be transported per unit time by the kidney tubules: {Formative2022}{VIP}

a. Depends on the rate at which the transport mechanism operates

b. Is directly related to the plasma concentration of the substance

c. Depends on the tubular transport maximum

d. Is dependent upon tubular load

e. Depends on the GFR

65- If the clearance of Substance X is greater than the clearance on inulin, it is most likely that substance X is: {Formative2022}{VIP}

a. Bound to tubular proteins b. Bound to plasma proteins c. Secreted d. Reabsorbed

66- The reabsorption of ..... ions and the resulting uptake of water by osmosis is stimulated by the hormone ..... {Merit}

a) Sodium, erythropoietin b) Sodium, ADH

c) Sodium, aldosterone d) Potassium, aldosterone

67- The appearance of glucose in the urine {Merit}

a) occurs normally.

b) indicates the presence of kidney disease.

c) occurs when the transport carriers for glucose become saturated.

d) is a result of hypoglycemia.

68- How does the composition of the filtrate change as it travels through the loop of Henle? {Merit}

a) In the ascending limb, the volume decreases, and in the descending limb, the concentration increases.

b) In the descending limb, the volume decreases, and in the ascending limb, the concentration decreases.

c) In the descending limb, the volume decreases, and in the ascending limb, the concentration increases.

d) In the ascending limb, the volume decreases, and in the descending limb, the concentration decreases.

69- What is the effect of antidiuretic hormone on the kidney tubules? {Merit}

a) It causes Na+ to be absorbed from the filtrate into the tubular cells.

b) It causes the concentration of urine to decrease.

c) It causes the filtrate volume to increase.

d) It causes the walls of the collecting duct to become permeable to water.

70- How does the descending limb of the loop of Henle differ from the ascending limb? {Merit}

a) The descending limb is impermeable to water but permeable to sodium chloride.

b) The ascending limb is permeable to water but impermeable to sodium chloride.

c) The descending limb is permeable to water but impermeable to sodium chloride.

d) The ascending limb is permeable to both water.

71- Concerning Henle's loop which of the following is correct: {Merit}

a) the descending limb is impermeable to water.

b) the cortical type acts as a counter current multiplier.

c) the juxtamedullary type is shorter than the cortical type.

d) the descending limb is impermeable to sodium.

72- P cells of the collecting tubules are: {Merit}

a) Less in number than the intercalated cells

b) Responsible for hydrogen secretion

c) Responsible for bicarbonate reabsorption

d) Responsible for sodium and water reabsorption.

73- The U-shaped part of the nephron is: {Merit}

a) The glomerulus. b) The Proximal convoluted tubule.

c) The Loop of Henle. d) The Distal convoluted tubule.

74- The greatest fraction of filtered sodium is reabsorbed in what portion of the nephron? {Merit}

a) Proximal tubule b) Loop of Henle c) Distal tubule

d) Cortical collecting duct e) Medullary collecting duct

75- Chloride is reabsorbed actively in which part of the nephron {Merit}

a) proximal convoluted tubule b) Thick ascending limb of Loop of Henle

c) descending limb of Loop of Henle d) Thin ascending limb of Loop of Henle

76-Which is CORRECT about transport in loop of Henle? {Final2022}

a. in thick part active and in thin part passive

77- Hypertonic urine is excreted due to absorption of water in: {Merit}

a) Collecting ducts b) DCT

c) Ascending part of loop of Henle d) Descending part of loop of Henle

78- A patient is diagnosed with severe "central" diabetes insipidus caused by lack of antidiuretic hormone secretion. Which part of the tubule will have the lowest tubular fluid osmolarity in this patient? {Merit}

a) Proximal tubule b) Descending loop of Henle

c) Distal convoluted tubule d) Collecting duct

79- The following hormones acts on DCT of the kidney EXCEPT: {Merit}

a) Angiotensin II. b) Parathormone. c) ADH. d) Aldosterone.

80- What effect does aldosterone have? {Merit}

a) Increases the reabsorption of Na+ from the kidney tubules

b) Makes the kidney tubules more permeable to water

c) Catalyzes the formation of angiotensin I

d) Blocks the release of ADH

81- What is the purpose of ANP in urine production? {Merit}

a) Stimulate the reabsorption of Na+ b) Stimulate the reabsorption of Ca++

c) Inhibit the reabsorption of Na+ d) Stimulate the reabsorption of water

82- On which of the following does aldosterone exert its greatest effect? {Merit}

a) Glomerulus b) Proximal tubule c) The loop of Henle d) collecting duct

83- Several hormones regulate the tubular reabsorption of water and electrolytes at different sites in the nephron. Which of the following combination is correct? {Merit}

a) Angiotensin in distal tubule b) Aldosterone in collecting ducts

c) ADH in proximal tubule d) ANP in loop of Henle

84- Which of the following changes would you expect to find in a dehydrated person deprived of water for 24 hours? {Merit}

a) Decreased plasma renin activity

b) Decreased plasma antidiuretic hormone concentration

c) Increased plasma atrial natriuretic peptide concentration

d) Increased water permeability in the collecting duct

85- In the presence of ADH, the filtrate will be isotonic to plasma in the: {Merit}

a) Descending limb of the loop of Henle b) Ascending limb of the loop of Henle

c) Cortical collecting tubule d) Medullary collecting tubule e) Renal pelvis

86- Transport of glucose in the renal tubular cells occur via: {Merit}

a) Primary Active transport b) Concentration gradient c) Facilitated diffusion

d) Secondary active transport with sodium e) Secondary active transport with potassium

87- The greatest fraction of filtered phosphate is reabsorbed in what portion of the nephron? {Merit}

a) Proximal convoluted tubule b) Ascending limb of Loop of Henle

c) Early part of Distal convoluted tubule d) P-cells of the collecting duct

88- Regarding the transport of glucose by the renal tubules: {Merit}

a) In a healthy person, the distal tubules reabsorb all of the filtered glucose

b) Glucose is secreted into the urine in small quantities

c) The transport maximum of glucose is about 36 mg/min

d) Glucose transport by the renal tubules is linked to sodium transport

89- Na+ is reabsorbed from the basolateral surface of the renal epithelial cells by: {Merit}

a) Na+/H+ exchange. b) Na-glucose cotransport

c) Na+-K+ pump d) Facilitated diffusion

90- Amino acids are almost completely reabsorbed from the glomerular filtrate via active transport in the: {Merit}

a) Proximal convoluted tubule b) Loop of Henle.

c) Distal convoluted tubule. d) Collecting duct

91- Glucose is: {Merit}

a) Filtered, reabsorbed, and secreted b) Filtered, reabsorbed, and not secreted

c) Filtered, secreted, and not reabsorbed d) Filtered and neither reabsorbed or secreted

92-Phosphate reabsorption in the proximal convoluted tubules is: {Merit}

a) Due to active co-transport with calcium ions

b) Due to active co-transport with sodium ions

c) Due to passive diffusion down its electrochemical gradient

d) Inhibited by calcitonin

93- In which section of the renal tubule is most water reabsorbed. {Merit}

a) Descending limb of the loop of Henle b) Ascending limb of the loop of Henle

c) Collecting duct d) Proximal convoluted tubule

94- The cells of the proximal convoluted tubules: {Merit}

a) have very few mitochondria. b) have many brush border.

c) responding mainly to ADH. d) responding mainly to aldosterone.

e) are for reabsorption only.

95-Which of the following transporters in the kidney is a primary active transporter?

a) Na^+/K^+ ATPase b) Na^+/glucose cotransporter

c) Na^+/Cl^- cotransporter d) Na^+/H^+ exchanger

96-The Na^+/H^+ exchanger in the proximal tubule is classified as:

a) Primary active transport b) Secondary active transport

c) Passive facilitated diffusion d) Simple diffusion

97-Which segment of the nephron reabsorbs most filtered proteins via pinocytosis?

a) Loop of Henle b) Distal convoluted tubule c) Proximal tubule d) Collecting duct

98-Amino acids are primarily reabsorbed in which segment of the nephron?

a) Loop of Henle b) Proximal convoluted tubule

c) Distal convoluted tubule d) Collecting duct

99-Which of the following statements is TRUE about glucose reabsorption in the nephron?

a) Glucose is reabsorbed along the entire nephron

b) Glucose is only reabsorbed in the proximal convoluted tubule

c) Glucose is reabsorbed mostly in the distal tubule

d) Glucose reabsorption depends on aldosterone

100-Where is the majority of phosphate reabsorbed in the nephron?

a) Distal convoluted tubule b) Proximal convoluted tubule

c) Thick ascending limb d) Collecting duct

101-Approximately what percentage of filtered phosphate is reabsorbed in the proximal tubule?

a) 25% b) 50% c) 75% d) 90%

102-What percentage of filtered urea is reabsorbed passively in the proximal tubule?

a) 25% b) 50% c) 75% d) 90%

103-Which of the following substances is NOT reabsorbed via sodium cotransport in the proximal tubule?

a) Glucose b) Amino acids c) Lactate d) Urea

104-The mechanism responsible for urea reabsorption in the proximal tubule is primarily:

a) Active transport b) Facilitated diffusion c) Endocytosis

d) Passive reabsorption secondary to water reabsorption ("back diffusion")

105-Which hormone decreases phosphate reabsorption in the proximal tubule?

a) Aldosterone b) Parathyroid hormone (PTH) c) ADH d) Insulin

106-The Na^+/H^+ exchanger in the proximal tubule functions primarily as:

a) A sodium channel b) A secondary active antiporter

c) A primary active transporter d) A potassium channel

107-Which transporter facilitates sodium reabsorption by coupling with phosphate in the proximal tubule?

a) Sodium-phosphate cotransporter b) Na^+/K^+ ATPase

c) Sodium-glucose cotransporter d) None

108-The fluid leaving the proximal tubule and entering the loop of Henle is:

a) Hypotonic b) Hypertonic c) Isotonic d) Variable

109-The fluid entering the loop of Henle has an osmolarity of approximately:

a) 100 mOsm/kg b) 300 mOsm/kg c) 600 mOsm/kg d) 1200 mOsm/kg

110-In the thin descending limb of the loop of Henle, water reabsorption leads to an increase in filtrate osmolality up to:

a) 300 mOsm/kg b) 600 mOsm/kg c) 900 mOsm/kg d) 1200 mOsm/kg

111-Compared to the volume filtered at Bowman's capsule, the volume of fluid entering the loop of Henle is approximately:

a) The same b) Two-thirds c) One-third d) One-tenth

112-Water reabsorption in the thin descending limb occurs primarily by:

a) Active transport b) Sodium-coupled cotransport

c) Osmosis driven by medullary hyperosmolarity d) Endocytosis

113-Which statement about the thin descending limb is TRUE?

a) It actively reabsorbs sodium b) It is impermeable to water

c) It is permeable to water but impermeable to solutes d) It secretes potassium into the lumen

114-Which effect does the thin descending limb have on filtrate composition?

a) Increases osmolality while decreasing volume

b) Decreases osmolality while increasing volume

c) Actively secretes sodium d) Reabsorbs urea actively

115-Reabsorption of sodium chloride in the thin ascending limb occurs mainly by:

a) Active transport b) Passive diffusion c) Sodium-glucose cotransport d) Endocytosis

116-What generates the positive luminal potential in the thick ascending limb of Henle?

a) Active sodium reabsorption only b) Back diffusion of potassium into the lumen

c) Water reabsorption d) Urea secretion

117-Which of the following is NOT a high-threshold substance in the renal tubules?

a) Glucose b) Vitamins c) Amino acids d) Phosphate

118-Which of the following is an example of a non-threshold substance in the renal tubules?

a) Glucose b) Phosphate c) Amino acids d) Creatinine

119-At plasma glucose concentrations below 200 mg/dL, the filtered glucose:

a) Is partially reabsorbed b) Exceeds the transport maximum

c) Is completely reabsorbed d) Appears in urine

120-The plasma glucose concentration at which glucose begins to appear in urine is called:

a) Transport maximum b) Renal threshold for glucose c) Filtration rate d) Tubular load

121-If plasma glucose rises above the renal threshold (~200 mg/dL), what happens?

a) All glucose is still reabsorbed b) Glucose appears in urine (glucosuria)

c) Na^+ reabsorption increases d) Water reabsorption stops

122-Which of the following conditions does NOT lead to glycosuria?

a) Diabetes mellitus b) Phloridzine c) Alloxane d) Hyperthyroidism

123-Parathyroid hormone (PTH)

a)Decreases tubular reabsorption of Ca2+¬ .

b)Decreases tubular reabsorption of phosphate.

c)Secretion is decreased in patients with chronic renal failure.

d)Secretion is stimulated by a rise in plasma ionized Ca2+.

124-Vassopressin (ADH):

a) Secretion is increased if the osmotic pressure of plasma decreases.

b) Help reabsorption of 65% of the filtered water in the PCT.

c) Increases water reabsorption in the medullary portion of the CD.

d) Administration stops diuresis occuring in diabetes mellitus.

125-Where is the site of action of vasopressin?

a)Proximal tubule. b)Descending loop of Henle. c)Distal tubule. d)Collecting duct

126- Sodium is reabsorbed passively in:

a) PCT. b) DCT. c) Descending limb of loop of Henle.

d) Thin part of the ascending limb of loop of Henle

127- It is secreted in the distal segments of nephron but not in proximal convoluted tubule (PCT):

a) K+ ions. b) H+ ions. c) NH3 d) Urea.

128- Which one of the following is NOT secreted in distal convoluted tubule (DCT):

a)Urea. b)NH4. c)H+ ions. d) K+ ions

129-In response to an increase in GFR, the proximal tubule and the loop of Henle demonstrate an increase in the rate of Na+ reabsorption. This phenomenon is called:

a) Renal autoregulation. b) Glomerulotubular balance.

c) Mineralocorticoid escape. d) Tubuloglomerular feedback

130- All the following statements about PCT true, EXCEPT:

a) It reabsorbs most of Na+ in tubular fluid.

b) It reabsorbs most of K+ in tubular fluid.

c) It contain juxtaglomerular cells which secrete renin.

d) It reabsorbs most of HCO3- in tubular fliud.

131-A patient develops acute renal failure due to an autoimmune disorder affecting the proximal tubules. What percentage of the filtered load of sodium is reabsorbed by the proximal tubule?

a) 15% b) 25% c) 45% d) 65%

132-A 63-year-old hospitalized woman becomes oliguric and confused. Her blood glucose is found to be only 35 mg/dL. An IV access is obtained and an ampule of 50% dextrose is given followed by a continuous infusion of 10% dextrose. Most of the glucose that is filtered through the glomerulus undergoes reabsorption in which of the following?

a)Proximal tubule b)Descending limb of the loop of Henle

c)Ascending limb of the loop of Henle d)Distal tubule

133- The main driving force for water reabsorption by the proximal tubule epithelium is:

a) Active reabsorption of amino acids and glucose.

b) Active reabsorption of Na + .

c) Active reabsorption of water.

d) The high colloid osmotic pressure in the peritubular capillaries

134-Which of the following is correct with regard to glucose and amino acid?

a)Reabsorption is most marked in the distal convoluted tubule.

b)Transport from the lumen of the nephron depends on Na+ transport.

c)Transport is controlled by parathormone. d)Transport is blocked by aldosterone

135- Primary active secretion of H+ occurs by intercalated cells at the:

a) PCT. b) DCT. c) Think ascending loop of Henle. d) Descending loop of Henle.

136-Kidney functions do not include the renal tubular secretion of:

a) Urea. b) Uric acid. c) Ammonia. d) Sodium

137-At the luminal cell membrane of renal tubule, amino acids are reabsorbed by:

a) Simple diffusion. b) Facilitated diffusion. c) A Na+- cotransport. d) Pinocytosis

138- Where does over 60% of the reabsorption of water occur?

a) Proximal tubule. b) Descending loop of Henle (DLH).

c) Ascending loop of Henle (ALH). d) Distal tubule.

139- Which of the following substances undergoes filtration, reabsorption and secretion by the renal tubules?

a) Bicarbonate. b) Potassium. c) Amino acids. d) Phosphate

140- If the plasma glucose level is 150 mg/dl and GFR = 100 ml/min, the tubular load for glucose would be:

a)125 mg/min. b)1500 mg/min. c)150 mg/min. d)90 mg/ml.

141-Aldosterone acts on cortical collecting ducts to

a)Decrease K+ secretion. b)Increase K+ secretion.

c)Decrease water permeability. d)Increase water permeability

142-In late part of DCT:

a) 60% of NaCL in the glomerular filtrate is reabsorbed.

b) The tubular fluid is made hypertonic relative to plasma.

c) Aldosterone increase K+ ion reabsorption.

d) Water reabsorption is under control of ADH

143- In the proximal convoluted tubules (PCT) all of the following occur EXCEPT: {Merit}

a) active reabsorption of 65% of sodium.

b) reabsorption of 65% of water under the control of ADH.

c) complete active reabsorption of amino acids.

d) complete active reabsorption of glucose. e) secondary active H+ secretion.

144- In a person having normal GFR, most of the filtered water is reabsorbed in:

a) PCT. b) DCT. c) Loop of Henle. d) C.T.

145- Where is the luminal membrane that is least permeable to water in the presence of ADH?

a) Proximal tubule. b) Descending loop of Henle.

c) Ascending loop of Henle. d) Distal tubule.

146- Sodium reabsorption in PCT is: {Merit}

a) passive at the basolateral border.

b) by secondary active transport at the basolateral border.

c) by uniporter at the luminal border.

d) about 90% of the filtered amount. e) about 65% of the filtered amount.

147-Glucose is: {Merit}

a) reabsorbed actively at the DCT. b) reabsorbed passively at the PCT.

c) reabsorbed actively at the PCT. d) reabsorbed actively at the DCT & PCT.

e) filtered but neither reabsorbed nor secreted by the nephron.

148- Which of the following is reabsorbed passively in the PCT: {Merit}

a) Na+. b) K+. c) glucose. d) urea. e) amino acids.

149-The sodium-potassium pump is an example of

A. Facilitated diffusion B. Osmosis C. Active transport D. Simple diffusion

150-In the late segment of PCT, Na+ is reabsorbed primarily with …… to maintain electric neutrality.

A. Cl- B. K+ C. H+ D. HCO3-

151-In the distal tubules, sodium reabsorption is increased directly by increasing: {VIP}

a. Sympathetic nerve stimulation of the kidney. b. Atrial natriuretic hormone secretion.

c. ADH secretion. d. Aldosterone secretion.

152-The effect of ADH on the kidney is to : {VIP}

a) increase the permeability of the distal nephron to water b) increase the excretion of Na+

c) increase the excretion of water d) increase the diameter of renal artery

153-Potassium secretion in DCT will be increased if there is increase in {VIP}

a. Plasma hydrogen b. Plasma volume c. Plasma sodium conc

d. Plasma potassium conc e. Plasma calcium conc

154-the amount of potassium excreted by the kidney will decrease if {VIP}

a- distal tubular flow increase b circulating aldosterone level increase

c- dietary intake of potassium increase d- sodium reabsorption by distal nephron decrease

155-The greatest amount of hydrogen ion secreted by the proximal tubule is associated with : {VIP}

a. excretion of K ion b. excretion of H ion

c. reabsorption if Ca ion d. reabsorption of Hco3 ion

156-hydrogen ion is Not secreted in the following part of the nephron {VIP}

a- proximal convoluted tubule b- loop of henele c- distal convoluted tubule

d- collecting duct E- non of the above

157-Secretion of potassium ions into the urine is:

a. Increased with elevated level of antidiuretic hormone b. Decreased by aldosterone

c. Increase in acidosis

d. Associated with the reabsorption of sodium from the distal tubules and collecting ducts

158-Which of the following will produce the greatest increase in potassium secretion?

a)A decrease in urinary flow rates b)An increase in distal nephron sodium concentration

c)An increase in sympathetic nerve activity d)A decrease in renal blood flow

159- The renal tubular transport maximum for glucose is about:

a) 140 mg/min. b) 240 mg/min. c) 340 mg/min. d) 440 mg/min

160- Which of the following substances does not have a Tm value?

a) Albumin. b) Glucose. c) Phosphate. d) Urea.

161-If a person consumes a high-K+ diet, the majority of K+ excreted in the urine is derived from:

a)Glomerular filtrate. b)K+ that is not reabsorbed in the proximal tubule.

c)K+ secreted by the cortical collecting duct. d)K+ secreted by the medullary collecting duct

162- Which of the following causes increased tubular secretion of hydrogen ions?

a)A decrease in arterial PCO2. b)Adrenal cortical insufficiency.

c)Administration of a carbonic anhydrase inhibitor.

d)An increase in tubular sodium reabsorption.

163-Which of the following transport processes is driven indirectly by primary active transport?

a) Na^+/K^+ ATPase pumping Na^+ out b) Na^+-glucose cotransport in the proximal tubule

c) H^+-ATPase secretion d) ATP-driven Ca^{2+} pump

164-Atrial natriuretic peptide causes

a. Angiotensin II formation b. Increased sodium reabsorption

c. Decreased sodium reabsorption d. Decreased phosphate reabsorption

1. A	2. C	3. C	4. D	5. C
6. A	7. C	8. C	9. B	10.D
11.A	12.D	13.C	14.C	15.D
16.B	17.A	18.D	19.C	20.A
21.C	22.B	23.C	24.C	25.D
26.A	27.D	28.C	29.C	30.B
31.B	32.C	33.A	34.D	35.D
36.E	37.D	38.A	39.D	40.B
41.C	42.C	43.C	44.D	45.C
46.C	47.A	48.A	49.B	50.D
51.E	52.D	53.A	54.D	55.C
56.D	57.D	58.C	59.C	60.B
61.C	62.B	63.A	64.C	65.C
66.C	67.C	68.C	69.D	70.C
71.D	72.D	73.C	74.A	75.B
76.A	77.A	78.C	79.A	80.A
81.C	82.D	83.B	84.D	85.C
86.D	87.A	88.D	89.C	90.A
91.B	92.B	93.D	94.B	95.A
96.B	97.C	98.B	99.B	100. B
101. C	102. B	103. D	104. D	105. B
106. B	107. A	108. C	109. B	110. D
111. C	112. C	113. C	114. A	115. B
116. B	117. D	118. D	119. C	120. B
121. B	122. D	123. B	124. C	125. D
126. D	127. A	128. A	129. B	130. C
131. D	132. A	133. B	134. B	135. B
136. D	137. C	138. A	139. B	140. C
141. B	142. D	143. B	144. A	145. C
146. E	147. C	148. D	149. C	150. A
151. D	152. A	153. D	154. D	155. D
156. E	157. D	158. B	159. C	160. D
161. C	162. D	163. B	164. C	165.

Renal handling

1- All are true for renal handling of acids in metabolic acidosis EXCEPT {Merit}

a) Hydrogen ion secretion is increased

b) Bicarbonate reabsorption is decreased

c) Urinary acidity is increased

d) Urinary ammonia is increased

2-The countercurrent exchange in the vasa recta: {Formative2024}{Merit}

a) raises the concentration of Na+ in the blood leaving the kidneys.

b) maintains high concentrations of NaCl in the extracellular fluid.

c) removes Na+ from the extracellular fluid.

d) causes large quantities of Na+ to enter the filtrate.

3-The prime driving force for counter current multiplier system is: {Formative2024}

a) Urea recycling

b) Action of ADH via aquaporin channels

c) Reabsorption of Na+ in thick ascending limb

d) Medullary hyposmolarity

4- The ability of the kidney to excrete a concentrated urine will increase if..... {Final2021}

a. the permeability of the proximal tubule to water decreases.

b. the rate of blood flow through the medulla decreases.

c. the rate of flow through the loop of Henle increases.

d. the activity of the Na-K pump in the loop of Henle decreases.

5- The collecting ducts in the kidney..... {Final2021}{Merit}

a. Can actively transport water molecules into the urine.

b. Are the sites of most of renal water reabsorption.

c. Are impermeable to water by antidiuretic hormone (ADH).

d. Determine to a large extent the final osmolality of urine.

6- In the nephron, the osmolality of fluid in the : {Final2021}{3rd trial2022}

a. Tip of the loop of Henle is less than that of plasma.

b. Bowman's capsules is less than that in the distal tubules.

c. Collecting duct rises when vasopressin is being secreted.

d. Proximal convoluted tubule rises along its length.

7-In normal kidneys, which of the following is true of the osmolarity of renal tubular fluid that flows through the early distal tubule in the region of the macula densa? {Formative2024}{Merit}

a) Usually, isotonic compared with plasma

b) Hypertonic, compared with plasma, in antidiuresis

c) Usually, hypertonic compared with plasma

d) Usually, hypotonic compared with plasma

8- Regarding the filtrate when leaving PCT {Mid2023}

a) concentration of sodium in the filtrate is the same after leaving PCT

b) hypotonic

c) hypertonic

d) when inulin is given its concentration when leaving PCT is less than its concentration in plasma

9- With respect to the counter current system {Summar2022}{3rd trial2022}{Formative2022}

a- The loops of Henle act as counter current exchangers

b- Solute diffuse out of vessels conducting blood towards the systemic veins.

c- Water diffuses out of the ascending vessels

d- Counter current multiplier is done by vasa recta

10-The following is true about the Vasta Recta: {Formative2022}

a) Descending vasa recta reabsorbs water from the interstitial fluid

b) The descending capillaries loose Na and Urea

c) Plasma becomes hyperosmotic as it descends in vasa recta

d) Ascending capillaries loose water to interstitial fluid

11-Which is CORRECT about osmolarity? {Final2022}

a. Increase in collecting duct by secretion of ADH

12-The following is true about vasa recta ? {VIP}

a. Descending vasa recta reabsorbs water from interstitial fluid

b. The descending capillaries loose Na and urea

c. Plasma becomes hyperosmotic as it descend in Vasa recta

d. Ascending capillaries loose water to interstitial fluid

e. They are reasonable for countercurrent multiplayer mechanism

13-The tonicity of urine as it enters the renal collecting duct is {VIP}

a. Isotonic b. Hypotonic or isotonic but never hypertonic

c. Hypotonic d. Hypertonic e. Hypertonic or isotonic but never hypotonic

14- Vasa recta:

a. Is branch from afferent arterioles

b. Is responsible for countercurrent multiplier

c. Is responsible for increasing medullary osmolality

d. Is responsible for maintaining medullary osmolality

15-A trauma patient with multiple rib fractures requires intubation and mechanical ventilation. Mechanical ventilation causes an increase in the patient's vasopressin secretion and plasma levels. Which of the following is the effect of vasopressin on the kidney?

a)Increased diameter of the renal artery b)Increased excretion of Na+

c)Increased excretion of water d)Increased permeability of the distal nephron to water

16- Vasopressin (ADH): {Merit}

a) secretion is increased if the osmotic pressure of plasma decreases.

b) help reabsorption of 65% of the filtered water in the proximal convoluted tubule..

c) increases water reabsorption in the medullary portion of the collecting duct.

d) administration stops diuresis occurring in diabetes mellitus.

17- What structure contain a hypertonic urine in the absence of ADH?

a) Proximal tubule. b) Descending loop of Henle.

c) Ascending loop of Henle. d) Distal tubule.

18- The fluid leaving the DCT may be {Merit}

a) Hypotonic or hypertonic but never isotonic b) Hypotonic or isotonic but never hypertonic

c) Hypotonic, isotonic or hypertonic d) Isotonic or hypertonic but never hypotonic

19- Which of the following parts of the nephron shows the highest osmolarity: {Merit}

a) PCT. b) DCT. c) thick part of the ascending limb of loop of Henle.

d) cortical collecting ducts. e) medullary collecting ducts in presence of ADH.

20- The tubular fluid is iso-osmotic at the end of: {Merit}

a) proximal tubules. b) descending limb of loop of Henle.

c) ascending limb of loop of Henle. d) distal tubules.

e) medullary collecting tubules.

21-Vasa recta:

a)About 50% of renal blood flow passes through them.

b)Are also called counter current multiplier.

c)Has a function to set-up renal medullary longitudinal osmotic gradient.

d)Has a function to maintain renal medullar longitudinal osmotic gradient.

22- The vasa recta is also called:

a)Diluting segment. b)Equilibrium device.

c)Conter-current exchanger. d)Counter-current multiplier.

23- Urea is secreted in: {Merit}

a) PCT b) DCT c) Descending loop of Henle d) Ascending loop of Henle

24- The prime driving force for counter current multiplier system is: {Merit}

a) Medullary hyposmolarity b) Reabsorption of Na+ in thick ascending limb

c) Action of ADH via aquaporin channels d) Urea recycling

25- The counter-current mechanism of the nephron: {Merit}

a) is dependent on GFR. b) is necessary for producing concentrated urine.

c) can be inhibited by ADH. d) is a function of the cortical nephrons only.

e) both b & d are correct.

26- Medullary hyperosmolarity causes: {Merit}

a) increased Na+ reabsorption from the DCT. b) increased H+ secretion from the PCT.

c) increased H2O reabsorption from the collecting ducts. d) release of angiotensinogen.

e) dilution of urine

27- In the distal convoluted tubule: {Merit}

a) 60% of NaCl in the glomerular filtrate is reabsorbed.

b) the tubular fluid is made hypertonic relative to plasma.

c) aldosterone increase K+ ion reabsorption.

d) water reabsorption is under control of ADH..

28- Regarding water diuresis, the following is true, EXCEPT: {Merit}

a) is produced by drinking large amount of a hypotonic fluid..

b) produces urine of somolality far less than osmolality of plasma..

c) results from decreased water reabsorption by the proximal tubules and loop of Henle.

d) is independent of hyperosmolar concentration of medullary interstitium.

29- Concerning the renal tubules: {Merit}

a) the ascending limb of the loop of Henle is freely permeable to water.

b) at the tip of the loop of Henle in the renal medullary papillae, the osmolality of the tubular fluid is several times that of the glomerular filtrate.

c) in the presence of, ADH the fluid in the collecting duct is hypo-osmotic with respect to plasma.

d) the permeability of the cortical collecting duct to urea is under the control of ADH.

30-The ability of the kidney to excrete a concentrated urine will increase if: {VIP}

a.the permeability of proximal tubule to water decreased

b. The rate of blood flow through the medulla decreases

c. the rate of flow through loop of Henle increases

d. Activity of Na/K pump in loop of Henle increase

31-When a person is dehydrated, hypotonic fluid will be found in the : {VIP}

a) glomerular filtrate b) proximal tubule

c) descending lobe of loop of henle d) distal convoluted tubule

32-H+ secretion increased in the following case:

a. increasing PCO2 in extracellular fluid

b. decreasing in concentration of H+ the extracellular fluid.

c. low aldosterone d. Hyperkalemia

33-…… is the part of the nephron which doesn't share information of medullary hyperosmolarity

A. Thin ascending limb of the loop of Henle B. PCT

C. The medullary collecting tubule D. The thick ascending limb of loop of Henle

34- In normal kidneys, which of the following is true of the osmolarity of renal tubular fluid that flows through the early distal tubule in the region of the macula densa?

a. Usually isotonic compared with plasma

b. Usually hypotonic compared with plasma

c. Usually hypertonic compared with plasma

d. Hypertonic, compared with plasma, in antidiuresis

35- The juxtaglomerular apparatus:

a. Filter fluid from the glomerular cappillries.

b. Reabsorbs sodium and chloride from the tubules.

c. Contains macula densa. d. Collecting ducts.

36-The Juxtaglomerular apparatus:

a. Reabsorbs vital solutes from the glomerulus. b. Secrete aldosterone hormone.

c. Sensitive to low sodium concentration in filtrate.

d. Filters fluids from the glomerular capillaries

37- The collecting ducts in the kidney:

a. Can secrete water molecules activity into the urine

b. Are responsible for most of the reabsorption of water that occurs in the kidneys

c. Determine to a large extent the final osmolarity of urine

d. Are rendered impermeable to water by antidiuretic hormone

38- If the medullary thick ascending limb of loop of Henle is selectively destroyed this will result in reduced:

a. Absorption of water b. Passive transport of sodium

c. Active transport of sodium d. Passive transport of chloride

39-ADH causes……

A. Angiotensin formation B. Decreased sodium reabsorption

C. Decreased phosphate reabsorption D. Increased water reabsorption

40-A 22-year-old woman runs a 10-kilometer race on a hot day and becomes dehydrated. Her ADH levels are very high and her kidneys are functioning normally. In which part of the renal tubule is the most water reabsorbed?

a. Proximal tubule b. Loop of Henle c. Distal tubule d. Collecting tubule

41- Regarding the medullary collecting duct, all of the following are true EXCEPT:

a. It is highly permeable to water in presence of ADH

b. H+ is secreted by secondary active transport

c. Sodium is actively reabsorbed

d. Urea is transported by facilitated diffusion

42- The ability of the kidney to excrete concentrated urine will increase if which of the following occurs?

a) The glomerular capillary pressure increases

b) The flow of filtrate through the loop of Henle increases

c) The activity of the Na+-K+ pump in the loop of Henle increases

d) The permeability of the collecting duct to water decreases

43-A 25-year-old male athlete has just completed a long distance bicycle race during a hot, humid day. At the conclusion of the race, he provides a urine sample for testing. Assuming that

his fluid intake during the race was zero, in what portion of the nephron is the tubular fluid osmolarity the lowest?

a. Early part of distal tubules b. Thin descending limb of the loop of Henle

c. Medullary collecting duct d. Bowman's capsule e. proximal tubules

44-Sodium is reabsorbed along all parts of the nephron except:

a) Proximal tubule b) Thin descending limb of Henle

c) Thick ascending limb of Henle d) Distal convoluted tubule

45-What percentage of the filtered sodium is reabsorbed in the proximal convoluted tubule (PCT)?

A) 25% B) 50% C) 65% D) 80%

46-What percentage of filtered water is reabsorbed obligatorily in the proximal convoluted tubule (PCT)?

A) 25% B) 50% C) 65% D) 80%

47-What percentage of filtered water is reabsorbed obligatorily?

A) 50% B) 65% C) 87% D) 95%

48-Facultative water reabsorption accounts for approximately what fraction of filtered water?

A) 5% B) 13% C) 25% D) 50%

49-Approximately what percentage of filtered sodium is reabsorbed independently of plasma sodium levels?

a) 50% b) 75% c) 97% d) 100%

50-What is the approximate osmolarity of the interstitial fluid in the innermost medulla?

A) 300 mOsm/L B) 600 mOsm/L C) 900 mOsm/L D) 1,200 mOsm/L

51-The remaining ~3% of sodium reabsorbed in the distal nephron is primarily regulated by:

a) ADH b) Aldosterone c) ANP d) PTH

52-Sodium reabsorption in the PCT primarily depends on which mechanism?

A) Passive diffusion B) Primary active transport

C) Facilitated diffusion D) Endocytosis

53-Which segment of the nephron is primarily responsible for the majority of hydrogen ion secretion?

A) Loop of Henle B) Distal convoluted tubule

C) First half of proximal tubule D) Collecting duct

54-Which nephron segment reabsorbs ~15% of filtered water obligatorily?

A) Thick ascending limb of loop of Henle B) Thin descending limb of loop of Henle

C) Distal convoluted tubule D) Collecting duct

55-Where is the Na/K -ATPase pump located in the PCT cells?

A) Apical membrane B) Basolateral membrane C) Luminal surface D) Cytoplasm

56-The reabsorption of HCO_3^- in the proximal tubule is associated with which process?

A) Sodium excretion B) H^+ secretion C) Potassium reabsorption D) Water retention

57-Facultative water reabsorption occurs in which part of the nephron?

A) PCT B) Thin descending limb

C) Distal nephron and collecting duct D) Loop of Henle

58-Facultative water reabsorption is regulated by:

A) Aldosterone B) ADH (vasopressin) C) ANP D) Renin

59-Which hormone facilitates urea movement in the inner medullary collecting duct?

A) Aldosterone B) ADH (vasopressin) C) ANP D) PTH

60-Which type of ADH receptor is found in the kidney?

A) V1 receptor B) V2 receptor C) V3 receptor D) Alpha-1 receptor

61-Activation of the cAMP pathway by ADH leads to:

A) Closure of aquaporin II channels B) Opening of aquaporin II channels

C) Closure of aquaporin I channels D) Opening of aquaporin I channels

62-ADH-induced water reabsorption primarily occurs in which nephron segment?

A) Proximal tubule B) Thick ascending limb of loop of Henle

C) Distal convoluted tubule and collecting duct D) Glomerulus

63-The reabsorption of urea in the inner medullary collecting duct contributes to:

A) Sodium excretion B) Medullary hyperosmolarity

C) Potassium reabsorption D) pH regulation

64-The fluid leaving the proximal tubule is:

A) Hyperosmotic B) Hypoosmotic C) Iso-osmotic D) Variable

65-Concentrated urine is produced when:

A) Circulating ADH levels are low B) Circulating ADH levels are high

C) Sodium intake is low D) Aldosterone is low

66-Which of the following conditions can lead to high ADH and concentrated urine?

A) Water deprivation B) Hemorrhage C) SIADH D) All of the above

67-The counter-current multiplier system is located in which part of the nephron?

A) Cortical nephron loop of Henle B) Juxtamedullary nephron loop of Henle

C) Distal convoluted tubule D) Collecting duct

68-The primary purpose of the counter-current multiplier system is to:

A) Maintain medullary hyperosmolarity B) Create medullary hyperosmolarity

C) Excrete urea D) Dilute plasma

69-What is the main function of the vasa recta in the kidney?

A) Create medullary hyperosmolarity B) Maintain medullary hyperosmolarity

C) Reabsorb glucose D) Secrete urea

70-How does a high-protein diet affect urine concentration?

A) Decreases the kidney's ability to concentrate urine

B) Increases the kidney's ability to concentrate urine

C) Has no effect on urine concentration

D) Increases urine volume only

71-Normally, the lowest urine osmolality is in the:

a)Beginning of DCT. b)End of DCT. c)Cortical CT. d)Tip of the loop of Henle

72- Which of the following results in an increased osmotic gradient in the medulla of the kidney?

a) Administration of a diuretic drug that inhibits Na + reabsorption by thick ascending limb cells.

b) Drinking a liter of water. c) Long loops of Henle. d) Low dietary protein intake

73-In addition to increasing the permeability of the collecting duct to water, ADH increases the permeability of the collecting duct to which of the following?

a) Ammonium b) Potassium c) Sodium d) Urea

74-Which of the following leads to decreased Na+ reabsorption by the kidneys?

a)An increase in central blood volume. b)An increase in GFR.

c)An increase in plasma aldosterone level. d)An increase in renal sympathetic nerve activity

75- The urine osmolality may be decreased by:

a) High medullary blood flow. b) High protein diet.

c) High plasma ADH level. d) Decreased urine flow rate

76-A patient with uncontrolled hypertension is placed on a new diuretic targeted to the Na+ reabsorption site from the basolateral surface of the renal epithelial cells. Which of the following transport processes is the new drug affecting?

a)Na+/H+ exchange b)Na+-glucose cotransport c)Na+-K+ pump d)Solvent drag

77-A 60-year-old woman is always thirsty and wakes up several times during the night to empty her bladder. Plasma osmolality is 295 mOsm/kg H2O (normal range, 281 to 297 mOsm/kg H2O), urine osmolality is 100 mOsm/kg H2O, and plasma vasopressin levels are higher than normal. The urine is negative for glucose. The most likely diagnosis is:

a)Diabetes mellitus. b)Diuretic drug abuse.

c)Nephrogenic diabetes insipidus. d)Neurogenic diabetes insipidus.

78-In a person with severe central diabetes insipidus (deficient production or release of vasopressin), urine osmolality and flow rate is typically about:

a)50 mOsm/kg H2O, 18 L/day. b)50 mOsm/kg H2O, 1.5 L/day.

c)300 mOsm/kg H2O, 1.5 L/day. d)300 mOsm/kg H2O, 18 L/day.

79-A 69-year-old male patient develops acute renal failure of prerenal origin, which has increased renin secretion by the kidney. A stimulus for increasing renal renin secretion is an increase in which of the following?

a)Mean blood pressure b)Sympathetic nerve activity

c)Circulating angiotensin II d)Atrial natriuretic peptide secretion

80-The effect of dietary protein on urine concentration is primarily due to:

A) Changes in aldosterone secretion

B) Increased urea production contributing to medullary hyperosmolarity

C) Increased sodium intake D) Decreased water intake

81-Excess ADH in SIADH leads to:

A) Polyuria and dehydration B) Water retention and hyponatremia

C) Increased potassium excretion D) Hypernatremia

82-The main problem in SIADH is:

A) Excess aldosterone secretion B) Excess ADH secretion

C) Deficiency of ADH D) Renal failure

1. B	2. B	3. C	4. B	5. D
6. C	7. D	8. A	9. B	10.C
11.A	12.C	13.C	14.D	15.D
16.C	17.B	18.B	19.E	20.A
21.D	22.C	23.D	24.B	25.B
26.C	27.D	28.C	29.B	30.B
31.D	32.A	33.B	34.B	35.C
36.C	37.C	38.C	39.D	40.A
41.B	42.C	43.A	44.B	45.C
46.C	47.C	48.B	49.C	50.D
51.B	52.B	53.C	54.B	55.B
56.B	57.C	58.B	59.B	60.B
61.B	62.C	63.B	64.C	65.B
66.D	67.B	68.B	69.B	70.B
71.A	72.C	73.D	74.A	75.A
76.C	77.C	78.A	79.B	80.B
81.B	82.B	83.	84.	85.

Renal Acid-Base Control Mechanisms

1- Which of the following has a function in maintaining the acid-base balance of the body? {Summar2022}

a- Juxtaglomerular cells b- Principle cells c- intercalated cells d- Macula densa

2- Which cell is responsible for acid-base balance ? {Final2022}

a. Intercalated cell

3- Secretion of H+ ions in the PCT is associated with: {Final2023}

a. Bicarbonate reabsorption b. Reabsorption of K c. Decrease PH below 4.4

4- Renal acid secretion is affected by all of the following Except: {Final2021}{Formative2022}

a. PCO2. b. K+ c. Aldosterone. d. Ca+2

5- More hydrogen is secreted in..... {Final2021}

a. alkalosis. b. administration of diamox. c. hypokalemia. d. hyperventilation.

6- In the renal tubules, bicarbonate is responsible for buffering hydrogen ion in..... {Final2021}

a. Proximal convoluted tubule. b. Loop of Henle

c. Distal convoluted tubule. d. Collecting ducts.

7- Which of the following buffers is a weak buffer in blood but highly effective in renal tubules {Final2021}

a. Bicarbonate buffer. b. Phosphate buffer. c. Protein buffer. d. Physiological buffers.

8-All are correct describing the acid base balance state in a patient suffering from excessive loss gastric HCL EXCEPT:

a) Renal compensation may occur through increase HCO3 excretion

b) Respiratory center may be inhibited to compensate this condition

c) pH may be near 7.45

d) Arterial blood gas changes are similar to that in severe diarrhea

9- Urinary buffers.... {Final2021}

a. Carbonic anhydrase in the DCT increases the buffering capacity.

b. Phosphate buffer system plays a role in buffering in the distal tubule and the collecting ducts.

c. The lowest pH of fluid in the tubular lumen occurs across the PCT.

d. Phosphate buffer system plays a role in buffering in the PCT.

10- The most important buffer in the ECF and plasma is: {Final2023}{Final2022}{3rd trial2022}

a. Bicarbonate buffer. b. Phosphate buffer. c. Protein buffer. d. Hemoglobin.

11-The most important buffer in the body is:

A. Carbonic acid / bicarbonate system B. Reduced / oxidized haemoglobin system

C. Proteinic acid / proteinate system D. The phosphate system (i.e. NaH2PO4/Na2HPO4)

12-The major buffer of CO2 in the blood is:

a. Haemoglobin b. Albumin c. Bounded to Chloride ions d. Bicarbonate ions

13- Bicarbonate buffer system is the most effective buffer system in: {Formative2022}

a) Extracellular fluid. b) Intracellular fluid c) Kidney. d) Cerebrospinal fluid.

14-The first line of defence against acid base shift is: {Formative2024}

a) Adipose tissue b) Renal system c) Blood buffers d) Respiratory system

15- Fluid entering the loop of Henle from the proximal convoluted tubule has a higher: {Merit}

a) Osmolarity than plasma b) Na+ than plasma

c) Glucose than plasma d) Ammonia than plasma

16- Ammonia secretion mechanism in the kidney helps in: {Merit}

a) Water balance b) Acid-base balance

c) Temperature regulation d) Maintaining oncotic pressure

17- The secretion of Ammonia depends upon the: {Merit}

a) pH of tubular fluid b) pH of blood c) Renal blood flow d) Plasma flow

18- Most important buffer system presents in the distal convoluted tubule: {Merit}

a) Bicarbonate b) Phosphate c) Protein d) Ammonia

19- Reabsorption of filtered bicarbonate or formation of new bicarbonate: {Merit}

a) Is enhanced by carbonic anhydrase inhibitors b) Contributes to excretion of titratable acid

c) Is reduced during acidosis d) Is accomplished by net secretion of Na+

20- In cases of acidosis, the kidneys {Merit}

a) excrete urine with pH below 4 b) increase the reabsorption of ammonia

c) increase the reabsorption of titratable acids d) increase the formation of bicarbonate

21-Incases of acidosis, the kidney can……

A. Form titratable acids B. Decrease reabsorption of NaHCO3

C. Decrease reabsorption of filtered buffers D. Decrease secretion of H+

22 - Which of the following occurs during acidosis: {Merit}

a) Urine pH becomes 6.5 b) Ammonia excretion is increased

c) Bicarbonate reabsorption is decreased. d) Hydrogen secretion is decreased

23-Types of buffers include {Merit}

a- Acetate buffer b- Bicarbonate buffer c- Lactate buffer d- Pyruvate buffer

24-Which out of the following conditions will not cause respiratory alkalosis? {Merit}

a) Fever b) Anxiety c) Laryngeal obstruction d) Salicylate toxicity

25- Which of the following is a cause of metabolic alkalosis? {Merit}

a) salicylate poisoning b) chronic renal failure c) diarrhea d) hyperaldosteronism

26-Cause of metabolic acidosis:

a)Asphaxia. b)Diabetes mellitus. c)Hyperventilation. d)Ingestion of NaHCO3

27-Cause of metabolic alkalosis:

a)Asphaxia. b)Diabetes mellitus. c)Hyperventilation. d) Ingestion of NaHCO3

28-Cause of respiratory acidosis:

a)Asphaxia. b)Diabetes mellitus. c)Hyperventilation. d)Chronic renal disease.

29-Cause of respiratory alkalosis:

a)Diabetes mellitus. b)Hyperventilation.

c)Regurgitation of gastric contents. d)Chronic renal disease

30-Ammonia adaptation means …..

A. Ability of ammonia to buffer excess acids

B. Ability of ammonia to be secreted as long as H+ is secreted

C. Ability of ammonia to transformed into NH4Cl

D. Decrease secretion of ammonia with excess acid secretion

31-The most important buffer in the renal tubules is:

a. Plasma proteins b. Phoshate buffer c. Haemoglobin d. Biocarbonate buffer

32-The most important buffer in the intracellular fluid is:

a. Biocarbonate buffer. b. Phoshate buffer. c. Plasma proteins. d. Haemoglobin

33-Which of the following buffers act as alkali reserve?

a. Protein buffer b. Bicarbonate buffer c. Phosphate buffer d. Hemoglobin buffer

34-Which of the following organs are mainly responsible for maintaining normal acid base balance?

a. Lungs and heart b. Kidney and liver c. Lungs and kidney d. Kidney and spleen

35- The following findings are found in metabolic acidosis , EXCEPT: {Merit}

a) decreased respiratory rate b) decreased excretion of $NaHCO_3$.

c) increased excretion of NH_4Cl in urine d) increased excretion of titratable acid in urine

36- In metabolic acidosis ,the primary abnormality is : {Merit}

a) CO_2 excess. b) bicarbonate deficiency c) bicarbonate excess d) Na^+ deficiency

37- Regarding acidification of urine by the renal tubule: {Merit}

a) K^+ is normally reabsorbed from the tubular fluid in exchange for H^+ secretion

b) the H^+ reacts with NaH_2PO_4 in the tubular fluid to give Na_2HPO_4 .

c) H^+ is secreted into tubular fluid by P cells lining the distal tubule and collecting duct.

d) most of H^+ secreted in tubular fluid in cases of acidosis is in the form of ammonium salts .

38-Normally, most of the H^+ is excreted by the kidney in the form of: {Merit}

a) HCO_3^- b) phosphate ion c) NH_4^+. d) titratable acid

39- Which of the following conditions is most likely to cause acidosis? {Merit}

a) severe persistent vomiting b) severe diarrhea

c) excessive sweating d) frequent drinking of sodium bicarbonate .

40- A 20-year old lady with diabetes mellitus was admitted to Kasr-Aini hospital. She was hyperpenic and dehydrated. Urine analysis showed glucosuria and Ketonuria .Blood report showed : [HCO3-] = 20 mmol / L P CO2 = 34 mmHg pH = 7.38 The most likely diagnosis is: {Merit}

a) metabolic acidosis with partial renal compensation

b) respiratory acidosis with complete renal correction

c) metabolic acidosis with complete respiratory compensation

d) respiratory alkalosis with partial renal compensation

41-In renal tubules, bicarbonate is responsible for buffering hydrogen ion in :. {VIP}

A) proximal convoluted tubules B) loop of henle

C)Distal convoluted tubules D) collecting tubules

42-The following changes are likely to be observed in hyperaldosteronism, EXCEPT:

a)Hyperkalemia. b)Metabolic alkalosis.

c)Decreased hematocrite. d)Increased ECF volume

1. C	2. A	3. A	4. D	5. C
6. A	7. B	8. D	9. B	10.A
11.A	12.D	13.A	14.C	15.D
16.B	17.B	18.D	19.B	20.D
21.A	22.B	23.B	24.C	25.D
26.B	27.D	28.A	29.B	30.B
31.B	32.D	33.B	34.C	35.A
36.B	37.D	38.C	39.B	40.C
41.A	42.A	43.	44.	45.

Micturition and Micturition reflex

1-First desire to void urine (micturate) occurs when urinary bladder contains about:

a)100 ml urine. b)150 ml urine. c)400 ml urine. d)700 ml urine

2- Micturition reflex is: {Final2023}

a. Absent in children

b. its Center in the medulla

c. Can be easily voluntarily inhibited if the urinary bladder volume reaches 150 ml

d. Efferent fibers are sympathetic only

3-In the normal bladder, micturition is: {3rd trial2022}{Final2022}{Formative2022}

a-Initiated by the pelvic nerves b-Initiated at a volume of 600mL

c-Significantly affected by sympathetic nerves d-Not facilitated at the level of the brain stem

4- Which of the following injury will lead to a large volume, hypotonic bladder? {Final2021}{VIP}

a. Transection of afferent pelvic nerve from bladder.

b. Transection of both afferent pelvic nerve & efferent parasympathetic nerve to bladder.

c. Transection of brain stem above midbrain.

d. Transection of brain stem above pons.

5- In the normal bladder, micturition is {Summar2022}

a- Initiated by the pelvic nerves b- at a volume of 600mL

c- Significantly affected by sympathetic nerves d- not facilitated at the level of the brain stem

6- Initial micturition is felt when the bladder gets: {Merit}

a) Just before urination b) 150ml of urine c) 400ml of urine d) 600ml of urine

7-The first desire for micturition is felt at a bladder volume of about …..ml

A. 150 B. 90 C. 300 D. 500

8- About Higher control of micturition:

a. Several centers located in the hypothalamus

b. They keep the micturition reflex always excited

c. They prevent micturition by continuous contraction of the internal urethral sphincter

d. Strong facilitative and inhibitory centers in the brain stem, mainly in the pons.

9- Micturition may be inhibited if: {Merit}

a) the sympathetic nerves carrying afferent information from bladder to spinal cord are cut

b) the pelvic nerves are cut

c) parasympathomimetic drug like prostigmine is administered

d) adrenergic antagonist is administered

10-Cutting the sympathetic nerve to the urinary bladder causes all the following, EXCEPT:

a)Loss of tone in the internal uretheral sphincter.

b)Loss of tone in the external uretheral sphincter.

c)Loss of pain sensation from the bladder. d)Infertility in the male

1. B	2. C	3. A	4. B	5. A
6. B	7. A	8. D	9. B	10.B